

Chapter 3 :

Diode Applications

3.1 Introduction

In chapter two, some common devices (PN junction based devices) have been discussed. In this chapter, circuit analysis with those devices has covered. Different types of diode circuits are described elaborately. The most common circuits covered in this chapter are: rectifier circuits, filter circuits, clipper circuits, clamper circuits and Zener diode regulator circuits. Their operations and mathematic analysis have been discussed. Moreover, the response of diode to small signal AC and large signal AC have also been discussed. Most of the electronic circuits require power. A simple power supply circuit has also been designed and described.

3.2 Analysis of DC Circuits Containing Diode

Let us consider a very simple diode circuit which consists of a resistor R and a diode D connecting in series [Fig.3-1(a)]. To power up the circuit a DC voltage source is connected. We are interested to find the value of current, I , in the circuit and the voltage drop across the resistor, V_R . Although this is a very basic circuit, it has a lot of applications. To calculate the value of V_R , first we have to find the value of I in this circuit. Depending on the required precision, the applied voltage level and the value of R , the circuit can be analyzed in four different ways to find out the value of I .

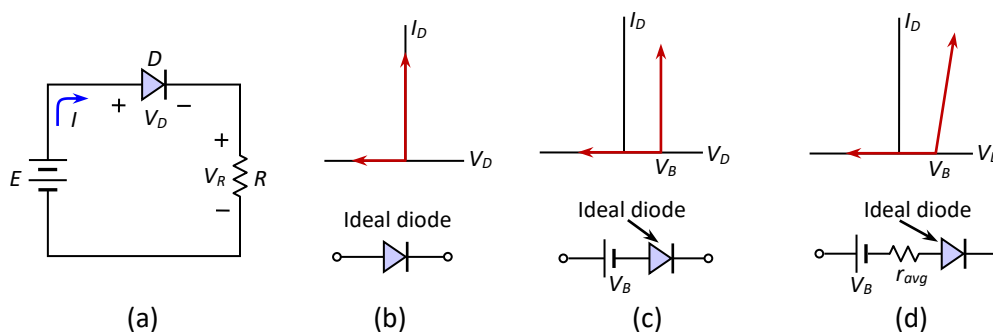


Fig.3-1: (a) Basic diode circuit and (b)-(d) diode equivalents used to analyze the circuit

3.2.1 Method 1 (Ideal Diode Model)

If the value of applied voltage is very high compared to the barrier voltage V_B of the diode and the value of R is large compared to the diode internal resistance, we can consider the diode as an ideal device. Then, the diode voltage drop can be neglected and the value of I can be calculated by using Ohm's law as,

$$I = \frac{E}{R} \quad (3-1)$$

Thus, the voltage drop across the resistor will be

$$V_R = I \times R = \frac{E}{R} \times R = E \quad (3-2)$$

The diode equivalent circuit and its characteristic curve used in this analysis are shown in Fig.3-1(b). This type of analysis will not give accurate results, but if the above mentioned conditions are fulfilled the results are acceptable.

3.2.2 Method 2 (Constant Voltage Model)

Although, the value of internal resistance of a diode is less than the external circuit resistance, sometimes the barrier voltage is not so small compared to the applied voltage. In that case, we have to consider the equivalent circuit and characteristic curve of Fig.3-1(c). So, applying KVL in this circuit we get,

$$\begin{aligned} E - V_D - V_R &= 0 \\ \text{or,} \quad E - V_B - IR &= 0 \\ \therefore I &= \frac{E - V_B}{R} \end{aligned} \quad (3-3)$$

Now, the value of V_R will be,

$$V_R = IR = \frac{E - V_B}{R} R = E - V_B \quad (3-4)$$

This method is most commonly used for diode circuit with low to medium voltage levels.

3.2.3 Method 3 (Piecewise Linear Model)

Now we will consider a situation where none of the barrier voltage and the internal resistance of the diode can be neglected. For this case, like the value of V_B , the value of internal resistance, r_{avg} , of the diode must be considered. The equivalent circuit and the diode characteristic curve for this situation are given in Fig.3-1 (c). So, using KVL we get,

$$\begin{aligned} E - V_D - V_R &= 0 \\ \text{or,} \quad E - V_B - Ir_{avg} - IR &= 0 \quad [\because V_D = V_B + Ir_{avg}] \\ \therefore I &= \frac{E - V_B}{R + r_{avg}} \end{aligned} \quad (3-5)$$

Now, the value of V_R will be,

$$V_R = IR = \frac{E - V_B}{R + r_{avg}} R \quad (3-6)$$

Although, this method will give the more accurate result than the first two methods, we have not considered the nonlinearity of the diode characteristic yet.

3.2.4 Method 4 (Exact Diode)

If we want to consider all the parameters (as used in method 3) as well as the nonlinearity of the diode characteristic, we have to use the graphical method or analytical method using Shockley's equation. In the graphical method, the graph of the equation $E - V_D - V_R = 0$ has to be plotted on the diode I - V characteristic curve of the diode. The value of the circuit current and diode voltage drop, V_D , will be determined from the intersecting point. To use this method, the I - V characteristic curve of the diode must be provided or we have to determine it first. In analytical method, the circuit current and voltage levels can be determined by solving the Shockley's equation. These methods are discussed elaborately later. Among all these analysis methods the first two approximations are used frequently.

Example 3-1

For circuit of Fig.3-2, determine V_D , V_R and I considering the above mentioned three equivalent circuits of the diode. Assume a Si diode with $V_B = 0.7$ V, and $r_{avg} = 100 \Omega$.

Solution:

In the given circuit, the diode is forward biased by $E = 12$ V.

Method 1 (Ideal diode):

Using Equ.(3-1),
$$I = \frac{E}{R} = \frac{12 \text{ V}}{2.2 \text{ k}} \approx 5.46 \text{ V [Ans.]}$$

Using Equ.(3-2),
$$V_R = I \times R = 12 \text{ V [Ans.]}$$

Thus,
$$V_D = E - V_R = 0 \text{ V [Ans.]}$$

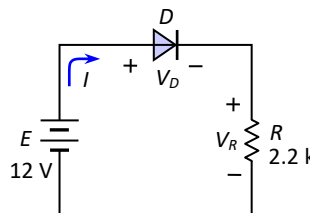


Fig.3-2: Diode circuit for Example 3-1

Method 2 (Diode with barrier voltage):

For Si diode,
$$V_D = 0.7 \text{ V [Ans.]}$$

Using Equ.(3-3),
$$I = \frac{E - V_B}{R} = \frac{12 \text{ V} - 0.7 \text{ V}}{2.2 \text{ k}} \approx 5.14 \text{ V [Ans.]}$$

Using Equ.(3-4),
$$V_R = E - V_B = 12 \text{ V} - 0.7 \text{ V} = 11.3 \text{ V [Ans.]}$$

Method 3 (Diode with barrier voltage and resistance):

Using Equ.(3-5),
$$I = \frac{E - V_B}{R + r_{avg}} = \frac{12 \text{ V} - 0.7 \text{ V}}{2.2 \text{ k} + 100 \Omega} \approx 4.91 \text{ V [Ans.]}$$

Using Equ.(3-6),
$$V_R = IR = 4.91 \text{ V} \times 2.2 \text{ k} = 10.80 \text{ V [Ans.]}$$

Method 4 (Diode with barrier voltage, resistance and nonlinear property):

As the I - V characteristic of the diode is not given, we cannot use the graphical method. Moreover, as the value of I_S is not given, we cannot use the Shockley's equation too.

Comments: Among the three methods used to solve the circuit, the last one gives the most accurate results.

Example 3-2

For the circuit of Fig.3-3, calculate the values of I_D , I_{R1} , V_o , and the exact diode voltage drop, V_D . Assume the bulk resistance of the diode $r_D = 10 \Omega$, $I_S = 10^{-9} \text{ mA}$, and $T = 27^\circ\text{C} = 300 \text{ K}$.

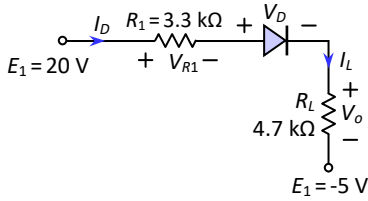


Fig.3-3: Diode circuit for Example 3-2

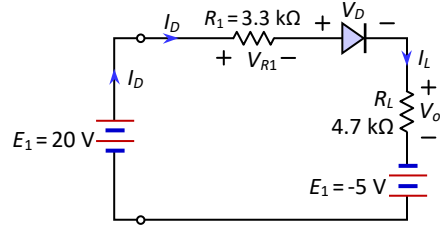


Fig.3-4: The circuit of Fig.3-3 is redrawn

Solution:

The given diode circuit is redrawn in Fig.3-4, using batteries. From the polarity of the batteries, it is clear that the connected voltages (batteries) will add up and the total voltage will be,

$$E_T = 20 \text{ V} + 5 \text{ V} = 25 \text{ V}$$

Here, we have to determine the exact voltage drop across the diode. So, we have to use Shockley's equation for diode current. If I_D mA is the current in the circuit, the voltage applied across the PN junction of the diode will be,

$$V_D = 25 \text{ V} - I_D(R_1 + r_D + R_L) \quad [\text{considering all current in mA}]$$

or,

$$V_D = 25 \text{ V} - I_D(3.3 + 0.01 + 4.7) = 25 \text{ V} - 8.01I_D$$

Now using this value into Shockley's equation we get,

$$I_D = (10^{-9} \text{ mA}) \left(e^{(25 \text{ V} - 8.01I_D)/(0.025875 \text{ V})} - 1 \right)$$

Here, I_D is the only unknown variable. But as this variable is on both sides of the expression and on the right side it is on the power of e , it will be complicated to solve this expression analytically! On the other hand, we can solve it using iteration method.

The solution of this expression has been done using iteration method and found,

$$I_D = 3.050552624 \text{ mA} [\text{Ans.}]$$

$$V_{R1} = I_D R_1 = (3.050552624 \text{ mA}) \times (3.3 \text{ k}\Omega) = 10.066823659 \text{ V} [\text{Ans.}]$$

$$V_{RL} = V_o = I_D R_L = (3.050552624 \text{ mA}) \times (4.7 \text{ k}\Omega) = 14.3375973328 \text{ V} [\text{Ans.}]$$

The voltage across the diode will be,

$$V_D = E_T - I_D R_1 - I_D R_L$$

$$\therefore V_D = 25 \text{ V} - (10.066823659 \text{ V} + 14.3375973328 \text{ V}) = 0.595579008 \text{ V} [\text{Ans.}]$$

Let's check the result:

The voltage drop across the bulk resistance (inside the diode) be,

$$V_D = E_T - I_D R_1 - I_D R_L$$

$$V_{rD} = I_D r_D = (3.050552624 \text{ mA}) \times (10 \Omega) = 0.03050552624 \text{ V}$$

Therefore, the voltage across the PN junction is,

$$0.595579008 \text{ V} - 0.03050552624 \text{ V} = 0.56507348176 \text{ V}$$

Putting this value into Shockley's equation,

$$I_D = (10^{-9} \text{ mA})(e^{(0.56507348176 \text{ V})/(0.025875 \text{ V})} - 1) = (10^{-9} \text{ mA})(e^{22.00966183574879} - 1) \\ = 3,050,552,829.5324443817 \times 10^{-9} \text{ mA} \approx 3.050552829 \text{ mA [checked]}$$

Comments: In this example, we find the applications of Shockley's equation. To get accurate results, more digits have been considered in the calculation than usually used.

3.3 Analysis of Small-Signal Diode Circuit

Generally speaking, electronic devices can be employed in two distinct kinds of operation: **small signal** and **large signal**. In small signal applications, the current and voltage change in the device occurs over a very limited range of its I - V characteristic curve. In practice, we can consider small signal operation when the current and voltage change over a linear portion of the device characteristic curve.

Consider the circuit shown in Fig.3-5(a), containing a DC voltage source, E , and an AC voltage source $e = A \sin(\omega t)$. Here, the peak value A is assumed to be small compared to E . The total voltage applied to the diode circuit is $v(t) = E + A \sin(\omega t)$ which is called an AC voltage with a DC

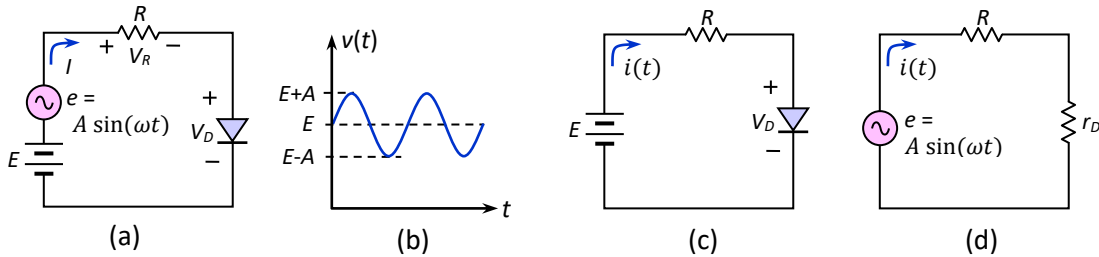


Fig.3-5: (a) AC voltage with a DC level of E (b) Small signal diode circuit (c) AC equivalent circuit

level of E volts. It is sketched in Fig.3-5(b). Note that $v(t)$ has the maximum value of $(E + A)$ and the minimum value of $(E - A)$. We want to determine the voltage across and the current through the diode.

To analyze this circuit we can use *superposition theorem*. Although superposition can be applied only when the circuit components are linear, assuming small signal operation, we are justified in applying it to the present problem.

Eliminating the AC voltage source from Fig.3-5(a), we obtain a circuit like Fig.3-5(c). Now using Equ. (3-3) and considering a Si diode we have $I = (E - 0.7)/R$. Using this current, we can, then, find the AC resistance as $r_D = 26 \text{ mV}/I$.

Eliminating the DC voltage source, the AC equivalent circuit is shown in Fig.3-5(d). Note that in the AC equivalent circuit the diode is represented by a single resistance $r_D \Omega$. Thus, the AC current can be calculated directly, using Ohm's law, as,

$$i = \frac{e}{R + r_D} = \frac{A \sin(\omega t)}{R + r_D} = \frac{A}{R + r_D} \sin(\omega t) \quad (3-7)$$

The AC voltage drop across the diode, v_D , can be calculated as $v_D = i \times r_D$ or by applying the voltage divider rule as,

$$v_D = \frac{r_D}{R + r_D} e = \frac{r_D A \sin(\omega t)}{R + r_D}$$

Combining the results of DC and AC analyses, we can find the total current and voltage drop in the diode as,

$$i(t) = I + i = \frac{E - 0.7}{R} + \frac{A}{R + r_D} \sin(\omega t) \quad (3-8)$$

and

$$v_D(t) = 0.7 + \frac{r_D A \sin(\omega t)}{R + r_D} \quad (3-9)$$

Once we have calculated the total current, we can also calculate the voltage drop across the resistor as $v_R(t) = i(t) \cdot R$.

If we neglect r_D , the AC signal will not drop across the diode. It will be fully developed across the external resistor. This is possible only when the diode has a DC bias voltage like E .

Example 3-3

As shown in the Fig.3-6(a), a silicon diode is biased with 6 V DC and then an AC signal $e = 2 \sin(\omega t)$ is applied. The bulk resistance of the diode is 1Ω . Calculate the current through and the voltage drop across the diode. Sketch the current versus time graph.

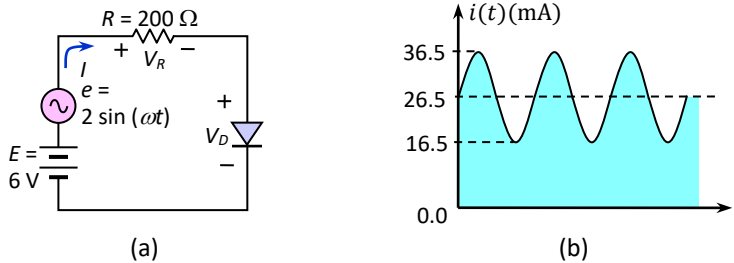


Fig.3-6: (a) Circuit for Example 2-3, (b) Sketch of the current of the circuit

Solution:

Using Equ.(3-9) we find,
$$v_D(t) = 0.7 + \frac{r_D A \sin(\omega t)}{R + r_D}$$

Using Equ.(3-3) we find,
$$I = \frac{E - 0.7}{R} = \frac{6 - 0.7}{200} \text{ A} = 26.5 \text{ mA}$$

So, diode AC resistance will be
$$r_D = \frac{26 \text{ mV}}{I} + r_B = \frac{26 \text{ mV}}{26.5 \text{ mA}} + 1 \Omega = 2 \Omega$$

Now using Equ.(3-8) and Equ.(3-9) we will calculate the total current and diode voltage.

$$i(t) = I + i = \frac{6 - 0.7}{200} + \frac{2}{200 + 2} \sin(\omega t) = 26.5 \text{ mA} + 10 \sin(\omega t) \text{ mA}$$

and
$$v_D(t) = 0.7 \text{ V} + \frac{r_D A \sin(\omega t)}{R + r_D} = 0.7 \text{ V} + \frac{2 \times 2}{200 + 2} \sin(\omega t) = 0.7 + 0.02 \sin(\omega t) \text{ V}$$

The sketch of the current versus time graph is shown in Fig.3-6(b).

Comments: Here the diode is forward biased by the DC voltage ($E = 6 \text{ V}$) and the AC current passes through this forward biased diode. If the swing of the AC current is small, and a linear part of the diode I - V characteristic is used, the shape of the AC current will almost same as that of the AC voltage.

3.4 Load Line Analysis of Diode Circuit

Analysis of diode circuit can be performed graphically using the I - V characteristic curve of the diode. Although this method is not frequently used in practice, it gives the exact solution that will happen practically in the circuit. Consider a diode circuit as shown in the following figure (Fig.3-4a).

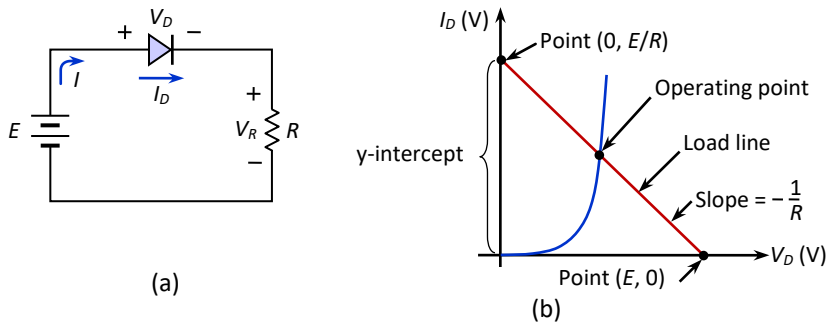


Fig.3-7: (a) DC diode circuit, (b) Load line and operating point

By applying Kirchhoff's voltage law we get, $E - V_D - I_D R = 0$. Solving this equation for I_D , we get,

$$I_D = -\frac{V_D}{R} + \frac{E}{R} \quad (3-10)$$

In this equation, V_D and I_D are variables and E and R are constants. This is an equation of straight line that is similar to

$$y = mx + c \quad (3-11)$$

where, m is the slope of the line and c is the y -intercept.

Further comparison of equations (3-10) and (3-11) reveals that the slope m corresponds to $-1/R$ while y -intercept c corresponds to E/R . We conclude that the graph of Equ.(3-10) is a straight line on $I_D - V_D$ axes and has slope $-1/R$ and I_D -intercept of E/R . This line is called the DC **load line**. Although we can plot this line using the y -axis intercept and the slope ($-1/R$), it is little bit complicated as it uses slope. Rather we will plot this line as below.

When $I_D = 0$, $V_D = E$, these give a point on the x -axis (actually V_D axis).

When $V_D = 0$, $I_D = E/R$, give another point on the y -axis (actually I_D axis).

Now, connecting these two points we can draw the straight line of Equ.(3-11) on $I_D - V_D$ coordinate system.

The significance of this load line is that it gives every possible combination of the current I_D and voltage V_D in the give circuit. As the value of current I_D is the load for the diode and this line shows all possible values of I_D , it is called **load line**. Alternately, as the slope of this line is $-1/R$ and for some cases R is called the load resistance, this line has such a name.

To solve the circuit using graphical method we have to draw the load line on the same graph of the I - V characteristic curve of the diode. The intersecting point of the load line and the diode I - V characteristic curve will give the values of current (I_D) in the circuit and the voltage drop (V_D) across the diode. This intersecting point is called the **operating point** or **quiescent point (Q-point)** or **bias point**. The method is elaborated further in the following example.

Example 3-4

Using the I - V characteristic curve of Fig.3-8(b), determine the value of V_D and I_D for the circuit of Fig.3-8(a).

Solution:

First of all, we have to draw the load-line on the given I - V characteristic curve of the diode.

Applying KVL in the given circuit we get,

$$E - V_D - I_D R = 0$$

When $I_D = 0$,

$$V_D = E = 5 \text{ V}$$

and when $V_D = 0$,

$$I_D = \frac{E}{R} = \frac{5 \text{ V}}{430 \Omega} = 11.63 \text{ mA}$$

If we put these two points on the given $I_D - V_D$ coordinate system, we get the load line as shown in the Fig.3-8(b). The intersecting point of this load line and the I - V characteristic curve gives the Q -point. From this Q -point we get,

$$I_D = 10.22 \text{ mA [Ans.]}$$

and

$$V_D = 0.73 \text{ V [Ans.]}$$

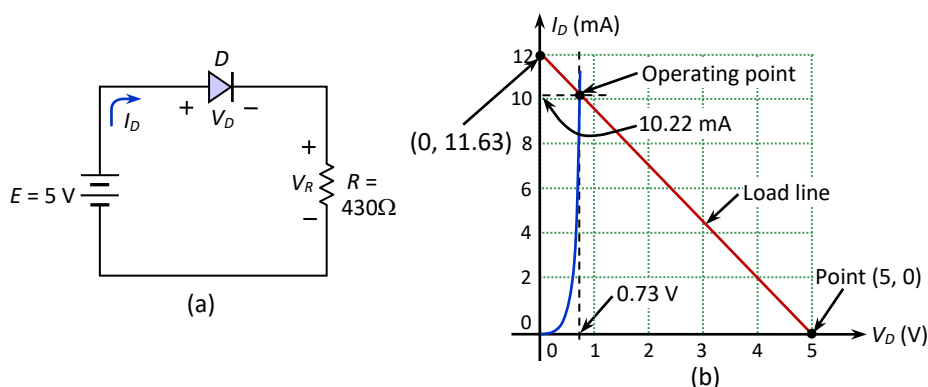


Fig.3-8: (a) Diode circuit for Example 2.2, (b) Load-line, operating point and value of I_D and V_D for the given circuit

Comments: Here the values of I_D and V_D have been calculated graphically. Graphical method should give better results, but reading the values from the intersecting point introduces error. To overcome this problem upto a certain level, we have to use a very large graph paper with more details.

Example 3-5

Repeat Example 2-4 using the approximate equivalent model for the silicon semiconductor diode.

Solution:

Using the same procedure, described in Example 3-4, the load line and the operating point have been drawn on the approximate characteristics of the diode. Here also the operating point is $(0.73, 10.22 \text{ mA})$. So, $I_D = 10.22 \text{ mA}$ [Ans.] and $V_D = 0.73 \text{ V}$ [Ans.]

Comments: Here a constant barrier voltage has been considered. So the value of V_D will be same whatever the position of operating point. For better understanding, the diode equivalent circuit is given in the same graph.

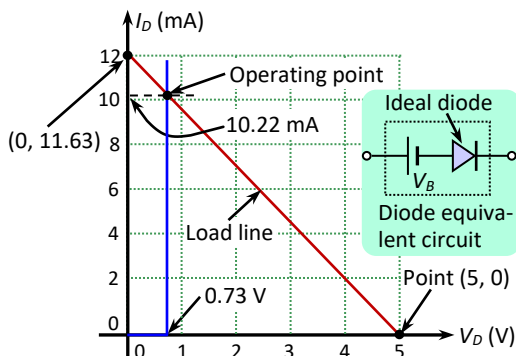


Fig.3-9: Load-line, operating point and value of I_D and V_D for the given circuit

Example 3-6

Repeat Example 3-4 using the ideal diode model.

Solution:

Using the same procedure as described in Example 3-4, the load line and the operating point have

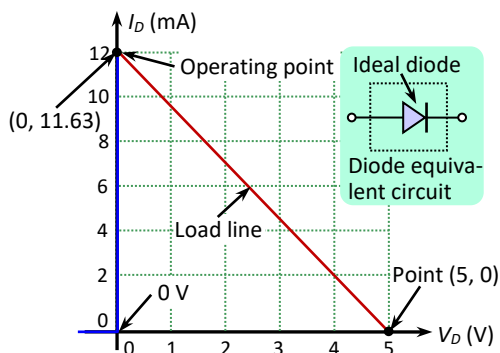


Fig.3-10: Load-line, operating point and value of I_D and V_D for the given circuit

been drawn on the ideal approximate characteristics of the diode (Fig.3-10). Here, the operating point is (0, 11.63 mA). Therefore, $V_D = 0$ V [Ans.], and $I_D = 11.63$ mA [Ans.]

Comments: The analytical method to solve this circuit will be easier than graphical method. Here, we have used graphical method to observe the position of the operating point.

3.5 Diode as a Switch

Already we have got the concept that diode can be used as a switch. Actually diode is a voltage controlled one-way switch as it conducts when forward biased and block the current when reverse biased. A diode can be used as a switch only for DC voltage or current. If an AC voltage is applied in a diode circuit, without any DC bias, only one half-cycle of the AC voltage will pass through the diode. This operation is called rectification and will be discussed later. But if an AC voltage is applied to a diode circuit with a DC bias voltage the whole AC signal will pass through the diode as explained in section 3.3. Thus, a diode can be used as a switch only for DC. As shown in Fig.3-6, the controlling of the ON/OFF state of the diode is done by the polarity of the applied voltage. If the applied voltage forward biases the diode, we can say the diode is ON (considering ideal diode) as shown in Fig.3-11(a). On the other hand, if the applied voltage reverse biases the diode, it behaves as an OFF switch [Fig.3-11(b)].

There is another way in which we can determine the state of a diode. If the conventional current (positive current) of a circuit tries to flow in the same direction as the arrow-head of the diode indicate, it will work as a closed switch and current will flow. If the conventional current of a circuit tries to flow in the opposite direction to the arrow-head of the diode, it will behave as an open switch and current will not flow. Note the direction of current and diode arrow of the Fig.3-11 to understand the description. Thus, we can say a diode is a one-way switch.

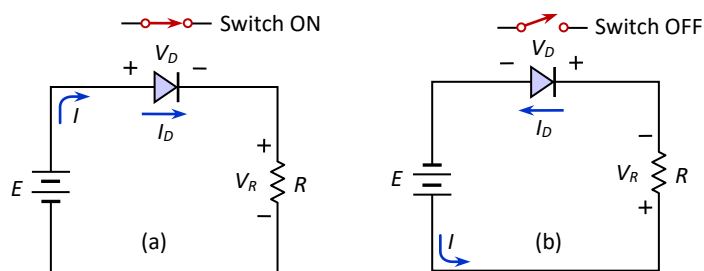


Fig.3-11: Diode as a switch (a) Forward biased- ON switch (b) Reversed bias- OFF switch

Practical Application of Diode as a Switch

Most of the DC-powered loads (like TV, Mobile, digital telephone set etc.) has polarity. That is the positive terminal of a power source (like a batter, power supply) has to be connected to the +ve terminal of the load, and the negative terminal of the

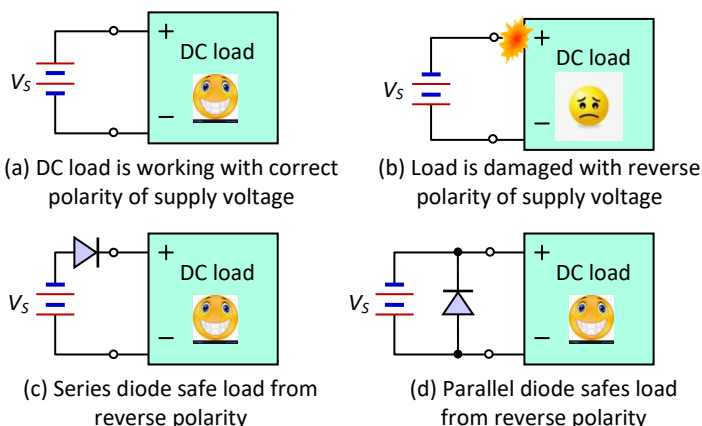


Fig.3-12: Reverse polarity protection using diodes

power source has to be connected to the –ve terminal of the load. The load will be damaged if the power is connected in reverse polarity as described in Fig.3-12. If the power is supplied from an external source, there has a chance of mistake and the risk of damaging the loads! This situation is illustrated in Fig.3-12(b). This problem can be overcome using diodes. The diode may be used in series or in parallel to the load.

In case of series diode, if the external power source is connected with wrong polarity (-ve voltage to +ve terminal), the diode will be reverse biased and will work as an open circuit [Fig.3-12(c)]. So the load will be protected from being damaged. But if the external power is connected with correct polarity (+ve voltage to +ve terminal), the diode will be forward biased and work as a short circuit. So the load will get power and will run properly. But remember that due to the barrier voltage of the diode, V_B amount of voltage (0.7 V for Si diode, and 0.3 V for Ge diode) will be dropped in the diode. The load will get $V_S - V_B$ voltage less than the supply voltage.

In case of parallel connected diode, if the external power source is connected with wrong polarity (-ve voltage to +ve terminal), the diode will be forward biased and will bypass the reverse current from the load. So the load will be protected from being damaged. But if the external power source is connected with correct polarity (+ve voltage to +ve terminal), the diode will be reverse biased and work as an open circuit. In this case the diode has no effect on the operation of the load. Here, if the supply is connected incorrectly, a huge amount of current will flow through the diode and will produce a huge power loss. So this parallel diode protection circuits are used only for very low power loads.

Example 3-7

For the diode circuit of Fig.3-13(a), calculate the output voltage (V_o) and output current (I_o) considering the barrier voltages of the diode.

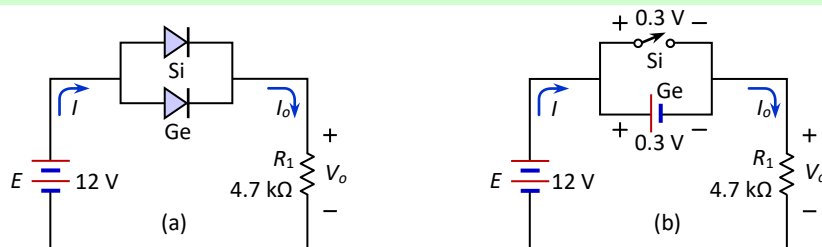


Fig.3-13: (a) Diode circuit for Example and (b) Equivalent circuit

Solution:

As the positive terminal of the battery is connected to the anodes of the diodes, the applied voltage will turn both diodes ON. We also know that, the voltage across parallel components must be the same. However, if both diodes were ON, the 0.7 V drop across the silicon diode would not match the 0.3 V across the germanium diode. When the supply is turned ON the voltage across both diodes will increase. But at 0.3 V the Ge diode will be ON and 0.3 V will remain almost constant across it. Thus, the maximum voltage across the Si diode will be 0.3 V. So, the Si diode will never be ON. Rather, it will remain always OFF for this circuit. Thus, the output voltage which is the voltage across the R_1 resistor will be,

$$V_o = V_{R1} = 12 \text{ V} - 0.3 \text{ V} = 11.7 \text{ V [Ans.]}$$

Therefore, the current through this resistor will be,

$$I_o = I_{R1} = \frac{V_{R1}}{R_1} = \frac{11.7 \text{ V}}{4.7 \text{ k}\Omega} \approx 2.49 \text{ mA [Ans.]}$$

Comments: For 0.3 V across the Si PN junction, a very small current will flow through it. If we neglect this current, we can consider the Si diode as open.

Example 3-8

Fig.3-14 shows a 12 V battery charging circuit. v_i is a sinusoidal voltage with 24 V peak amplitude. Find the portion of each cycle during which the diode conducts and the battery gets charge. Also, find the peak value of the diode current and the maximum reverse-bias voltage that appears across the diode. Assume the internal resistance of the battery and source is zero.

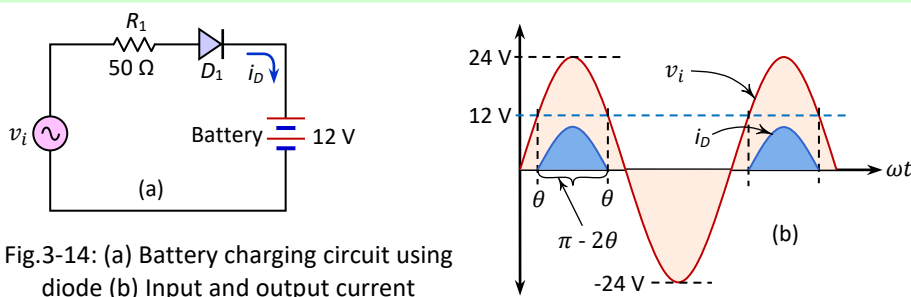


Fig.3-14: (a) Battery charging circuit using diode (b) Input and output current

Solution:

The diode conducts when v_i exceeds 12 V, as shown in Fig.3-14(b). For the negative half-cycle, the diode is completely OFF. Let the conduction angle be θ that is the point where the diode conducts. The value of θ can be calculated as,

$$v_i = 12 \text{ V}$$

$$\text{or,} \quad 24 \sin(\omega t) = 24 \sin(\theta) = 12$$

$$\text{or,} \quad \sin(\theta) = \frac{12}{24} = 0.5$$

$$\text{or,} \quad \theta = \sin^{-1}(0.5) \approx 30^\circ$$

Therefore, the diode will conduct during $180^\circ - 2\theta = 180^\circ - 2 \times 30^\circ = 120^\circ$, i.e., one third of a cycle.

The peak value of the diode current can be calculated as,

$$I_D = \frac{24 \text{ V} - 12 \text{ V}}{50 \Omega} \approx 0.24 \text{ A [Ans.]}$$

The maximum reverse voltage across the diode occurs when v_i is at its negative peak value. For negative half-cycle, v_i will add up with the battery voltage. Therefore,

$$\text{Maximum reverse bias voltage (PIV)} = v_{i(\text{peak})} + V_{\text{Batt}} = 24 \text{ V} + 12 \text{ V} = 36 \text{ V [Ans.]}$$

Comments: Actually this circuit is a half-wave rectifier which will be discussed in Section 3.7.

3.6 Logic Gate Implementation using Diodes (OR gate and AND gate)

Using the ON/OFF switching property of diodes different logic gates can be implemented. To get some idea two basic gates are implemented and explained here.

OR Gate:

The circuit diagram for an OR gate is shown in Fig.3-15 and the equivalent circuits for different input logic combinations are shown in Fig.3-16. At first let the input values be $A = 0$ and $B = 0$. In this case, the anode and cathode of both diodes are connected to ground. Thus, they will not be forward biased and will behave as open circuit as shown in Fig.3-16(a). So no current will flow through the resistor R and the output voltage will be zero, i.e., Logic '0'. In digital electronics, low voltage is considered as Logic '0' (or LOW) and high level of voltage is considered as Logic '1' (or HIGH). Here, we have considered Logic '1' = 12 V.

For logic input combinations $A = 0, B = 1$ and $A = 1, B = 0$ one of the diodes is forward biased and one of them is reverse biased as shown in Fig.3-16(b) and (c). The reverse biased diodes have no effect in the circuit. Current will flow through the forward biased diodes and voltage will be developed across the resistor R . Deducing the diode's voltage drop (0.7 V) the output voltage will be $V_o = 11.3$ V. That is, we get Logic '1' or HIGH output.

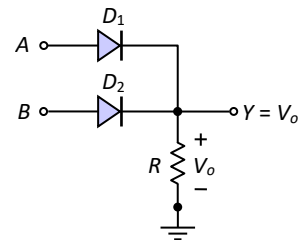


Fig.3-15: Circuit diagram of an OR gate using diodes

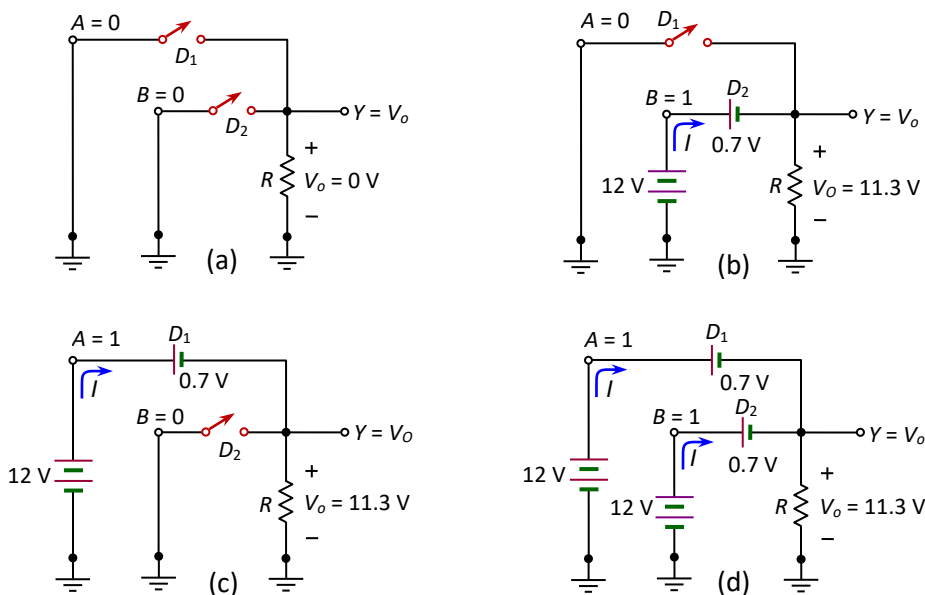


Fig.3-16: Equivalent circuits for different input logic combinations

Similarly, for the input combination $A = 1$ and $B = 1$ both diodes are forward biased and the output voltage will be $V_o = 11.3 \text{ V}$, which is Logic 1. The outputs found for different input combinations are summarized in the Table 3-1 (called **Truth Table** in digital electronics). We find that the output is LOW only when both inputs are LOW and when any one or both inputs are HIGH the output is HIGH. This type of logic operation is called **OR operation** and circuit is called **OR Gate**.

Table 3-1: Truth Table of OR gate (circuit of Fig.3-15)

Input combinations				Output	
A		B		Y	
Logic value	Circuit value	Logic value	Circuit value	Logic value	Circuit value
0	0 V	0	0 V	0	0 V
0	0 V	1	12 V	1	11.3 V
1	12 V	0	0 V	1	11.3 V
1	12 V	1	12 V	1	11.3 V

AND Gate:

The circuit diagram of an AND gate is shown in Fig.3-17 and the equivalent circuits for different input logic combinations are shown in Fig.3-18(a) to 3-18(d). At first, let us consider the input values be $A = 0$ and $B = 0$. In this case, the diodes will be forward biased and current will flow through the diodes as shown in Fig.3-8(a). Most of the supply voltage will be dropped across resistor R and the output voltage will only 0.7 V (forward voltage drop of a diode). This low voltage is considered as Logic '0' or LOW.

For logic input combinations $A = 0, B = 1$ and $A = 1, B = 0$ one of the diodes is forward biased and the other is reverse biased as shown in Fig.3-18(b) and 3-18(c). The reverse biased diodes have no effect in the circuit as they are open. Current will flow through the forward biased diodes and the supply voltage will be dropped across the resistor R . Thus, we will get output $V_o = 0.7 \text{ V}$, i.e., Logic '0'.

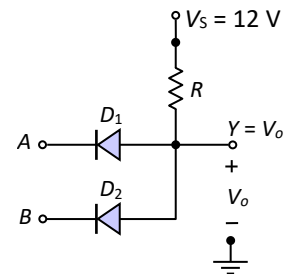


Fig.3-17: Circuit diagram of an AND gate using diodes

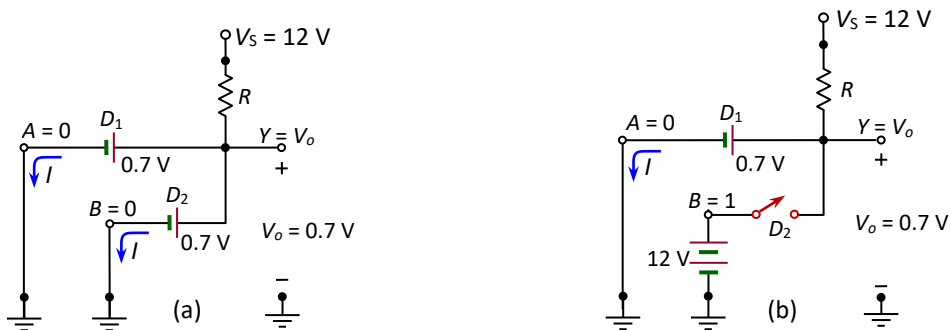


Fig.3-18: Equivalent circuits of AND Gate for different input logic combinations

Now, for the input combination $A = 1$ and $B = 1$ both diodes are reversed biased and they behave as open circuit. In this case, if the output terminal is open, no current will flow through resistor R . Hence, $V_R = 0$ and the full 12 V will be available at the output terminal. That is, the output is Logic '1' or HIGH. If we connect loads at the output terminal current will flow through R and the output

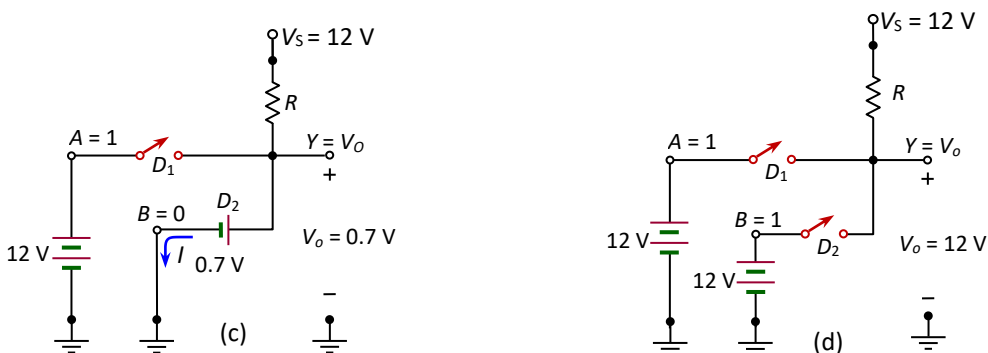


Fig.3-18: Equivalent circuits of AND Gate for different input logic combinations

voltage will decrease from 12 V. This is called **loading effect** and can reduce the output level but not less than an acceptable range. The outputs found for different input combinations are summarized in Table 2.2. From the table, we find that the output is *HIGH* only when both input are *HIGH* and the output is *LOW* when any of the inputs or both inputs are *LOW* which is the operation of an AND gate.

Table 3-2: Truth Table of AND gate (circuit of Fig.3-17)

Input combinations				Output	
A		B		Y	
Logic value	Circuit value	Logic value	Circuit value	Logic value	Circuit value
0	0 V	0	0 V	0	0.7 V
0	0 V	1	12 V	0	0.7 V

1	12 V	0	0 V	0	0.7 V
1	12 V	1	12 V	1	12 V

Example 3-9

Fig.3-19 shows an AND gate, with a voltage source (10 V) in series with the output load. The logic level of this circuit is: HIGH = 10 V to 10.5 V and LOW = 0 to 1 V. Determine the output level for the input levels shown in the figure. Also determine the current flowing through the load resistance.

Solution:

The network is redrawn in Fig.3-20 with the state of the diodes. As the logic level of input B is LOW (= 0 V), and 10 V is connected to the anode of D_2 (through 1 k resistor), this diode will be forward biased. Assuming $V_{B2} = 0.7$ V, the current through the resistor R will be,

$$I = \frac{10 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = 9.3 \text{ mA}$$

The logic level of A is HIGH (= 10 V) which is applied to the cathode of diode D_1 and the anode voltage of this diode is 0.7 V. So this diode will be reverse biased or OFF. With this situation the output voltage of the circuit will be 0.7 V.

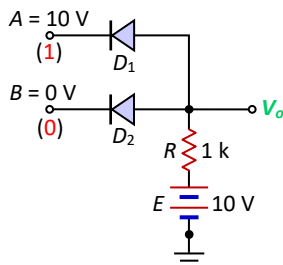


Fig.3-19: AND gate for Example 3-9

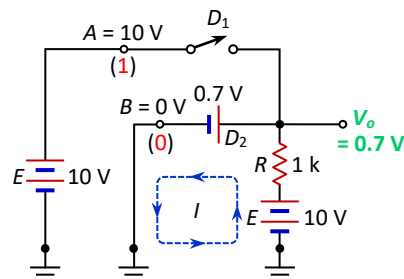


Fig.3-20: Circuit for given logic states

Comments: Actually this is another combination of AND gate. Study all other combinations.

3.7 Rectifier Circuits

In electronic circuits four types of converters are used. The circuits that convert AC power into DC power are called **rectifiers**. On the other hand, the circuits that convert DC power into AC power are called **inverters**. Examples of inverters are **uninterruptable power supply (UPS)** used in computer and **instant power supply (IPS)** which is used in our house during load shedding. There are two other types of converters. These are **DC-to-DC converter** and **AC-to-AC converter**. The DC-to-DC converter is used to convert high voltage to low voltage or vice versa. Examples of DC-to-DC converters are – buck-converter, boost converter, etc. The AC to AC converter is mainly used to convert the frequency of AC power, e.g., Cycloconverter. In this topic the construction and operation of only rectifiers will be discussed elaborately.

There are two types of rectifiers. They are *half-wave rectifier* and *full-wave rectifier*. Full-wave rectifier may be of two types – *two-diode full-wave rectifier* and *full-wave bridge rectifier*.

3.7.1 Half-wave Rectifier

Generally, rectifiers are used to develop DC power-supply. At first, the 220 V AC (single phase supply voltage in Bangladesh) power is reduced to the required value using a step-down transformer. The AC output of the transformer is converted into DC voltage using the rectifiers. The half-wave rectifier is the simplest form of all rectifiers and it consists of a single rectifier diode as shown in the Fig.3-21(a). The circuit is redrawn in Fig.3-21(b), with a sinusoidal voltage source, in place of the transformer.

The output of the transformer has both positive and negative half-cycle. For the positive half-cycle (0 to $T/2$) the diode D_1 is forward biased and current flows through the load resistance R_L . Alternately, we can say, for +ve half-cycle, as the conventional current flows along the direction of the arrow of the diode symbol, current will flow. If we consider an ideal diode, the whole of the positive half-cycle will be developed across the load resistance as shown in Fig.3-22(a).

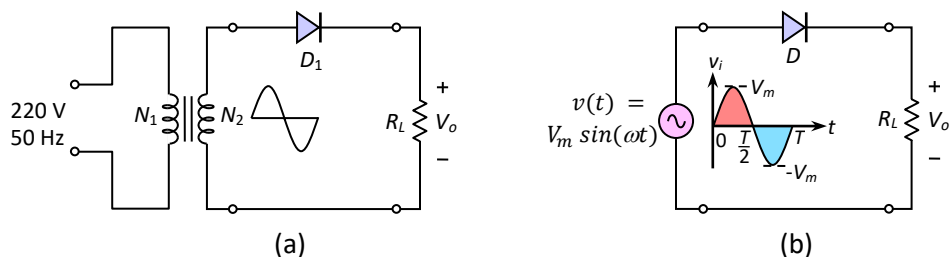


Fig.3-21: (a) Half-wave rectifier circuit with transformer,
(b) half-wave rectifier with equivalent AC source

For the negative half-cycle ($T/2$ to T), diode D_1 will be reverse biased and current will not flow through it. As there is no current, the voltage across the load resistance R_L will be zero [Fig.3-22(b)]. Here, we have neglected the reverse current of the diode. Alternately we can say as the direction of the suggested conventional current is opposite to the arrow of the diode current will not flow. Thus, we will not find any voltage at the output for the negative half-cycle. If we think of the

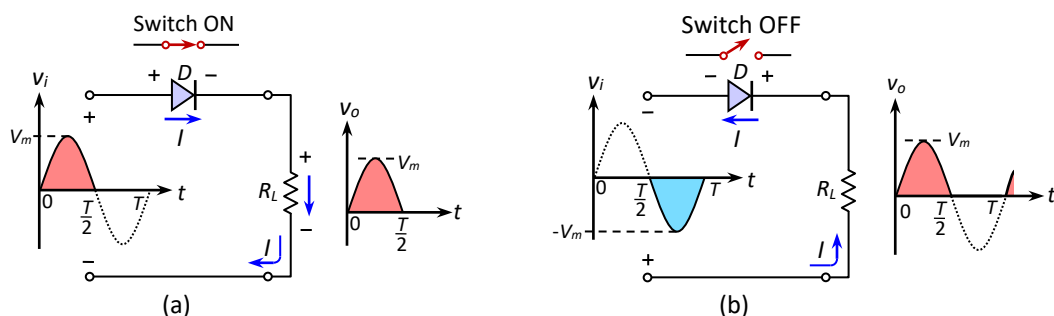


Fig.3-22: (a) Circuit behavior during +ve half-cycle, and
(b) Circuit behavior during -ve half-cycle

direction of the current flow through the load, we find that the current flows only in one direction, i.e., downward. Hence, we can say the current is no more alternating, i.e., the rectifier has

converted the AC power into DC power. The half-wave rectified waveform is shown in Fig.3-23(a). But this output voltage is not pure DC. It is called **pulsating DC**. It has DC voltage as well as some AC voltage. The AC voltage can be removed using filter circuit and it is discussed later. If we consider the barrier voltage of the diode, the output voltage will be $V_i - V_D$ and the waveform will be as shown in Fig.3-23(b).

Peak Inverse Voltage (PIV)

The **peak inverse voltage (PIV)** equals the maximum value of the voltage by which the diodes of a rectifier are reverse biased. This is the value of the maximum reverse bias voltage that the rectifier diodes must withstand, i.e., the V_{BR} (reverse breakdown voltage) value of the diode must be greater than the PIV value. For the half-wave rectifier we find that the diode is reverse biased by the peak voltage of the AC input. Thus the PIV rating of the diode of the half wave rectifier should be,

$$PIV = V_m = \sqrt{2} \times V_{rms} \quad (3-12)$$

For example, if we use a 12 V transformer in the circuit of Fig.3-21(a), the PIV rating of the diode should be $\sqrt{2} \times 12 \text{ V} = 17 \text{ V}$. The reverse breakdown voltage of the diode used in this circuit must be greater than 17 V.

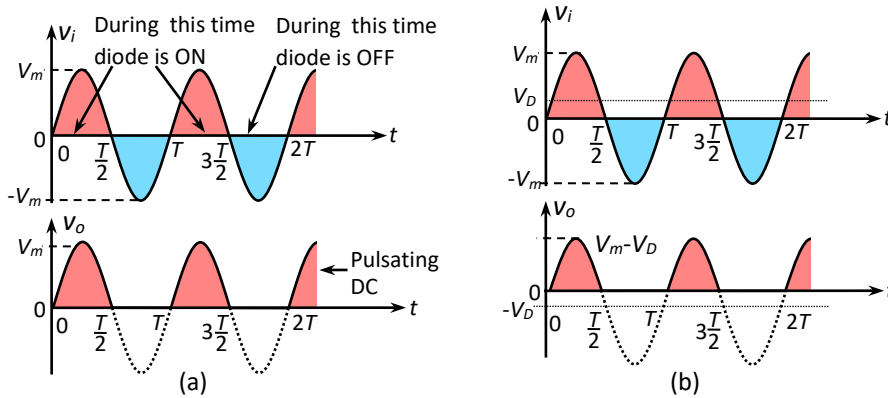


Fig.3-23: Input and output of half-wave: (a) Without considering V_D , and (b) With considering V_D output considering V_D

Calculation of Average and RMS value

The peak value of the output current of the rectifier is given by (neglecting diode voltage drop)

$$I_{om} = \frac{V_m}{r_s + r_D + R_L} \quad (3-13)$$

Where, V_m = the peak input voltage = peak output voltage of the transformer

r_s = source resistance = transformer secondary resistance for this case

r_D = diode DC resistance

The peak value of the output voltage is,

$$V_{om} = \frac{V_m}{r_s + r_D + R_L} \times R_L \quad (3-14)$$

But in most of the practical cases, $r_s + r_D \ll R_L$. So we can neglect $r_s + r_D$.

Thus, from Equ.(3-13) we get,
$$I_{om} = \frac{V_m}{R_L} \quad (3-15)$$

and,
$$V_{om} = V_m \quad (3-16)$$

As the output voltage follows the sinusoidal shape of the input AC voltage the instantaneous output voltage and current will be,

$$i_o = \frac{V_m}{R_L} \sin(\omega t) \text{ [for + ve half – cycle]} \quad (3-17)$$

and,
$$v_o = V_{om} \sin(\omega t) \text{ [for + ve half – cycle]} \quad (3-18)$$

The pulsating output of the rectifier has both DC value and AC value. The DC value is also called average value. The output voltage is produced only for the positive half-cycle (0 to $T/2$) of input voltage. Thus, the average or DC value of the output voltage will be

$$V_{dc} = V_{avg} = \frac{1}{T} \int_0^{T/2} V_{om} \sin(\omega t) d(\omega t) = \frac{1}{T} \int_0^{T/2} \frac{V_m \times R_L}{r_s + r_D + R_L} \sin(\omega t) d(\omega t)$$

Here, T is the time of a complete cycle. The angle of a complete cycle is 2π , and $T/2$ corresponds to π . So, the above equation can be written using angles instead of time.

$$V_{dc} = V_{avg} = \frac{1}{2\pi} \frac{V_m \times R_L}{(r_s + r_D + R_L)} \int_0^{\pi} \sin(\theta) d\theta \text{ } [\because \omega t = \theta]$$

or,
$$V_{dc} = \frac{V_m \times R_L}{2\pi(r_s + r_D + R_L)} [-\cos \theta]_0^{\pi}$$

or,
$$V_{dc} = \frac{V_m \times R_L}{2\pi(r_s + r_D + R_L)} [-(-1) + 1]$$

or,
$$V_{dc} = \frac{V_m \times R_L}{2\pi(r_s + r_D + R_L)} \times 2$$

or,
$$V_{dc} = \frac{V_m \times R_L}{\pi(r_s + r_D + R_L)}$$

$\therefore$
$$\boxed{V_{dc} = V_{avg} = \frac{V_m \times R_L}{\pi(r_s + r_D + R_L)}} \quad (3-19)$$

If $r_s + r_D \ll R_L$, the value of average or DC voltage will be,

$$\boxed{V_{dc} = V_{avg} = \frac{V_m}{\pi}} \quad (3-20)$$

Here, we have not considered the forward voltage drop of the diode (V_D). If we consider this voltage drop V_m will decrease by V_D and the average value will be,

$$\boxed{V_{dc} = V_{avg} = \frac{(V_m - V_D) \times R_L}{\pi(r_s + r_D + R_L)}} \quad (3-21)$$

or, neglecting $r_s + r_D$

$$V_{dc} = V_{avg} = \frac{(V_m - V_D)}{\pi} \quad (3-22)$$

For silicon diode the value of V_D is approximately 0.7 V and for Ge diode it is 0.3 V.

RMS Value Calculation

The RMS value can be calculated as

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^{T/2} (V_m \sin(\omega t))^2 d(\omega t)} = \sqrt{\frac{1}{T} \int_0^{T/2} \left(\frac{V_m \times R_L}{r_s + r_D + R_L} \sin(\omega t) \right)^2 d(\omega t)}$$

or,

$$V_{rms} = \frac{1}{\sqrt{T}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\int_0^{T/2} \{\sin(\omega t)\}^2 d(\omega t)}$$

Again considering angle instead of time we get

$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\int_0^\pi \sin^2 \theta d\theta}$$

or,

$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\int_0^\pi \frac{1}{2} (1 - \cos 2\theta) d\theta}$$

or,

$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^\pi}$$

or,

$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\frac{1}{2} \pi}$$

∴

$$V_{rms} = \frac{V_m \times R_L}{2(r_s + r_D + R_L)} \quad (3-23)$$

Again, if $r_s + r_D \ll R_L$ the RMS value will be,

$$V_{rms} = \frac{V_m}{2} \quad (3-24)$$

If we consider the voltage drop (V_D) of the diode the RMS value will be,

$$V_{rms} = \frac{(V_m - V_D) \times R_L}{2(r_s + r_D + R_L)} \quad (3-25)$$

or, neglecting $r_s + r_D$,

$$V_{rms} = \frac{(V_m - V_D)}{2} \quad (3-26)$$

Example 3-10

A half-wave rectifier circuit is designed using a 12 V transformer. If the transformer secondary resistance (r_s) is $10\ \Omega$, diode internal resistance (r_D) is $50\ \Omega$ and the load resistance is $1\ \text{k}\Omega$, calculate the DC output voltage and RMS output voltage i) neglecting V_D and ii) considering V_D for a Si diode.

Solution:

The maximum value of the transformer output voltage, which is the input voltage to the rectifier, is

$$V_m = \sqrt{2} \times 12\ \text{V} \approx 17\ \text{V}$$

i) Neglecting V_D :

DC output voltage or the average output voltage is,

$$\text{Using Equ.(3-19), } V_{dc} = \frac{V_m \times R_L}{\pi(r_s + r_D + R_L)} = \frac{17\ \text{V} \times 1000\ \Omega}{\pi(10 + 50 + 1000)\ \Omega} \approx 5.11\ \text{V (Ans.)}$$

Using Equ.(3-23) the RMS output voltage can be calculated as,

$$V_{rms} = \frac{V_m \times R_L}{2(r_s + r_D + R_L)} = \frac{17\ \text{V} \times 1000\ \Omega}{2(10 + 50 + 1000)\ \Omega} \approx 8.02\ \text{V (Ans.)}$$

ii) Considering $V_D = 0.7\ \text{V}$ for the Si diode:

Using Equ.(3-21), DC output voltage is,

$$V_{dc} = \frac{(V_m - 0.7) \times R_L}{\pi(r_s + r_D + R_L)} = \frac{(17 - 0.7)\ \text{V} \times 1000\ \Omega}{\pi(10 + 50 + 1000)\ \Omega} \approx 4.49\ \text{V (Ans.)}$$

Using Equ.(3-25), RMS output voltage is,

$$V_{rms} = \frac{(V_m - 0.7) \times R_L}{2(r_s + r_D + R_L)} = \frac{(17 - 0.7)\ \text{V} \times 1000\ \Omega}{2(10 + 50 + 1000)\ \Omega} \approx 7.69\ \text{V (Ans.)}$$

Comments: The DC or average value of a half-wave rectifier is always greater than the RMS value.

3.7.2 Full-wave Rectifier

In half-wave rectifier, as discussed above, the load gets power only during one half-cycle. That's why it has low DC and RMS value. To supply power to the load during both the positive and negative half-cycles, we have to use full-wave rectifier. There are two types of full-wave rectifiers as described below.

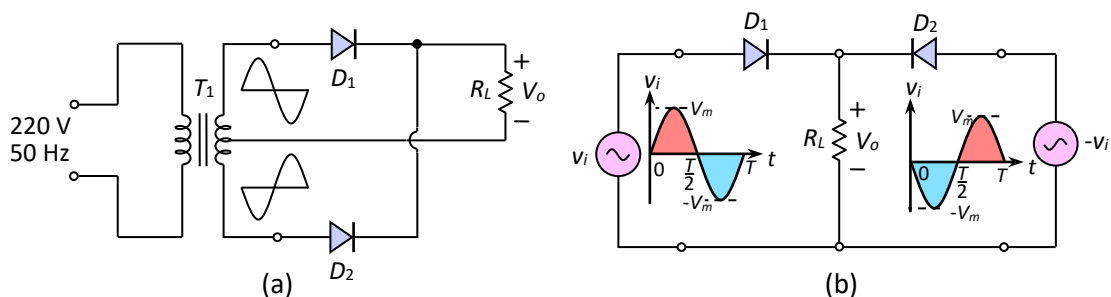


Fig.3-24: (a) Two-diode full-wave rectifier, (b) Equivalent circuit of the two half-wave rectifier

Two-diode Full-wave Rectifier

The two-diode rectifier circuit is also called **full-wave rectifier with centre-tapped transformer**. The circuit diagram of a **two-diode rectifier** is shown in Fig.3-24(a). It uses two rectifier diodes and its input voltage is supplied from a transformer (T_1). The transformer must have a tape at the centre of the secondary coil. For this reason, this type of transformer is called **centre-tapped** transformer. The circuit is essentially a combination of two half-wave rectifiers, each energized by half of the transformer secondary as shown in Fig.3-24(b).

A centre-tape transformer has the property that for the positive half cycle of input, the top-most terminal will be positive w.r.t. centre-tape, and the bottom-most terminal will be negative w.r.t. the centre-tape (Fig.3-25). For the negative half-cycle of input the polarity will be just opposite. Actually, the top-most terminal and the centre-tape will behave as one secondary coil, and the centre-tape with the bottom-most terminal behaves as another secondary coil. Both the secondary coils will produce $v_i = V_m \sin(\omega t)$. Considering this, we can redraw the full-wave rectifier as in Fig.3-24(b).

During the positive half-cycle ($t = 0$ to $T/2$ or $\theta = 0$ to π), the upper terminal of transformer secondary coil is positive, as illustrated in Fig.3-25. This positive terminal is connected to the anode of D_1 , and the negative centre-tape is connected to the cathode of D_1 via R_L . Consequently, D_1 is forward biased, and the load current, I_L , flows from +ve terminal of the secondary coil through D_1 , through R_L (downward), and back to the transformer centre-tap. During this time, the -ve voltage at the lower terminal of the transformer secondary coil causes diode D_2 to be reverse biased. Thus, D_1 behaves as short-circuit and D_2 behaves as open circuit as shown in Fig.3-25. If the diode voltage drop is neglected the whole input voltage (transformer output voltage) will be developed across the load R_L with upper terminal positive.

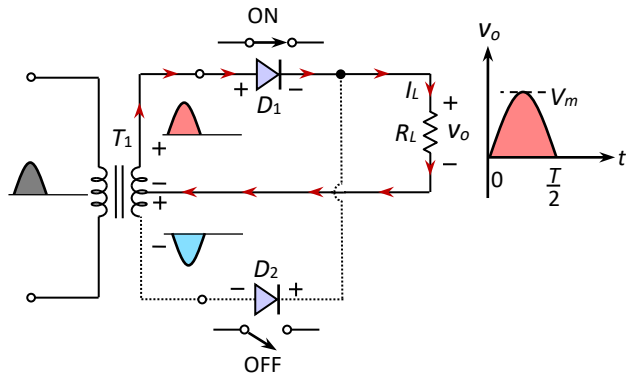


Fig.3-25: Circuit status for positive half-cycle

During the negative half-cycle of the input ($t = \frac{T}{2}$ to T or $\theta = \pi$ to 2π), the (bottom-most) positive terminal of the transformer secondary coil will be connected to the anode of diode D_2 and the negative centre-tape will be connected to the cathode of D_2 via R_L . Thus, D_2 will be forward biased and behaves as short circuit as illustrated in Fig.3-26. Now, the load current flows from bottom of the transformer secondary through D_2 , through R_L (from top to bottom), and back to the transformer centre-tape. As the polarity of the voltage of the

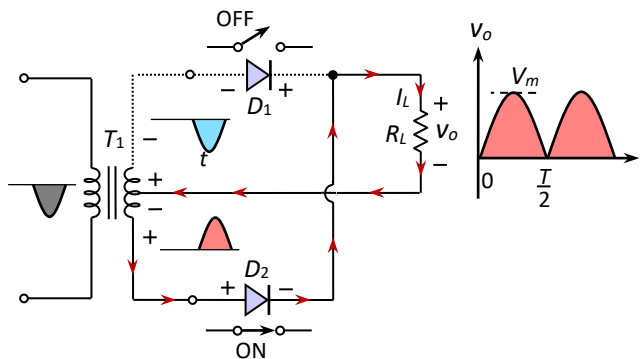


Fig.3-26: Circuit status for negative half-cycle

upper half of the secondary has just reversed, diode D_2 will now be reverse biased and it will behave as open circuit. In this way, the whole negative half-cycle will be developed across the load resistance R_L with upper terminal positive again.

The direction of load current is always downward. That is, load is getting full-wave DC voltage. Like the half-wave rectifier, the output is not pure DC but pulsating DC. If we neglect the voltage drop of the diodes the output waveform will be as shown in Fig.3-27(a) and if the diode voltage drop is considered the output waveform will be like Fig.3-27(b). The peak output voltage will be $V_m - V_D$. For Si diode we can consider, $V_D = 0.7$ V and for Ge diode we can take $V_D = 0.3$ V.

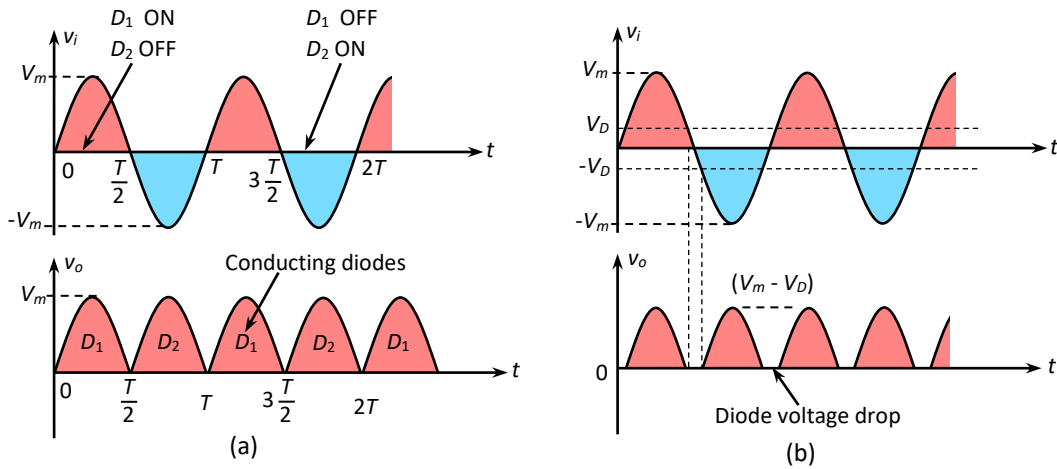


Fig.3-27: (a) Full-wave rectified output neglecting V_D , and
(b) Full-wave rectified output considering V_D

Peak Inverse Voltage (PIV)

Let us consider the situation shown in Fig.3-25. Here, the upper-half and lower-half of the transformer secondary produce a maximum voltage of V_m with respect to the centre-tape. The maximum voltage of the upper-half has been dropped across the load. Hence, the maximum load voltage is V_m . Now, if we apply KVL in the loop containing the OFF diode, D_2 , we get (starting from the top of R_L and moving along the direction indicated by the arrows through D_2)

$$-V_m - V_m + V_{D2} = 0 \quad \text{or,} \quad V_{D2} = 2V_m$$

Thus,

$$\boxed{\text{PIV} = 2V_m} \quad (3-27)$$

If we consider the circuit of Fig.3-26, we will find that D_1 will also be reverse biased by $2V_m$. So, the PIV for D_1 will also be $2V_m$.

Example 3-11

For the full-wave rectifier of Fig.3-28, i) Calculate the peak output current, and the PIV rating of the diodes. ii) Draw the input waveforms to the diodes and the output waveforms. For both cases assume Si diodes with $V_B = 0.7$ V.

Solution:

i) The RMS input voltage to the step-down transformer is 220 V and its step-down ratio is 20: 1. Therefore, the RMS voltage at the secondary of the transformer, which is also the input voltage of the diodes, will be,

$$v_{2(RMS)} = \frac{220 \text{ V}}{20} = 11 \text{ V}$$

The peak value of this voltage will be,

$$V_{2m} = V_{RMS} \times \sqrt{2} = 11 \text{ V} \times \sqrt{2} \approx 15.56 \text{ V}$$

This is a rectifier with two diodes and a centre tap transformer. Only one diode conducts per cycle. Therefore, the peak output voltage of the rectifier (across the load) will be,

$$V_{om} = 15.56 \text{ V} - 0.7 \text{ V} = 14.86 \text{ V}$$

The peak output current will be,

$$I_{om} = \frac{V_{om}}{R_L} = \frac{14.86 \text{ V}}{470 \Omega} = 31.62 \text{ mA [Ans.]}$$

The PIV rating of the diodes can be calculated using Equ.(3-) as,

$$PIV = 2V_m = 2V_{2m} = 2 \times 15.56 \text{ V} = 31.12 \text{ V [Ans.]}$$

ii) The input and output waveforms are drawn in Fig.3-29.

Comments: For the +ve half-cycle diode D_1 will conduct and will drop 0.7 V and for the -ve half-cycle diode D_2 will conduct and will drop another 0.7 V. These voltage drops are shown using the dashed lines on the input waveform [Fig.3-29(b)].

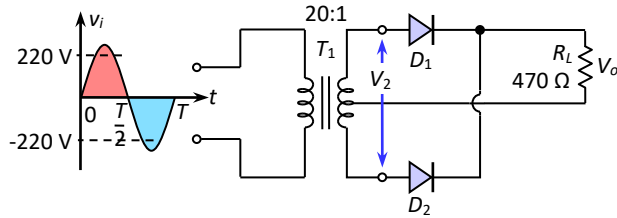


Fig.3-28: Full-wave rectifier for Example 2-5

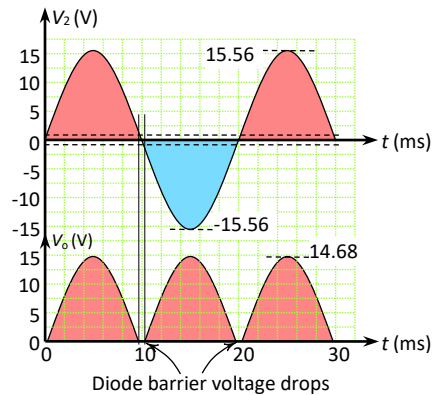


Fig.3-29: Input and output waveforms of Fig.3-28

3.7.3 Full-Wave Bridge Rectifier

If we want to use two-diode rectifier, we must use a centre-tapped transformer, also called full-wave transformer. Sometimes full-wave transformer may not be available then we have to use bridge rectifier that does not need centre-tapped transformer. Moreover, we can produce 24 V using a 12 V centre-tapped transformer by using a bridge-rectifier circuit. For that we have to use the top-most and the bottom-most terminals of the secondary leaving the centre-tape open.

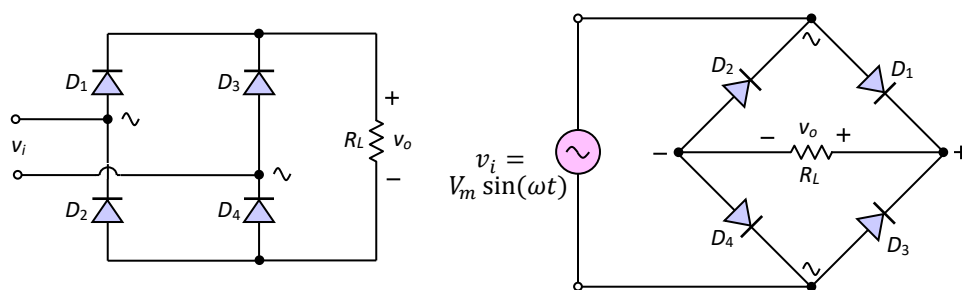


Fig.3-30: (a) Full-wave bridge rectifier, (b) Alternative circuit of bridge rectifier

The circuit diagram of a bridge rectifier is shown in Fig.3-30(a). Fig.3-30(b) shows another configuration of the bridge rectifier that is also commonly used. The bridge rectifier consists of four diodes connected with their arrowhead symbols all point toward the positive output terminal of the circuit. Among the four corners of a bridge rectifier two are AC input terminals and two are DC output terminal. The points at which anode of one diode and the cathode of another diode are connected are AC input terminals. The terminal where the cathodes of two diodes are connected is the positive terminal of DC output, and the point at which the anodes of two diodes are connected is the negative output terminal. Check the diode connections and compare with this description.

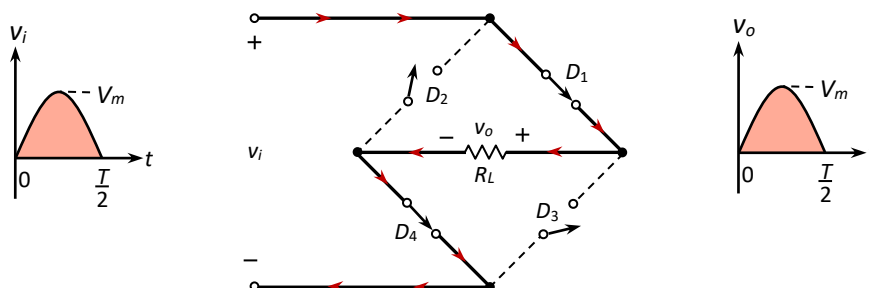


Fig.3-31: Circuit status for the positive half-cycle of input AC voltage

During the positive half-cycle of input voltage, diodes D_1 and D_4 become forward biased and behave as ON switch as shown in Fig.3-11(c). Thus, load current flows from the positive input terminal through D_1 to R_L , and then through R_L and D_4 back to the negative input. Note that the direction of load current through R_L is from positive to negative terminal. During this time, the positive input terminal is applied to the cathode of D_2 , and the negative output is connected at the anode of D_2 . So, D_2 is reverse biased during the positive half-cycle of AC input. Similarly, D_3 has the negative input at its anode and the positive output at its cathode during this time. Thus, D_3 is also reverse biased. Hence, during the positive half-cycle of the input AC, diodes D_1 and D_4 are forward biased, and D_2 and D_3 are reversed biased.

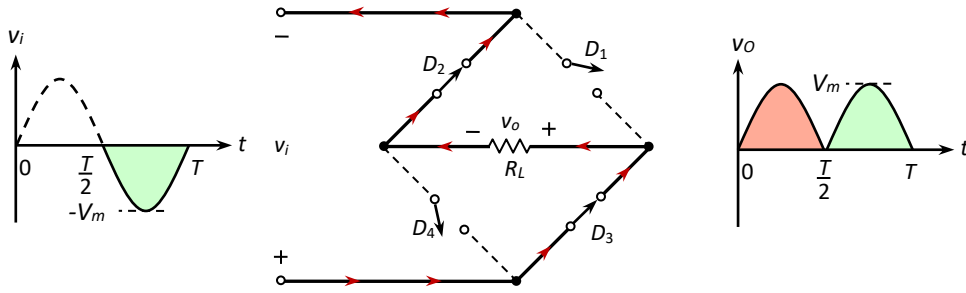


Fig.3-32: Circuit status for the negative half-cycle of input voltage

As shown in Fig.3-32, during the negative half-cycle of the input, diodes D_2 and D_3 are forward biased while D_1 and D_4 are reverse biased. For this negative half-cycle, the load current flows in the same direction (positive to negative) via D_2 and D_3 .

It is seen that during both half-cycles of the input, the load current always flows in the same direction (from right to left) making the right terminal of R_L always positive. Both positive and negative half-cycles are found at the output. The negative half-cycles are inverted, so that the output is a series of continuous positive half-cycles as shown in Fig.3-27.

Peak Inverse Voltage (PIV)

Let us consider the situation as shown in Fig.3-31. Here, we will consider the maximum voltage developed at the AC input and the maximum voltage developed across the load. Both of them are V_m . Now, we will apply KVL in the loop containing the OFF diodes, D_2 and D_3 . Let's move from the upper terminal of input, diode D_2 , load resistance R_L , diode D_3 and finally through the AC source to the starting point (loop complete).

$$-V_{D2} + V_m - V_{D3} + V_m = 0 \quad \text{or,} \quad V_{D2} + V_{D3} = 2V_m$$

$$\text{or,} \quad V_{D2} = V_{D3} = V_m \quad [\because V_{D2} = V_{D1}]$$

Thus,

$$\boxed{\text{PIV} = V_m}$$

(3-28)

If we consider the circuit of Fig.3-32, we will find the same result for D_1 and D_4 .

Example 3-12

For the full-wave rectifier of Fig.3-33, i) Calculate the peak output current, and the PIV rating of the diodes. ii) Draw the input waveforms to the rectifier and the output waveforms. For both cases assume Si diodes with $V_B = 0.7$ V.

Solution:

i) The RMS input voltage to the step-down transformer is 220 V and its turn ratio is 55:2. Therefore, the RMS voltage at the transformer secondary coil, which is also the input voltage of the diodes, will be,

$$v_{2(\text{RMS})} = \frac{(220 \text{ V}) \times 2}{55} = 8 \text{ V}$$

The peak value of this voltage will be,

$$V_{2(\text{peak})} = V_{\text{RMS}} \times \sqrt{2} = 8 \text{ V} \times \sqrt{2} \approx 11.31 \text{ V}$$

As the rectifier is a full-wave bridge rectifier, two diodes conduct per cycle. Therefore, the peak output voltage of the rectifier (across the load) will be,

$$V_{o(\text{peak})} = 11.31 \text{ V} - 2 \times 0.7 \text{ V} = 9.91 \text{ V}$$

The peak output current will be,

$$I_{o(\text{peak})} = \frac{V_{o(\text{peak})}}{R_L} = \frac{9.91 \text{ V}}{1 \text{ k}\Omega} = 9.91 \text{ mA [Ans.]}$$

The PIV rating of the diodes can be calculated using Equ.(3-28) as,

$$\text{PIV} = V_m = V_{2(\text{peak})} = 11.31 \text{ V [Ans.]}$$

ii) The input and output waveforms are drawn in Fig.3-34.

Comments: For the +ve half-cycle diodes D_1 & D_4 will conduct and will drop 1.4 V and for the -ve half-cycle diodes D_2 & D_3 will conduct and will drop another 1.4 V. These voltage drops are shown using the dashed lines in the input wave-form.

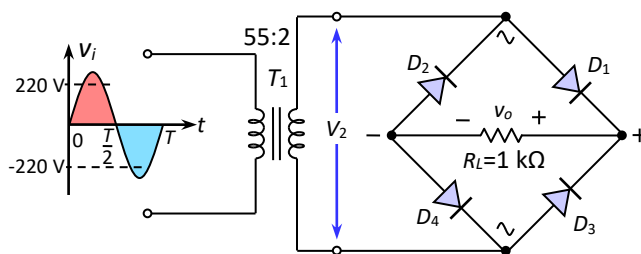


Fig.3-33: Full-wave bridge rectifier for Example 2-5

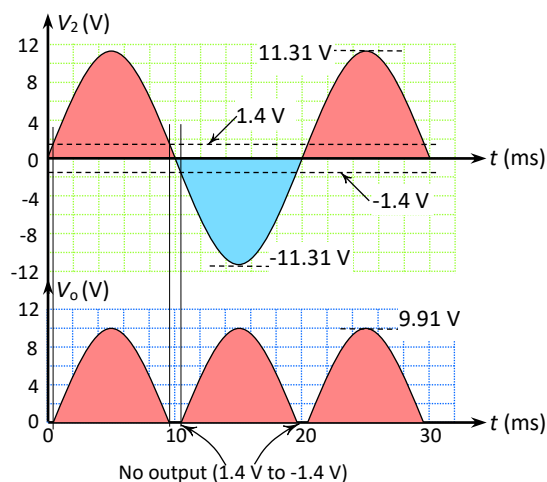


Fig.3-34: Input and output waveform of Example 2-3

Calculation of Average Output Voltage

As discussed for the half-wave rectifier, the output voltage follows the sinusoidal shape of the input AC voltage. The instantaneous output voltage and current will be same as described for half-wave

rectifier. But for the full-wave rectifier, the output voltage is produce for the positive half-cycle (0 to $T/2$) as well as for the negative half-cycle ($T/2$ to T). Thus, to find the average or DC value of the output voltage, we have to integrate the wave from 0 to T .

$$V_{dc} = V_{avg} = \frac{1}{T} \int_0^T V_{om} \sin(\omega t) d(\omega t) = \frac{1}{T} V_{om} \int_0^T \sin(\omega t) d(\omega t)$$

Considering angle instead of time ($T = 2\pi$) and using Equ.(3-14) we can write,

$$V_{dc} = V_{avg} = \frac{1}{2\pi} \frac{V_m \times R_L}{r_s + r_D + R_L} \int_0^{2\pi} \sin(\theta) d\theta \quad [\because \omega t = \theta]$$

$$\text{or,} \quad V_{dc} = \frac{V_m \times R_L}{2\pi(r_s + r_D + R_L)} \left[\int_0^{\pi} \sin(\theta) d\theta + \int_{\pi}^{2\pi} \sin(\theta) d\theta \right]$$

$$\text{or,} \quad V_{dc} = \frac{V_m \times R_L}{2\pi(r_s + r_D + R_L)} \{ [-\cos \theta]_0^{\pi} + [-\cos \theta]_{\pi}^{2\pi} \}$$

$$\text{or,} \quad V_{dc} = \frac{V_m \times R_L}{2\pi(r_s + r_D + R_L)} \{ -(-1) + 1 + \{ -(-1) + 1 \} \}$$

$$\text{or,} \quad V_{dc} = \frac{V_m \times R_L}{2\pi(r_s + r_D + R_L)} \times 4$$

$$\therefore \quad \boxed{V_{dc} = V_{avg} = \frac{2V_m \times R_L}{\pi(r_s + r_D + R_L)}} \quad (3-29)$$

If $r_s + r_D \ll R_L$ the value of average or DC voltage will be,

$$\therefore \quad \boxed{V_{dc} = V_{avg} = \frac{2V_m}{\pi}} \quad (3-30)$$

Here, we have not considered the forward voltage drops of the diodes. If we consider the voltage drops of two diodes, the average value will be,

$$\boxed{V_{dc} = V_{avg} = \frac{2(V_m - 2V_D) \times R_L}{\pi(r_s + r_D + R_L)}} \quad (3-31)$$

$$\text{or, neglecting } r_s + r_D \quad \boxed{V_{dc} = V_{avg} = \frac{2(V_m - 2V_D)}{\pi}} \quad (3-32)$$

For a silicon diode the value of V_D can be approximately consider 0.7 V and for germanium diode 0.3 V. As the current passes through two diodes, in bridge rectifier the total voltage drop is $2V_D$.

RMS Value Calculation

The RMS value or effective value can be calculated as,

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T (V_{om} \sin(\omega t))^2 d(\omega t)} = \frac{1}{\sqrt{T}} V_{om} \sqrt{\int_0^T \sin^2(\omega t) d(\omega t)}$$

or, using Equ.(3-14),
$$V_{rms} = \frac{1}{\sqrt{T}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\int_0^T (\sin(\omega t))^2 d(\omega t)}$$

Again considering angle instead of time we get,

$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\int_0^{2\pi} \sin^2 \theta d\theta}$$

or,
$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\int_0^{2\pi} \frac{1}{2} (1 - \cos 2\theta) d\theta}$$

or,
$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\int_0^{\pi} \frac{1}{2} (1 - \cos 2\theta) d\theta + \int_{\pi}^{2\pi} \frac{1}{2} (1 - \cos 2\theta) d\theta}$$

or,
$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\frac{1}{2} \left[\left(\theta - \frac{1}{2} \sin 2\theta \right) \right]_0^{\pi} + \frac{1}{2} \left[\left(\theta - \frac{1}{2} \sin 2\theta \right) \right]_{\pi}^{2\pi}}$$

or,
$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\frac{1}{2} \pi + \frac{1}{2} \pi}$$

or,
$$V_{rms} = \frac{1}{\sqrt{2\pi}} \frac{V_m \times R_L}{r_s + r_D + R_L} \sqrt{\pi}$$

∴
$$\boxed{V_{rms} = \frac{1}{\sqrt{2}} \frac{V_m \times R_L}{(r_s + r_D + R_L)}} \quad (3-33)$$

Again, if $r_s + r_D \ll R_L$ the RMS value will be,

$$\boxed{V_{rms} = \frac{V_m}{\sqrt{2}}} \quad (3-34)$$

If we consider the voltage drop of the diode the RMS value will be,

$$\boxed{V_{rms} = \frac{1}{\sqrt{2}} \frac{(V_m - 2V_D) \times R_L}{(r_s + r_D + R_L)}} \quad (3-35)$$

or, neglecting $r_s + r_D$
$$\boxed{V_{rms} = \frac{(V_m - 2V_D)}{\sqrt{2}}} \quad (3-36)$$

Note that in Equ.(3-31), (3-32), (3-35) and (3-36) we have considered the diode voltage drop as $2V_D$. This is only for the bridge rectifier. But for the two-diode rectifier it would be V_D .

Example 3-13

A full-wave bridge rectifier is shown in Fig.3-35. If the diode resistance $r_D = 50 \Omega$, and the source resistance $r_s = 100 \Omega$, and $V_D = 0.7 \text{ V}$ calculate the RMS and average output voltage i) With considering all parameters, and ii) Considering ideal condition.

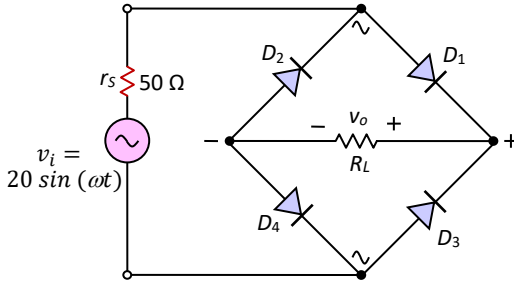


Fig.3-35: Fullwave bridge rectifier for Example 3-12

Solution:

(i) Without considering V_D : The input AC voltage is $v_i = 20 \sin(\omega t)$. So, the maximum voltage of this sinusoidal input is

$$V_m = 20 \text{ V}$$

Using Equ.(3-31),

$$V_{avg} = \frac{2(V_m - 2V_D) \times R_L}{\pi(r_s + r_D + R_L)}$$

or,

$$V_{avg} = \frac{2 \times (20 \text{ V} - 1.4 \text{ V}) \times 1000 \Omega}{\pi(100 \Omega + 50 \Omega + 1000 \Omega)} \approx 10.3 \text{ V [Ans.]}$$

Using Equ.(3-35),

$$V_{rms} = \frac{1}{\sqrt{2}} \frac{(V_m - 2V_D) \times R_L}{(r_s + r_D + R_L)}$$

or,

$$V_{rms} = \frac{1}{\sqrt{2}} \frac{(20 \text{ V} - 1.4 \text{ V}) \times 1000 \Omega}{(100 \Omega + 50 \Omega + 1000 \Omega)} \approx 11.44 \text{ V [Ans.]}$$

ii) Considering ideal condition:

Using Equ.(3-30),

$$V_{avg} = \frac{2V_m}{\pi} = \frac{2 \times 20}{\pi} \approx 12.73 \text{ V [Ans.]}$$

Using Equ.(3-34),

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{20}{\sqrt{2}} \approx 14.14 \text{ V [Ans.]}$$

Comments: In case of small values of V_m and R_L , the diode voltage drops and the other unwanted resistances may cause a significant voltage drops.

3.7.4 Ripple Factor of Rectified Output

Although the function of a rectifier is to produce DC voltage from AC input, none of the half-wave and full-wave rectifier can produce pure DC output. The output is pulsating and it has both DC and AC value. The DC value is useful for us, but the AC component is not useful in this case. Ripple factor is a measure of how properly a rectifier can produce DC voltage compared to the AC voltage. Hence, **ripple factor**, r , is defined as the ratio of the RMS value of the AC voltage to the DC value or average value. Mathematically, ripple factor is defined as,

$$r = \frac{V_{rrms}}{V_{dc}} = \frac{\text{RMS value of ripple (or AC component) of output}}{\text{Average or DC value of output voltage}} \quad (3-37)$$

The output of a rectifier, v_o , is the sum of DC voltage (V_{dc}) and AC ripple voltage (v_r). That is

$$v_o = V_{dc} + v_r$$

The RMS value of the total rectified output is,

$$V_{rms} = \sqrt{\left[\frac{1}{T} \int_0^T v_o^2 dt \right]} = \sqrt{\left[\frac{1}{T} \int_0^T (V_{dc} + v_r)^2 dt \right]}$$

$$\text{or,} \quad V_{rms} = \sqrt{\frac{1}{T} \int_0^T V_{dc}^2 dt + \frac{1}{T} \int_0^T v_r^2 dt + \frac{1}{T} \int_0^T 2V_{dc}v_r dt}$$

$$\text{or,} \quad V_{rms} = \sqrt{\frac{1}{T} V_{dc}^2 \int_0^T dt + \left(\sqrt{\frac{1}{T} \int_0^T v_r^2 dt} \right)^2 + \frac{1}{T} 2V_{dc} \int_0^T v_r dt} \quad (3-38)$$

As the Integration over a full cycle of an AC signal is zero, Therefore,

$$\int_0^T v_r dt = 0 \quad (3-39)$$

And, $\left(\sqrt{\frac{1}{T} \int_0^T v_r^2 dt} \right)$ is the RMS value of the AC component or the ripple. That is,

$$\sqrt{\frac{1}{T} \int_0^T v_r^2 dt} = V_{rrms} \quad (3-40)$$

Putting the values of Equ.(3-39) and Equ.(3-40) into Equ.(3-38) we get,

$$V_{rms} = \sqrt{\frac{1}{T} V_{dc}^2 [t]_0^T + V_{rrms}^2 + 2 \frac{1}{T} V_{dc} \times 0}$$

$$\text{or,} \quad V_{rms} = \sqrt{\left(\frac{1}{T} V_{dc}^2 \times T \right) + V_{rrms}^2 + 0}$$

$$\text{or,} \quad V_{rms} = \sqrt{V_{dc}^2 + V_{rrms}^2}$$

$$\text{or,} \quad V_{rms}^2 = V_{dc}^2 + V_{rrms}^2$$

$$\text{or,} \quad V_{rrms}^2 = V_{rms}^2 - V_{dc}^2$$

$$\therefore \quad V_{rrms} = \sqrt{V_{rms}^2 - V_{dc}^2} \quad (3-41)$$

Putting the value of V_{rrms} from Equ.(3-41) into Equ.(3-37), we get the ripple factor as,

$$r = \frac{\sqrt{V_{rms}^2 - V_{dc}^2}}{V_{dc}} = \sqrt{\frac{V_{rms}^2 - V_{dc}^2}{V_{dc}^2}}$$

or,

$$r = \sqrt{\frac{V_{rms}^2}{V_{dc}^2} - 1} \quad (3-42)$$

Ripple Factor of Half-Wave Rectifier

For half-wave rectifier $V_{rms} = V_m/2$ and $V_{dc} = V_m/\pi$. Therefore the value of ripple factor for this rectifier is

$$r = \sqrt{\left(\frac{\frac{V_m}{2}}{\frac{1}{\pi}V_m}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2}\right)^2 - 1} = 1.21$$

The value of ripple factor of half-wave rectifier is greater than 1. It means that the half-wave rectified output has more AC component than DC value.

Ripple Factor of Full-Wave Rectifier

For full-wave rectifier $V_{rms} = V_m/\sqrt{2}$ and $V_{dc} = 2V_m/\pi$. Therefore the value of ripple factor for this rectifier is

$$r = \sqrt{\left(\frac{\frac{V_m}{\sqrt{2}}}{\frac{2}{\pi}V_m}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} = 0.48$$

The value of ripple factor of full-wave rectifier is about 0.48. It means that the AC value of the full-wave output is a little bit lower than the DC value.

3.7.5 Efficiency of Rectification

The target of rectification is to produce DC power from AC power. Here, AC power is the input to the rectifier circuit and DC power developed across the load is the output. For any system the ration of output power to the input power is called efficiency. Thus, the efficiency of the rectifier will be,

$$\eta = \frac{P_{dc}}{P_{ac}} \quad (3-43)$$

P_{dc} can be calculated from the DC output voltage and the load resistance as $P_{dc} = V_{dc}^2 / R_L$. As the AC current flows from the AC source to the load resistance through the diode, the total resistance faced by the AC current is $(r_s + r_D + R_L)$. Thus, the AC input power will be $P_{ac} = V_{rms}^2 / (r_s + r_D + R_L)$. Therefore, the efficiency of the rectifier will be,

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{V_{dc}^2 / R_L}{V_{rms}^2 / (r_s + r_D + R_L)} \quad (3-44)$$

Where, r_s = source resistance = transformer secondary resistance

r_D = diode DC resistance

R_L = load resistance

V_{dc} = DC output of the rectifier

V_{rms} = RMS value of the input voltage

As the value of $r_s + r_D$ is very small compared to R_L , we can neglect this value. In that case the efficiency of the rectifier will be,

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{V_{dc}^2/R_L}{V_{rms}^2/R_L} = \frac{V_{dc}^2}{V_{rms}^2} = \left(\frac{V_{dc}}{V_{rms}}\right)^2 \quad (3-45)$$

Efficiency of Half-Wave Rectification

We know that the RMS value of the AC input voltage is $V_m / \sqrt{2}$, but for the case of half-wave rectifier the AC source supply power only during the positive half-cycle. During the negative half-cycle no power is taken from the source. That's why the AC input power is also half-wave rectified and its RMS value will be $V_m/2$. Hence, the efficiency of the half-wave rectifier, η_H , will be

$$\eta_H = \left(\frac{V_{dc}}{V_{rms}}\right)^2 = \left(\frac{V_m}{\pi} / \frac{V_m}{2}\right)^2 = \left(\frac{2}{\pi}\right)^2 \approx 0.41 = 41\%$$

Efficiency of Full-Wave Rectification

In case of full-wave rectification the AC source supplies power in both positive and negative half-cycle. Now, the RMS value of AC input will be $V_m / \sqrt{2}$ and we have found the DC value of full-wave rectified output is $2V_m / \pi$. Therefore, the efficiency of the full-wave rectifier, η_{FW} , will be

$$\eta_F = \left(\frac{V_{dc}}{V_{rms}}\right)^2 = \left(\frac{2V_m}{\pi} / \frac{V_m}{\sqrt{2}}\right)^2 = \left(\frac{2\sqrt{2}}{\pi}\right)^2 = 2 \times \eta_H \approx 0.81 = 81\%$$

Efficiency of full-wave rectifier is double of that of the half-wave rectifier.

3.7.6 Transformer Utilization Factor (TUF)

To design any power supply, it is necessary to determine the rating of the transformer. TUF is a useful parameter to determine the rating of a transformer. TUF is defined as the ratio of the DC power delivered to the load to the rated power of the transformer. That is,

$$\text{TUF} = \frac{\text{DC power delivered to the load}}{\text{AC power or rating of the transformer secondary}} \quad (3-46)$$

$$\text{or,} \quad \text{TUF} = \frac{P_{dc}}{P_{ac\text{-rated}}} \quad (3-47)$$

For a half-wave rectifier, DC power is

$$P_{dc} = V_{dc} \times I_{dc} = \frac{V_m}{\pi} \times \frac{V_m}{\pi R_L} = \frac{V_m^2}{\pi^2 R_L}$$

Transformer rated output is (neglecting all voltage drops from transformer to the load)

$$P_{ac\text{-rated}} = V_{rms} \times I_{rms} = \frac{V_m}{\sqrt{2}} \times \frac{V_m}{2R_L} = \frac{V_m^2}{2\sqrt{2}R_L}$$

[$\because$ output current from the transformer is half wave rectified, it supplies only for one half-cycle.]

$$\therefore \text{TUF}_H = \frac{V_m^2}{\pi^2 R_L} \times \frac{2\sqrt{2}R_L}{V_m^2} = \frac{2\sqrt{2}}{\pi^2} = 0.287$$

Similarly, for the full-wave rectifier,

$$\text{TUF}_F = \frac{4V_m^2}{\pi^2 R_L} \times \frac{2R_L}{V_m^2} = \frac{8}{\pi^2} = 0.812$$

Thus, we find that in case of half-wave rectifier only 28.7% power of the transformer will be supplied to the load. Whereas in case of full-wave rectifier, 81.2% of the transformer rated power will be supplied to the load.

Example 3-7

Find the output voltage across the load resistance, R_L of the circuit in Fig.3-36. Also determine the PIV rating for each diode.

Solution:

For the positive half-cycle, diode D_1 will be reverse biased and D_2 will be forward biased. This situation will be valid for $v_i > 0.7$ V. The situation is illustrated in Fig.3-37(a). The current flowing through D_2 will be

$$I_{D2} = \frac{25 \text{ V} - 0.7 \text{ V}}{(2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega) \parallel 1.1 \text{ k}\Omega} = \frac{25 \text{ V} - 0.7 \text{ V}}{(5.5 \text{ k}\Omega) \parallel 1.1 \text{ k}\Omega} = \frac{24.3 \text{ V}}{0.92 \text{ k}\Omega} \approx 26.41 \text{ mA}$$

This current will be divided between $(R_L + R_1)$ and R_2 . Using current divider rule, we can calculate I_L as,

$$I_L = \frac{(26.41 \text{ mA}) \times 1.1 \text{ k}\Omega}{(2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega) + 1.1 \text{ k}\Omega} \approx 4.4 \text{ mA}$$

Thus, the output voltage across the load resistance (R_L) will be,

$$V_o = V_L = I_L R_L = (4.4 \text{ mA}) \times 2.2 \text{ k}\Omega = 9.68 \text{ V}$$

The maximum reverse voltage across the diode D_1 can be calculated using KVL as,

$$V_{D1} - 0.7 \text{ V} - V_o = 0$$

$$\therefore V_{D1} = 0.7 \text{ V} + 9.68 \text{ V} = 10.38 \text{ V}$$

Thus, the required PIV of the diode D_1 should be 10.38 V.

For the negative half-cycle, diode D_2 will be reverse biased and D_1 will be forward biased. This

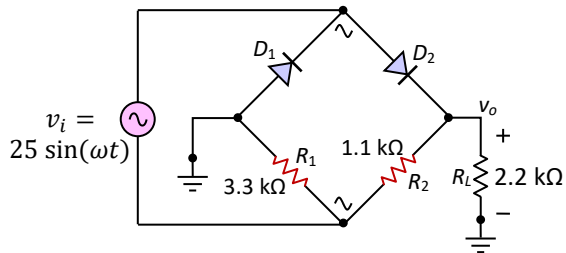


Fig.3-36: Bridge circuit for Example 3-14

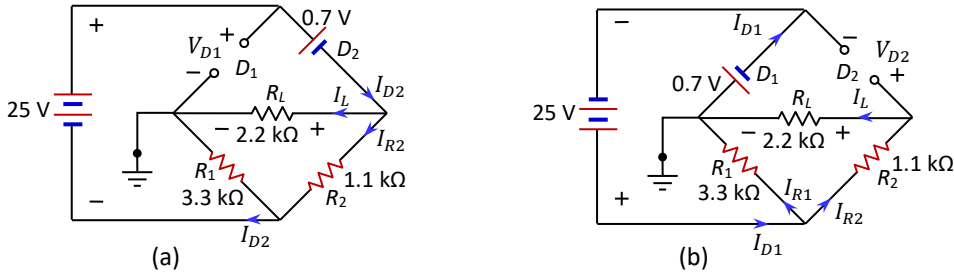


Fig.3-37: Circuit state for: (a) positive peak voltage, and (b) negative peak voltage

situation will be valid for $v_i < -0.7$ V. The situation is illustrated in Fig.3-37(b). The current flowing through D_1 will be,

$$I_{D1} = \frac{25 \text{ V} - 0.7 \text{ V}}{3.3 \text{ k}\Omega \parallel (2.2 \text{ k}\Omega + 1.1 \text{ k}\Omega)} = \frac{25 \text{ V} - 0.7 \text{ V}}{3.3 \text{ k}\Omega \parallel (3.3 \text{ k}\Omega)} = \frac{24.3 \text{ V}}{1.65 \text{ k}\Omega} \approx 14.73 \text{ mA}$$

This current will be divided between $(R_L + R_2)$ and R_1 . Using current divider rule, we can calculate I_L as,

$$I_L = (I_{R_L} + I_{R_2}) = \frac{(14.73 \text{ mA}) \times 3.3 \text{ k}\Omega}{3.3 \text{ k}\Omega + (2.2 \text{ k}\Omega + 1.1 \text{ k}\Omega)} \approx 7.365 \text{ mA}$$

Thus, the output voltage across the load resistance (R_L) will be,

$$V_o = V_L = I_L R_L = (7.365 \text{ mA}) \times 2.2 \text{ k}\Omega = 16.203 \text{ V}$$

The maximum reverse voltage across the diode D_2 can be calculated using KVL as,

$$\begin{aligned} V_{D2} - 0.7 \text{ V} - V_o &= 0 \\ V_{D2} &= 0.7 \text{ V} + 16.203 \text{ V} = 16.903 \text{ V} \end{aligned}$$

So the required PIV of the diode D_2 should be 16.903 V.

Comments: The output voltage can also be calculated using voltage divider rule as,

$$V_o = V_L = \frac{(25 \text{ V} - 0.7 \text{ V}) \times 2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 1.1 \text{ k}\Omega} \approx 16.2 \text{ V}$$

3.8 Filters

In general, a **filter** is a circuit that blocks a particular range of frequencies and passes some other range of frequencies. In case of rectifier, the function of the filter circuit is to block the AC component and to pass the DC component. That is, the filter circuit minimizes the ripple component and converts the pulsating output voltage into pure DC voltage. When a filter is connected at the output of a rectifier, it can be used as a DC power supply.

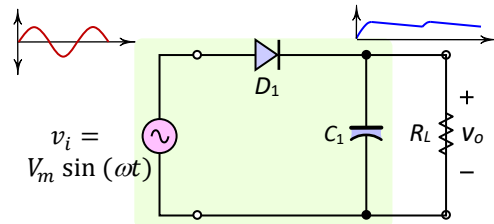


Fig.3-38: Half-wave rectifier circuit with shunt capacitor filter

There are different types of filter circuits. Some of the commonly used filter circuits are described here.

3.8.1 Shunt Capacitor Filter

In this circuit a large value capacitor, C_1 is connected at the output terminals of the rectifier, that is, across the load resistor, R_L (as shown in Fig.3-38). Here, we have considered the half-wave rectifier first and the full-wave rectifier will be covered later. The operation of this circuit depends on the storage capacity of the capacitor. The charging, discharging of the capacitor and the variation of output voltage is shown in Fig.3-39. In Fig.3-39(a), the dashed line represents AC input voltage (v_i) to the rectifier (to the left terminal of the diode). On the other hand the solid line is the output voltage after filtered by the capacitor.

For the positive half-cycle, diode D_1 is forward biased and the capacitor is charged by the current flowing through this diode. During this charging, only the internal resistance of the transformer (r_s) and the diode DC resistance (r_D) are effective. So the charging time constant is $\tau_c = (r_s + r_D) \times C_1$. As the value of resistance ($r_s + r_D$) is very small, the time constant is also very small. Thus, the capacitor charges immediately. As the AC input voltage (v_i) [dashed line in Fig.3-39(a)] increases to its peak value, the capacitor is also charged to this peak value (V_m). After this point, v_i decreases from the peak value but the capacitor voltage remains at V_m . Thus, the diode, D_1 gets reverse biased (as $v_i < V_m$). If there were no load resistor (R_L) across the capacitor, its output voltage would have been always the peak value (V_m). But if R_L is connected at the output, during the time the diode is reverse biased, the capacitor discharges through the load resistance R_L . Hence, the capacitor voltage decreases gradually during this time. As the value of load resistance is comparatively larger, the discharging time constant $\tau_d = R_L \times C_1$, is also larger. Hence, the capacitor discharges slowly as shown in Fig.3-39. After some time, the next positive half-cycle of v_i occurs. When v_i exceeds the capacitor voltage (little bit less than V_m due to discharging through load) the diode becomes forward biased and the capacitor is charged to the peak voltage again. The process is repeated continuously and the output voltage across the load becomes smoother than the pulsating waveform, as shown in Fig.3-39 by the solid line.

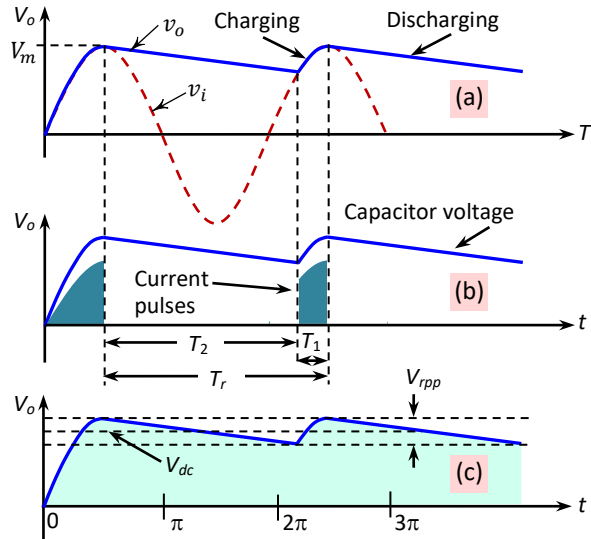


Fig.3-39: (a) Output waveform (b) output waveform with current pulses, and (c) ripple on the output waveform

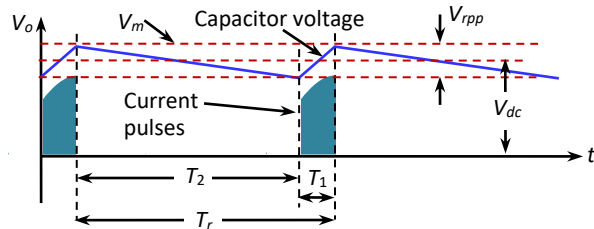


Fig.3-40: Approximate saw-tooth output waveform of capacitor filter

The process is repeated continuously and the output voltage across the load becomes smoother than the pulsating waveform, as shown in Fig.3-39 by the solid line.

Calculation of the Value of Capacitance

As discussed above, the capacitor filter can reduce the pulsating nature of the output but still there remains some ripples as shown in Fig.3-39(c). The ripple voltage at the output can be approximated as a saw-tooth wave under light load condition (large value of C and R_L) as shown in Fig.3-40. The ripple voltage has a peak-to-peak value of V_{rpp} centered around a DC level V_{dc} . Its time period (T_r) will be equal to the time period of the AC supply voltage, as the capacitor gets only one current pulse per one cycle.

The capacitor discharges during the time interval T_2 . The discharge rate is actually exponential, but for low discharge rate we can approximate it linear and the discharge current can be assumed to be equal to the average DC current I_{dc} . Thus, the amount of charge lost by the capacitor during discharge is given by,

$$Q_{dc} = I_{dc} \times T_2$$

This charge is filled up during the short time interval T_1 , by the ripple voltage V_{rpp} . Thus, using the formula $Q = CV$ we get,

$$Q_{ch} = V_{rpp} \times C$$

To maintain a constant voltage at the output these amounts of charge must be same. That is,

$$Q_{ch} = Q_{dc}$$

or,

$$V_{rpp} \times C = I_{dc} \times T_2$$

or,

$$V_{rpp} = \frac{I_{dc} \times T_2}{C} \quad (3-48)$$

As shown in Fig.3-40, the time period of the ripple (T_r) is equal to the time period of the AC voltage, that is, $T_r = T = T_1 + T_2$. But $T_1 \ll T_2$

$$\therefore T_2 \approx T = \frac{1}{f} \quad (3-49)$$

Where, f is the frequency of AC supply voltage. In Bangladesh it is 50 Hz. Now using Equ.(3-49) in Equ.(3-48) we get,

$$V_{rpp} = \frac{I_{dc}}{fC} = \frac{I_{dc}R_L}{fCR_L} = \frac{V_{dc}}{fCR_L} [\because I_{dc}R_L = V_{dc}] \quad (3-50)$$

From Fig.3-40, it is clear that

$$V_{dc} = V_m - \frac{V_{rpp}}{2} \quad (3-51)$$

Putting the value of V_{rpp} from Equ.(3-50) into Equ.(3-51), we get,

$$V_{dc} = V_m - \frac{V_{dc}}{2fCR_L}$$

or,

$$V_m = V_{dc} + \frac{V_{dc}}{2fCR_L} = V_{dc} \left(\frac{1 + 2fCR_L}{2fCR_L} \right)$$

$\therefore$

$$\boxed{V_{dc} = V_m \times \left(\frac{2fCR_L}{1 + 2fCR_L} \right)} \quad (3-52)$$

Now, we will approximate the output waveform as a triangular wave. The RMS value of a triangular wave is given by,

$$V_{rms} = \frac{V_{rpp}}{2\sqrt{3}} \quad (3-53)$$

Therefore, using Equ.(3-50) into Equ.(3-53) we get,

$$V_{rrms} = \frac{V_{dc}}{2\sqrt{3}fCR_L} \quad (3-54)$$

or,

$$\frac{V_{rrms}}{V_{dc}} = \frac{1}{2\sqrt{3}fCR_L}$$

According to the definition of ripple factor given in Equ.(3-37), V_{rrms}/V_{dc} is the ripple factor (r).

$\therefore$

$$r = \frac{1}{2\sqrt{3}fCR_L} \quad (3-55)$$

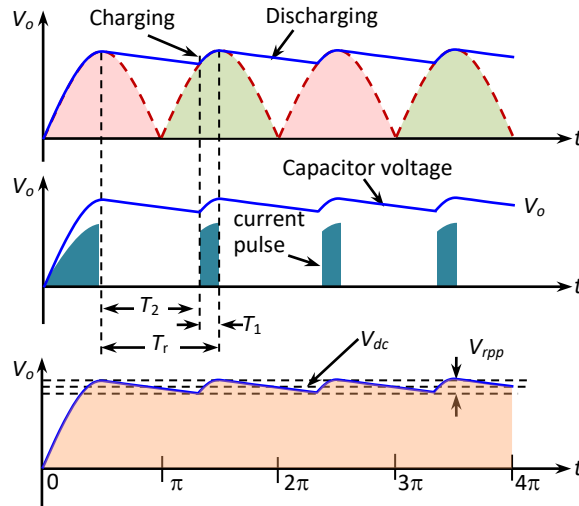


Fig.3-41: Output waveform of full-wave rectifier with capacitor filter

So far we have discussed the operation and performance of the capacitor filter with half-wave rectifier. For the full wave rectifier the operation process is same but the capacitor will be charged both in +ve and -ve half-cycles as shown in Fig.3-41. That is, the charging and discharging frequency will be double that of the half-wave rectifier. Thus, the value of $T_r = T/2$. Therefore, using the same analysis we will get,

$\therefore$

$$V_{dc} = V_m \times \left(\frac{4fCR_L}{1 + 4fCR_L} \right) \quad (3-56)$$

and the ripple factor will be,

$$r = \frac{1}{4\sqrt{3}fCR_L} \quad (3-57)$$

Therefore, we find that the ripple factor decreases with the increase of the values of C and R_L for both the half-wave and full-wave rectifications.

Example 3-15

In the rectifier circuit of Example 3-3, if a 470 μF capacitor is connected across the output, calculate the DC output voltage, ripple voltage, and the ripple factor considering an ideal diode and 50 Hz AC supply voltage.

Solution:

For the half-wave rectifier of Example 3-3, with the capacitor the DC output voltage can be calculated using Equ.(3-52),

$$V_{dc} = V_m \left(\frac{2fCR_L}{1 + 2fCR_L} \right) = 17 \text{ V} \left(\frac{2 \times 50 \times 470 \times 10^{-6} \times 1000}{1 + 2 \times 50 \times 470 \times 10^{-6} \times 1000} \right) \approx 16.65 \text{ V [Ans.]}$$

Peak-to-peak value of ripple voltage,

$$\text{Using Equ.(3-50), } V_{rpp} = \frac{V_{dc}}{fCR_L} = \frac{16.65\text{V}}{50 \times 470 \times 10^{-6} \times 1000} = 0.71 \text{ V [Ans.]}$$

RMS value of the ripple voltage can be calculated using Equ.(3-54)

$$V_{rms} = \frac{V_{dc}}{2\sqrt{3}fCR_L} = \frac{V_{rpp}}{2\sqrt{3}} = \frac{0.71\text{V}}{2\sqrt{3}} = 0.21 \text{ V [Ans.]}$$

The value of ripple factor will be,

$$\text{Using Equ.(3-55), } r = \frac{1}{2\sqrt{3} \times 50 \times 470 \times 10^{-6} \times 1000} = 1.23\% \text{ [Ans.]}$$

Comments: As discussed in Section 3.7.4, ripple factor of a half-wave rectifier is 121%. But due to the capacitor filter it has dropped to only 1.23%.

Example 3-16

Calculate the ripple factor for the power supply of Fig.3-42. Consider $R_L = 220 \Omega$ and diode voltage drop is 0.7 V.

Solution:

The turn ration of the step-down transformer is 11: 1. So the rms voltage at the transformer output will be,

$$v_{s(rms)} = 220 \text{ V}/11 = 20 \text{ V}$$

The peak secondary voltage will be,

$$v_{s(p)} = v_{s(rms)} \times \sqrt{2} = 20 \text{ V} \times \sqrt{2} \approx 28.28 \text{ V}$$

The peak value of unfiltered full-wave rectified voltage will be,

$$v_{rect(p)} = v_{s(p)} - 2 \times 0.7 \text{ V} = 28.28 \text{ V} - 1.4 \text{ V} \approx 26.88 \text{ V}$$

This voltage will be the maximum voltage of the filter capacitor, that is $V_m = v_{rect(m)} = 26.88 \text{ V}$.

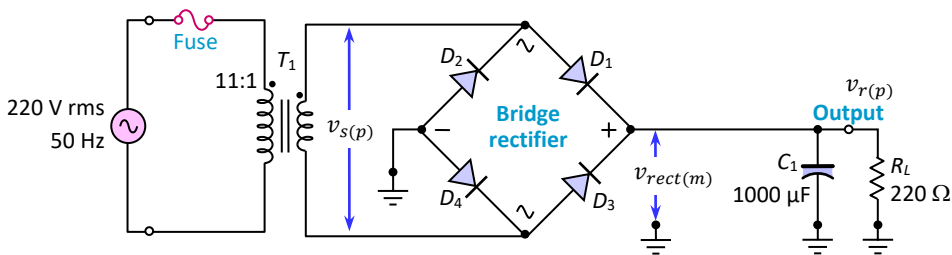


Fig.3-42: Full-wave bridge rectifier for Example 3-16

This circuit is a full-wave rectifier. So the DC voltage can be calculated using Equ.(3-56),

$$V_{DC} = 26.88 \text{ V} \times \left(\frac{4(50 \text{ Hz})(1000 \mu\text{F})(220 \Omega)}{1 + 4(50 \text{ Hz})(1000 \mu\text{F})(220 \Omega)} \right) \approx 26.28 \text{ V [Ans.]}$$

And the ripple factor can be calculated using Equ.(3-57),

$$r = \frac{1}{4\sqrt{3}fCR_L} = \frac{1}{4\sqrt{3}(50 \text{ Hz})(1000 \mu\text{F})(220 \Omega)} \approx 1.31\% \text{ [Ans.]}$$

Comments: Due to the very large value of the filter capacitor, the ripple factor is very small, and hence the DC value is very closed to the maximum voltage.

Example 3-87

A centre-tapped step-down transformer has turn ratio of 4:1. The primary is connected to an AC supply of 220 V, 50 Hz source and the secondary is connected to a two-diode full-wave rectifier circuit. The diode internal resistance is 100Ω and the load resistance is $1 \text{ k}\Omega$. Calculate the following

- AC output voltage of the transformer and the value of V_m for the rectifier
- DC voltage, current and power delivered to the load
- AC power consumed by the transformer
- Ripple factor
- PIV rating of the diodes

Solution:

i) Using the turn ratio and the input voltage, transformer output voltage can be calculated as,

$$\frac{N_1}{N_2} = \frac{V_1}{V_2} \quad \text{or,} \quad V_2 = \frac{N_2}{N_1} \times V_1 = \frac{1}{4} \times 220 \text{ V} = 55 \text{ V (rms)}$$

The RMS output of the transformer will be half of this value due to the CT. This voltage will be the input for the rectifier. That is,

$$V_i = \frac{55 \text{ V}}{2} = 27.5 \text{ V [Ans.]}$$

Thus, the maximum input voltage to the rectifier is,

$$V_m = 27.5 \times \sqrt{2} \approx 38.9 \text{ V [Ans.]}$$

ii) For this full-wave rectifier output DC voltage, current and power will be,

$$I_{dc} = \frac{2V_m}{\pi(r_D + R_L)} = \frac{2 \times 38.9 \text{ V}}{\pi(100 + 1000)\Omega} \approx 22.51 \text{ mA [Ans.]}$$

$$V_{dc} = I_{dc} \times R_L = 22.51 \text{ mA} \times 1000\Omega = 22.51 \text{ V [Ans.]}$$

$$P_{dc} = I_{dc}^2 \times R_L = (22.51 \text{ mA})^2 \times 1000\Omega \approx 0.51 \text{ Watt [Ans.]}$$

iii) We know that the efficiency of a full-wave rectifier is 81% [see Section 3.7.5]. So the input power to the transformer will be,

$$P_{ac} = \frac{0.51 \text{ Watt}}{80\%} = 0.64 \text{ Watt [Ans.]}$$

iv) The RMS output current will be,

$$I_{rms} = \frac{V_m}{\sqrt{2}(r_D + R_L)} = \frac{38.9 \text{ V}}{\sqrt{2}(100 + 1000)\Omega} \approx 25 \text{ mA}$$

The value of ripple factor

$$r = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{25 \text{ mA}}{22.51 \text{ mA}}\right)^2 - 1} = 0.48 \text{ [Ans.]}$$

v) As given by Equ.(3-28) PIV = $V_m = 27.5 \times \sqrt{2} \approx 38.9 \text{ V [Ans.]}$

Comments: The diodes used in this circuit should have the reverse breakdown voltage greater than the 38.9 V.

3.8.2 Series Inductor Filter

It consists of an inductor (also called choke as it chokes the AC current and passes the DC current) in series with the load resistor R_L as shown in Fig.3-43. The operation of such a filter depends on

the fundamental property of an inductor to oppose any sudden change in the current following through it. Since this inductor provides high impedance to the AC components of the filter output, it reduces their amplitude without affecting the DC component. Thus, an inductor filter reduces the ripple value as shown in Fig.3-44. Alternately, we can

say that when the current through the load resistance tends to rise above the average value ($2I_m/\pi$), the inductor opposes this rise and stores energy in magnetic field. On the other hand, when load current tends to fall below the average value the inductor prevents the current from falling down too much using its stored energy. In this way, the inductor tries to minimize variation of the load current and obviously the load voltage. In Fig.3-44, the red dashed line shows the full-wave rectifier output and the blue solid line shows the output after filtration.

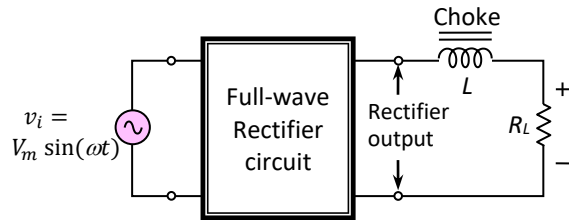


Fig.3-43: Rectifier with series inductor filter

The Fourier series for the full-wave rectified output is,

$$v_o = V_m \left(\frac{2}{\pi} - \frac{4}{3\pi} \cos 2\omega t - \frac{4}{15\pi} \cos 4\omega t \dots \dots \dots \right) \quad (3-58)$$

For this series inductor filter, the higher order harmonic components (the AC signals that have frequencies integer multiple of the input supply frequency, are called **harmonics**. The frequency of the input AC voltage is called the **fundamental frequency**. In Equ.(3-58), the fundamental frequency is ω and the harmonics are $2\omega, 4\omega$ and so on. Other than the 2nd harmonic produce very small output and can be neglected. Then, we can write,

$$v_o = V_m \left(\frac{2}{\pi} - \frac{4}{3\pi} \cos 2\omega t \right) = \frac{2V_m}{\pi} - \frac{4V_m}{3\pi} \cos 2\omega t \quad (3-59)$$

Now the output current is,

$$i_o = \frac{v_o}{Z} = \frac{v_o}{\sqrt{R_T^2 + X_L^2}} \quad (3-60)$$

where, R_T is the total resistance and X_L is the reactance of the choke (inductor L). The total resistance of the complete current loop is $R_T = (r_s + r_D + r_L + R_L)$. Here, r_L is the resistance of the choke. But generally, the value of $(r_s + r_D + r_L)$ is very small compared to R_L . Then neglecting $(r_s + r_D + r_L)$, we get $R_T = R_L$. The choke reactance ($X_L = \omega L$) has different values for different harmonic components. For the first component or the DC component as shown in Equ.(3-59) $\omega = 0$. So, we get $X_L = 0$. Thus, for the first DC component the value of the impedance will be $Z = R_T = R_L$. Then, from Equ.(3-59) and Equ.(3-60) we can write,

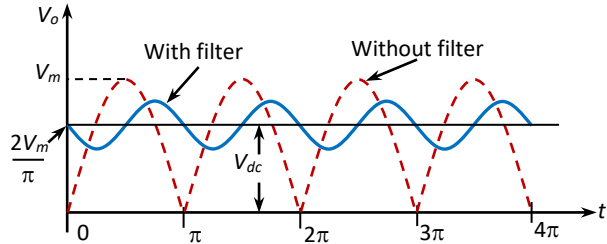


Fig.3-44: Output waveform of full-wave rectifier with inductor filter

$$I_{dc} = \frac{2V_m/\pi}{Z} = \frac{2V_m/\pi}{R_L} = \frac{2V_m}{\pi R_L} \quad (3-61)$$

or,

$$V_{dc} = I_{dc} \times R_L = \frac{2V_m}{\pi} \quad (3-62)$$

Equ.(3-62) produces the same DC value as we have determined for the full-wave rectifier before [see Equ.(3-30)].

Now for the 2nd component of Equ.(3-59), i.e., the 2nd harmonic of the output voltage, the value of the impedance will be (this is the only AC component, as we have neglected other higher order components)

$$Z = \sqrt{R_L^2 + (2\omega L)^2} \quad (3-63)$$

Dividing the 2nd term of Equ.(3-59) by Equ.(3-63), we get the AC output current or the ripple current (i_r) for inductor filter,

$$i_r = \frac{(4V_m/3\pi) \cos 2\omega t}{Z} = \frac{4V_m}{3\pi} \frac{1}{\sqrt{R_L^2 + (2\omega L)^2}} \cos(2\omega t - \phi) \quad (3-64)$$

Here, the -ve sign has been neglected, as it represents merely a phase change of 2nd harmonic. ϕ represents the phase shift of AC current and is given by $\phi = \tan^{-1} \left(\frac{2\omega L}{R_L} \right)$.

Equ.(3-64) gives the AC output current. The maximum value of this AC current is,

$$I_{rm} = \frac{4V_m}{3\pi} \frac{1}{\sqrt{R_L^2 + (2\omega L)^2}} \quad (3-65)$$

So, the RMS value of this AC current can be calculated dividing by $\sqrt{2}$,

$$I_{rrms} = \frac{4V_m}{3\pi\sqrt{2}} \frac{1}{\sqrt{R_L^2 + (2\omega L)^2}} = \frac{4V_m}{3\pi\sqrt{2}\sqrt{R_L^2 + (2\omega L)^2}} \quad (3-66)$$

Ripple Factor

In Section 3.7.4, we have defined ripple factor in terms of voltage. It can also be defined in the same way in terms of current,

$$r = \frac{\text{RMS value of ripple current}}{\text{DC current}} = \frac{I_{rrm}}{I_{dc}} \quad (3-67)$$

Using Equ.(3-61) and Equ.(3-66), we get

$$r = \frac{4V_m/(3\pi\sqrt{2}\sqrt{R_L^2 + (2\omega L)^2})}{2V_m/\pi R_L} = \frac{\sqrt{2}R_L}{3} \frac{1}{\sqrt{R_L^2 + (2\omega L)^2}}$$

or,

$$r = \frac{\sqrt{2}R_L}{3} \frac{1}{R_L\sqrt{1 + (2\omega L/R_L)^2}} = \frac{\sqrt{2}}{3} \frac{1}{\sqrt{1 + (2\omega L/R_L)^2}}$$

Neglecting 1 in the denominator we get,

$$r = \frac{\sqrt{2}}{3} \frac{1}{2\omega L/R_L}$$

or,

$$r = \frac{1}{3} \frac{R_L}{\sqrt{2}\omega L}$$

Finally,

$$r = \frac{R_L}{3\sqrt{2}\omega L} = \frac{R_L}{6\sqrt{2}\pi f L} \quad (3-68)$$

For $f = 50$ Hz,

$$r = \frac{R_L}{6\sqrt{2}\pi f L} = \frac{R_L}{1333L} \text{ [for } f = 50 \text{ Hz]} \quad (3-69)$$

Therefore, in case of series inductor filter, the ripple factor increases with the increase of R_L and decreases with the decrease of inductance L .

Example 3-9

- i) What should be the value of inductance of an inductor filter to be used in a full-wave rectifier operating at 50 Hz, if the ripple factor is not to exceed 5 % for $R_L = 100 \Omega$.
- ii) Repeat the above problem for the standard aircraft power frequency of 400 Hz.

Solution:

- i) Given: $f = 50$ Hz, $r = 5\% = 0.05$, and $R_L = 100 \Omega$

Using Equ.(3-69), $r = \frac{R_L}{1333L}$ or, $L = \frac{R_L}{1333r} = \frac{100 \Omega}{1333 \times 0.05} = 1.5 \text{ H [Ans.]}$

- ii) For $f = 400$ Hz,

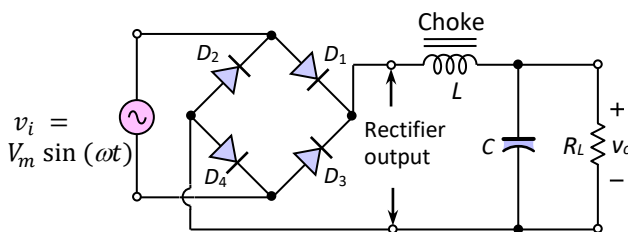
Using Equ.(3-68), $r = \frac{R_L}{6\sqrt{2}\pi f L}$ or, $L = \frac{R_L}{6\sqrt{2}\pi f r}$

or, $L = \frac{100 \Omega}{6\sqrt{2}\pi \times 400 \times 0.05} = 0.188 \text{ H [Ans.]}$

Comments: As the series inductor filter blocks the AC current (rate of change of current), the required value of inductance decreases with the increase in frequency.

3.8.3 LC Filter

We have already seen that the ripple factor of shunt capacitor filter decreases with the increase of load resistance R_L and that of the series inductor filter increases with the increase of load resistance R_L . Therefore, if we use both of them, i.e., a series inductor and a shunt capacitor then, the ripple factor will be independent of the load resistance, R_L . This type of filter is called *LC filter*. The circuit diagram of an *LC filter* is shown in Fig.3-45 at the output of a full-wave bridge rectifier.



4Fig.3-45: Rectifier with LC filter

The function of this circuit is twofold. At first, the inductor blocks the AC component and passes the DC component. But the inductor cannot block all of the AC voltage. The remaining AC voltage at the output is further filtered by the capacitor filter. The capacitor filter shorts the AC component to the ground and supply the DC value to the load resistance. The output waveform is shown in Fig.3-46. Here, the dashed line is full-wave rectified voltage and the solid line is the output voltage after filtering.

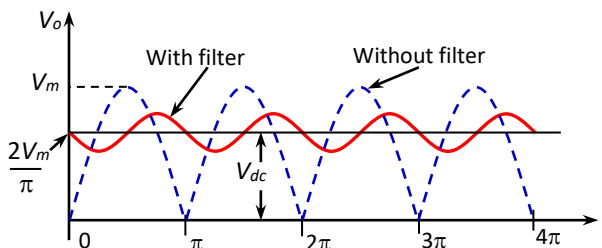


Fig.3-46: Output waveform of full-wave rectifier with LC filter

Ripple Factor Calculation

If we neglect the DC resistance of the inductor (r_L), it will behave as short circuit. Already we have seen that the Fourier series for the full-wave rectified

output has a DC component and many harmonics [Equ.(3-58)]. The DC component of the Fourier expression will easily pass through the inductor. Again, for this DC component the capacitor is open circuit. Thus, the full DC value $2V_m / \pi$ will be available across the load resistance. Hence, the DC output voltage is same as before, that is,

$$V_{dc} = \frac{2V_m}{\pi} \quad (3-70)$$

For the capacitor filter, we have seen that the ripple factor decreases with the increase of C and R_L . Thus, for lower value of ripple we must use large values of C and R_L . As the larger value of C will produce lower value of capacitive reactance, we can write $X_C \ll R_L$ for AC component. For this reason the impedance of the parallel combination of C and R_L will be approximately X_C .

Again, for the series inductor filter, we have seen that the ripple factor decreases with decrease of load resistance. Now, $X_L \gg R_L$. For this reason, we can conclude that X_C must be lower than X_L .

Now, if we consider the 2nd harmonic component, $(4V_m/3\pi) \cos 2\omega t$, the maximum value of the voltage across the capacitor (which is also the AC output or ripple output) will be (using voltage divider rule),

$$V_{rm} = \frac{4V_m}{3\pi} \times \frac{X_C}{X_C + X_L}$$

Thus, the RMS value of the AC output is

$$V_{rrms} = \frac{V_{rm}}{\sqrt{2}} = \frac{4V_m}{3\sqrt{2}\pi} \times \frac{X_C}{X_C + X_L}$$

The ripple factor will be,

$$r = \frac{V_{rrms}}{V_{dc}} = \left(\frac{4V_m}{3\sqrt{2}\pi} \times \frac{X_C}{X_C + X_L} \right) \div \left(\frac{2V_m}{\pi} \right)$$

$$\text{or,} \quad r = \frac{4V_m}{3\sqrt{2}\pi} \times \frac{X_C}{X_C + X_L} \times \frac{\pi}{2V_m} = \frac{\sqrt{2}}{3} \times \frac{X_C}{X_C + X_L}$$

Since $X_C \ll X_L$, we can write,

$$r = \frac{\sqrt{2}}{3} \times \frac{X_C}{X_L} = \frac{\sqrt{2}}{3} \times \frac{1/\omega C}{\omega L} = \frac{\sqrt{2}}{3(\omega L)(\omega C)} = \frac{\sqrt{2}}{3\omega^2 LC} \text{ [for any frequency } \omega]$$

For the 2nd order AC component

$$r = \frac{\sqrt{2}}{3(2\omega L)(2\omega C)} = \frac{1}{6\sqrt{2} \omega^2 LC} = \frac{1.194}{LC} \text{ [Considering } L \text{ in Henry, } C \text{ in } \mu F \text{ and } f = 50 \text{ Hz}]$$

$$\boxed{r = \frac{1}{6\sqrt{2} \omega^2 LC} = \frac{1.194}{LC}} \quad (3-71)$$

Now, the DC current $I_{dc} = 2V_m/\pi R_L$. Maximum value of 2nd harmonic current (AC current) is,

$$I_{2m} = \frac{4V_m}{3\pi X_L} = \frac{4V_m}{3\pi(2\omega L)}$$

I_{2m} should not exceed I_{dc} to prevent the negative current through the load.

Thus, at the critical condition,

$$I_{2m} = I_{dc} \quad \text{or,} \quad \frac{2V_m}{\pi R_L} = \frac{4V_m}{3\pi(2\omega L_c)}$$

$$\therefore \quad L_c = \frac{R_L}{3\omega} \quad (3-72)$$

This is the critical value of L for which the circuit can barely avoid negative value. But for practical case the value of L should be greater than L_c . A very good practice is to keep L , 25% higher than this critical value, L_c .

Example 3-109

A single phase full-wave rectifier uses a centre tapped 220-0-220 V, 50 Hz transformer. For a load current of 100 mA, design an LC filter so that the ripple factor is not more than 1%.

Solution:

Maximum input voltage to the rectifier is $V_m = \sqrt{2} \times V_{rms} = \sqrt{2} \times 220 \text{ V} \approx 311 \text{ V}$.

DC output voltage is,

$$\text{Using Equ.(3-70),} \quad V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 311 \text{ V}}{\pi} \approx 198 \text{ V}$$

Therefore the value of the load resistance is,

$$R_L = \frac{V_{dc}}{I_{dc}} = \frac{198 \text{ V}}{100 \text{ mA}} = 1.98 \text{ k}\Omega$$

For an LC filter we know the $X_C \ll R_L$. Let

$$10X_C = R_L \quad \text{or,} \quad 10 \times \frac{1}{\omega C} = R_L \quad \therefore C = \frac{10}{2\pi f R_L} = \frac{10}{2\pi(50)(1980)} \approx 16 \mu\text{F}, 200 \text{ V}$$

For 50 Hz AC frequency, using Equ.(3-71), the ripple factor will be,

$$r = \frac{1.194}{LC} \quad \text{or,} \quad L = \frac{1.194}{rC} = \frac{1.194}{0.01 \times 16} \approx 7.5 \text{ H [Ans.]}$$

$$L_c = \frac{R_L}{3\omega} = \frac{1980}{3 \times 2\pi f} = \frac{1980}{3 \times 2\pi \times 50} \approx 2 \text{ H [Ans.]}$$

Comments: The inductance used in the circuit should be greater than the critical inductance $L_c = 2 \text{ H}$.

3.9 Bleeder Resistor

Very often, a resistor (called **bleeder resistor**) is placed across the output filter of a power supply. By acting a pre-load on the power supply, it causes an initial voltage drop. Hence, when the actual load is connected, it causes a small amount of additional drop. In this way, the bleeder resistor reduces the change of the output voltage from no-load condition to full-load condition. When the

power supply is switched OFF, it provides a path for the filter capacitor to discharge through. That is why it is called a **bleeder resistor**.

Let us consider that the AC supply voltage, applied to a system, changes rapidly. We want to measure the level of AC voltage by measuring the output DC voltage of a power supply (consisting of a transformer, a rectifier circuit and a filter). If we use a DC volt meter, or an ADC (Analog to Digital Converter), they will draw a very small amount of current. Now consider that the AC supply decreases suddenly. But as the filter capacitor has no path to discharge, we cannot measure the decrease of AC voltage immediately. It will be possible to measure the change if we use the bleeder resistor.

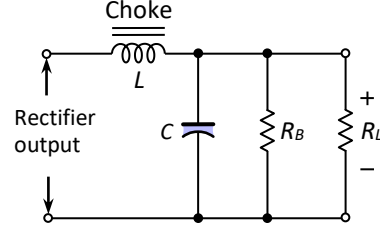


Fig.3-47: Bleeder resistor at the output of the filter and parallel to the load

3.10 Voltage Regulation

In an unregulated power supply, output voltage changes whenever input supply voltage or load current changes. The change in output voltage from no-load to full-load is called **voltage regulation**. When it is represented as a % of the no-load voltage, it is called **percentage regulation** or simply **regulation** (VR_L). As it is related to load current, it is also called **load regulation**. Mathematically it can be represented as,

$$\% \text{ regulation } (VR_L) = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 \quad (3-73)$$

Here, V_{NL} = no-load output voltage, and V_{FL} = full-load output voltage. A good power supply will have lower value of V_{NL} . It is related with the internal output resistance of the power supply. Generally, the power supply manufacturers declare the value of V_{FL} , I_{FL} and the internal resistance r_o . If we observe the equivalent circuit of a power supply (Fig.3-48), we find

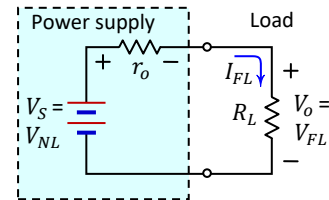


Fig.3-48: Equivalent circuit of a power supply with full-load

$$\begin{aligned} V_{NL} &= V_{FL} + I_{FL} r_o \\ \therefore VR_L &= \frac{V_{NL} - V_{FL}}{V_{FL}} = \frac{(V_{FL} + I_{FL} r_o) - V_{FL}}{V_{FL}} = r_o \left(\frac{I_{FL}}{V_{FL}} \right) \\ \therefore \boxed{VR_L = r_o \left(\frac{I_{FL}}{V_{FL}} \right)} \quad (3-74) \end{aligned}$$

From this expression it is clear that for a fixed value of V_{FL} , and I_{FL} a lower value of r_o gives better regulation.

So far, we have considered the variation of output voltage due to the variation of load current. But, the output voltage may also change due to the variation of the source AC voltage. This change of output voltage due to the change of source voltage is called **source regulation** (VR_S). In any country $\pm 10\%$ variation of the line voltage is an acceptable range. A power supply will provide V_{FL}

for the rated value of source voltage. If ΔV_o is the change in output voltage for the 10 % change in source voltage (ΔV_s), source regulation (VR_s) can be calculated as,

$$\% \text{ regulation } (VR_s) = \frac{\Delta V_o}{V_{FL}} \times 100 \quad (3-75)$$

$$\text{or, } VR_s = \frac{\text{Change of output voltage due to 10\% change in source voltage}}{V_{FL}} \quad (3-76)$$

A good power supply has lower values of both VR_L and VR_s .

Example 3-11

A certain 12 V regulator has a measured no-load output voltage of 12.2 V and a full-load voltage of 11.8 V. What is the load regulation expressed as a percentage?

Solution: Using Equ.(3-73),

$$\text{Load regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{12.2 \text{ V} - 11.8 \text{ V}}{11.8 \text{ V}} \times 100\% \approx 3.39\% [\text{Ans.}]$$

Comments: Actually, the load regulation depends on the net internal resistor of a voltage source. A lower value of internal resistance provides better (lower) VR_L .

3.10.1 Zener Diode as Voltage Regulator

The power supply that consists of simply a transformer, a rectifier circuit, and a filter circuit cannot provide good source regulation (constant voltage though the source voltage changes) and load regulation (constant voltage though the load current changes). We can improve the performance of this type of power supply, by using a voltage regulator circuit at the final stage (in between the filter and the load). Although a lot of complicated circuits are employed as voltage regulator, for a low power application, Zener diodes are commonly used. A simple regulator circuit consists of a Zener diode and a series resistor is shown in Fig.3-49. The unregulated voltage (V_s) is applied to the circuit and the regulated output voltage (V_o) is taken across the Zener diode. The regulated output voltage will be equal to the Zener breakdown voltage V_Z . Resistor R_1 limits the Zener diode current (I_Z) to the desired level. Using a suitable value of R_1 a suitable operating point of the Zener diode is set in between I_{ZK} and I_{ZM} [as shown in Fig.3-49(c)]

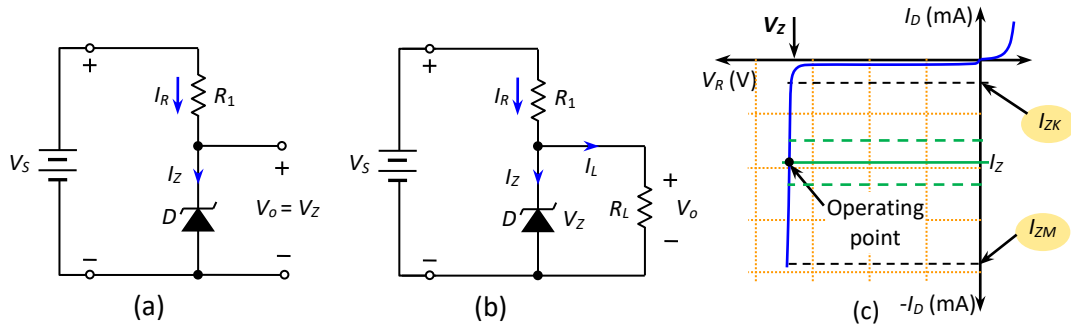


Fig.3-49: (a) Zener diode voltage regulator without load resistance, (b) Zener diode voltage regulator with load resistance, and (c) Operating point on the Zener diode characteristic curve

If the value of V_S increases the value of I_Z increases and the voltage drop V_R increases, but V_o remains constant. Similarly, if V_S decreases the value of I_Z decreases and the voltage drop V_R decreases and V_o remains constant again. But if the source voltage remains constant, and a load is connected across the Zener diode, depending on the load some current will be diverted from the Zener diode to the load. But the current through the resistor and hence voltage drop V_R will remain unchanged. Thus, the Zener output voltage will also remain constant. For this reason, the value of I_Z has to be set to a suitable level (in between I_{ZK} and I_{ZM} as shown in Fig.3-49(c).

If we don't know the value of I_L , it will be difficult to design the Zener current. Too low

value of designed I_Z may cause Zener current lower than I_{ZK} and too higher value of designed I_Z may cause unnecessary power loss, that is, inefficient circuit design. For this reason, a good circuit can be designed if the value of load current (I_L) and load voltage (V_L) is known beforehand.

Regulator for a Given Load

Here, we must know the output voltage (V_o) and load (output) current (I_o). From these values we can also find the value of load resistance $R_L = V_L / I_L = V_o / I_L$. The Zener regulator with a load resistor is shown in Fig.3-50(b). To select the power of the Zener diode, we must know the value of source voltage V_S . At first we have to choose a suitable Zener diode. We have to keep I_Z as minimum as possible. It may be just a bit greater than I_{ZK} . Then the current equation will be,

$$I_Z + I_L = \frac{V_S - V_Z}{R_1} \quad (3-77)$$

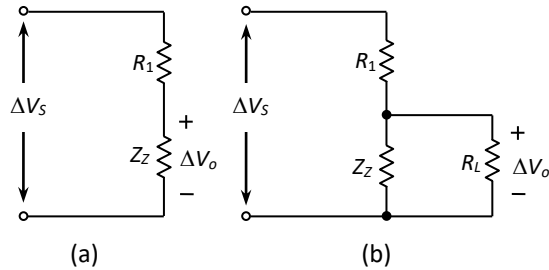


Fig.3-50: (a) Equivalent circuit of a Zener voltage regulator without load, (b) Equivalent circuit of a Zener voltage regulator with load.

If the load resistor is disconnected then, $I_L = 0$. In that case, as V_S is constant, the entire current will flow through the Zener diode. So, the maximum current rating of the Zener diode will be: $I_{ZM} = (V_S - V_Z)/R_1$.

Design summary

Given: Load current I_L and load voltage V_L .

Choose a Zener diode with $V_Z = V_L$. Now consider $I_Z = I_{ZK} + 2 \text{ mA}$. If I_{ZK} is not known, set $I_Z = 5 \text{ mA}$ (a common minimum value). Calculate $I_R = I_Z + I_L$ and find the value of R_1 from Equ.(3-77).

Zener diode power must be greater than,

$$P_Z = V_Z \times (I_Z + I_L) \quad (3-78)$$

and the power rating of the resistor should be,

$$P_R = (V_S - V_Z) \times (I_Z + I_L) = I_R^2 R \quad (3-79)$$

Regulator Performance

As the purpose of a regulator is to provide constant output voltage, the performance of Zener diode regulator is assessed by the **load regulation** and **source regulation**. The performance depends on the Zener diode dynamic impedance (Z_Z). The equivalent circuit of a Zener regulator is shown in Fig.3-50. Here, the Zener diode has been replaced with Z_Z .

For a change in source voltage ΔV_S the change in output voltage ΔV_o (source regulation) will be,

$$\Delta V_o = \frac{\Delta V_S \times Z_Z}{R_1 + Z_Z} \quad (3-80)$$

If a load resistor is connected in parallel with Z_Z , as shown in Fig.3-48(b), then

$$\Delta V_o = \frac{\Delta V_S \times (Z_Z || R_L)}{R_1 + (Z_Z || R_L)} \quad (3-81)$$

For a load current change of ΔI_L , the change in output voltage (load regulation) will be

$$\Delta V_o = \Delta I_L (Z_Z || R_L) \quad (3-82)$$

Generally, the filtered output of rectifier is used as the source voltage for a regulator. Obviously, it contains some ripple. This ripple is drastically attenuated at the output of the regulator. The ratio of the ripple voltage at the output to that at the input of a regulator is defined as **ripple rejection ratio**.

For this regulator, if the value of input ripple voltage is V_{ri} , the output ripple voltage will be

$$V_{ro} = \frac{V_{ri} \times (Z_Z || R_L)}{R_1 + (Z_Z || R_L)} \quad (3-83)$$

$$\text{or,} \quad \text{Ripple rejection ration} = \frac{V_{ro}}{V_{ri}} = \frac{(Z_Z || R_L)}{R_1 + (Z_Z || R_L)} \quad (3-84)$$

From the above discussion and equations, we can commend that the performance of a Zener diode regulator depends on both the value of R_1 and Z_Z . Higher value of R_1 and lower value of Z_Z will give better performance of the Zener diode voltage regulator.

Example 3-12

A Zener diode voltage regulator has to be designed for an output voltage of $V_o = 10\text{ V}$, and load current $I_L = 20\text{ mA}$. The source voltage is fixed and its value is $V_S = 18\text{ V}$. Design the regulator and draw the circuit diagram.

Solution:

As the value of reverse knee current I_{ZK} is not given, let us consider $I_Z = 5\text{ mA}$.

$$\therefore I_R = I_Z + I_L = 5\text{ mA} + 20\text{ mA} = 25\text{ mA}$$

$$\text{Using Equ.(3-77), } R_1 = \frac{V_S - V_Z}{I_R} = \frac{18\text{ V} - 10}{25\text{ mA}} = 320\ \Omega$$

Maximum current rating of the Zener diode should be $I_{ZM} = I_R = 25\text{ mA}$.

Power rating of the diode is $P_D = 25\text{ mA} \times 10\text{ V} = 25\text{ mW}$.

Power rating of the resistor is $P_R = I_R^2 \times R = (25\text{ mA})^2 \times (320\ \Omega) = 200\text{ mW}$ (use a quarter watt resistor).

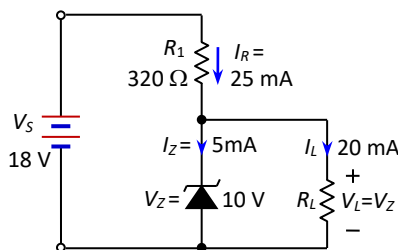


Fig.3-51: Zener voltage regulator of Example 3-21

Comments: The complete designed circuit is shown in Fig.3-49.

Example 3-132

- In the above Example 3-21, calculate the load current and load voltage if a $200\ \Omega$ load resistor is connected at the output.
- What will be the minimum value of R_L if the Zener has $I_{ZK} = 2\text{ mA}$?
- What will be the maximum value of R_L if a Zener diode is used with $I_{ZM} = 20\text{ mA}$?

Solution:

In the above Example 3-21, $V_S = 18\text{ V}$, $V_o = V_Z = 10\text{ V}$, $I_L = 20\text{ mA}$, and $I_Z = 5\text{ mA}$.

i) We know that for a Zener voltage regulator circuit, the output voltage is $V_o = V_Z = V_L$ and $I_L = V_L/R_L$.

Let's try this equation,

$$I_L = \frac{V_L}{R_L} = \frac{10\text{ V}}{200\ \Omega} = 50\text{ mA}$$

This answer is not correct. If 50 mA current flows through the load, it must come through the resistor, R_1 . Then the voltage drop across the resistor will be $V_{R1} = (50 \text{ mA}) \times (320 \Omega) = 16 \text{ V}$. And the voltage drop across the Zener diode and also the load resistance will be $V_Z = V_S - V_R = 18 \text{ V} - 16 \text{ V} = 2 \text{ V}$! This problem is due to the very small load resistance, R_L .

Actually in this case the Zener diode will not work in the breakdown region, and it will not work as a voltage regulator at all.

Well, for a given circuit where the values of V_S , V_Z and R_1 are already given, it may not work for a very low load resistance as we have seen here.

Actually, a Zener regulator will not work for a load if it tries to draw a current more than $(I_R - I_{ZK})$. In this case, the Zener diode will behave as an open circuit as shown in the Fig.3-52 and we have to find out the value of V_o or V_L using voltage divider rule.

Therefore, the correct value of V_L will be:

$$V_L = V_S \times \frac{V_L}{R_1 + R_L} = 18 \text{ V} \times \frac{200 \Omega}{320 \Omega + 200 \Omega} = 6.9 \text{ V}$$

So, $V_Z = 6.9 \text{ V}$ [Ans.]

As the applied voltage across the Zener diode is less than its breakdown voltage (10 V), it will not work in breakdown region.

ii) As the minimum current required for the Zener diode is $I_{ZK} = 2 \text{ mA}$, the maximum value of load current will be

$$I_{L(max)} = I_R - I_{ZK} = 25 \text{ mA} - 2 \text{ mA} = 22 \text{ mA}$$

Therefore the minimum value of load resistance will be

$$R_{L(min)} = \frac{V_Z}{I_{L(max)}} = \frac{10 \text{ V}}{22 \text{ mA}} = 455 \Omega \text{ [Ans.]}$$

iii) In the given circuit of Example 3-21, $I_R = 25 \text{ mA}$. If the maximum current capacity of the Zener diode is $I_{ZM} = 20 \text{ mA}$. The minimum

value of load current will be

$$I_{L(min)} = I_R - I_{ZM} = 25 \text{ mA} - 20 \text{ mA} = 5 \text{ mA}$$

Thus, the maximum value of the load resistance will be

$$R_{L(max)} = \frac{V_Z}{I_{L(min)}} = \frac{10 \text{ V}}{5 \text{ mA}} = 2 \text{ k}\Omega \text{ [Ans.]}$$

Comments: A Zener diode voltage regulator will work properly, if the load voltage (V_L), found from voltage divider rule by R_1 and R_L , is greater than the break down voltage of the Zener diode.

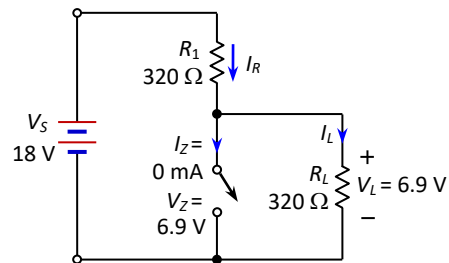


Fig.3-52: Zener diode will work as open circuit for very small load resistance

3.11 A Complete Power Supply

A power supply is a circuit that takes AC power as input and produces DC power as output (DC voltage and DC current). Often we need a simple DC power supply to operate electronic circuits. A good quality power supply will be designed and discussed in Chapter 18. Here, we will see how we can make a simple power supply by using the circuits studied so far. The circuit diagram of a simple power supply is shown in Fig.3-53. If we look at the circuit from left (input side) to right (output side), we find that it consists of a centre-tapped transformer, a rectifier section, a filter section and a regulator section. The transformer steps down 220 V AC supply into required AC voltage. Here, we have made a very common 5 V DC power supply. This type of power supply is mainly used in digital electronics. However, by changing the transformer's voltage rating and the Zener diode voltage, power supply of any value can be developed. The transformer must provide AC output voltage a little greater than the required DC voltage. That is why a 6 V CT transformer has been used here. To convert the 6 V AC, a two-diode rectifier has been used. If the transformer does not have a CT terminal we have to use the bridge rectifier. The output of the rectifier is pulsating DC. To convert this pulsating DC into pure DC an LC filter has been used. For simple desing, we can use a simple capacitor filter. Finally, to regulate the output voltage a Zener diode voltage regulator has been used. Here, the Zener diode breakdown voltage is 5.1 V. Zener diode voltage regulator is not suitable for high current-level. In that case, we need different types of regulator as discussed in Chapter 18.

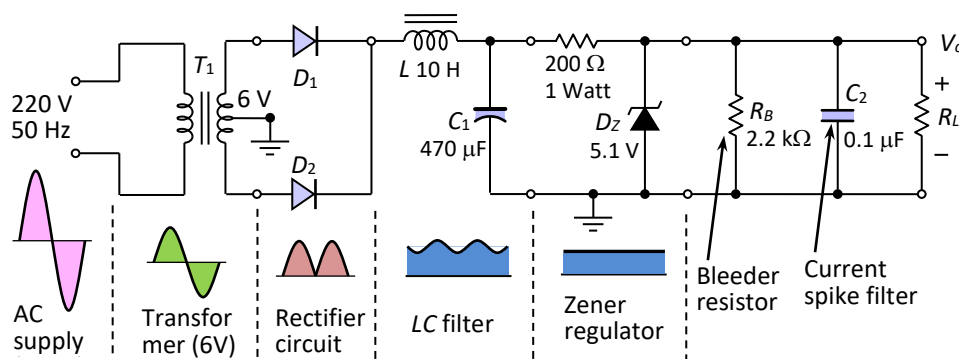


Fig.3-53: Different stages of a typical power supply with 5 V output

A $2.2\text{ k}\Omega$ bleeder resistor and a $0.1\text{ }\mu\text{F}$ ceramic disk capacitor have been used at the output. The function of the bleeder resistor has already been discussed. The function of the ceramic capacitor is to filter current spikes that may be generated in digital electronic circuits.

3.12 Voltage Multiplier

A voltage multiplier is a circuit which produces a higher value of DC output voltage than the AC voltage supplied to the input. Actually, a multiplier can produce output voltage which is equal to the integer-multiple of the peak value (V_m) of the AC voltage. That is why, it is called a voltage multiplier. High voltage can also be produced using a step-up transformer. But transformers are bulky and costly. Voltage multipliers are used where a very high voltage but low current is

necessary. Examples of practical applications are **photocopiers**, **bug zappers**, **CRT** (cathode ray tube) etc.

3.12.1 Half-wave Voltage Doubler

The circuit diagram of a half-wave voltage doubler is shown in Fig.3-54. It consists of two stages of series connected diode and capacitor circuits. For this reason it is also called a cascaded voltage doubler.

During the positive half-cycle of input voltage, D_1 becomes forward biased and D_2 becomes reverse biased. Current flows through D_1 and charges the capacitor C_1 up to the peak value of AC supply (V_m) as shown in Fig.3-55(a). During the negative half-cycle of input voltage, D_2 becomes forward biased and

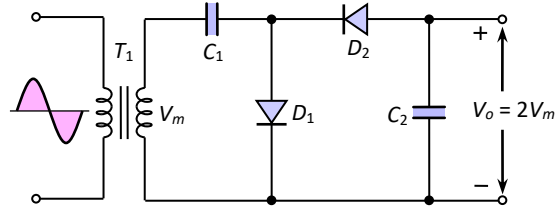


Fig.3-54: Circuit diagram of a half-wave voltage doubler

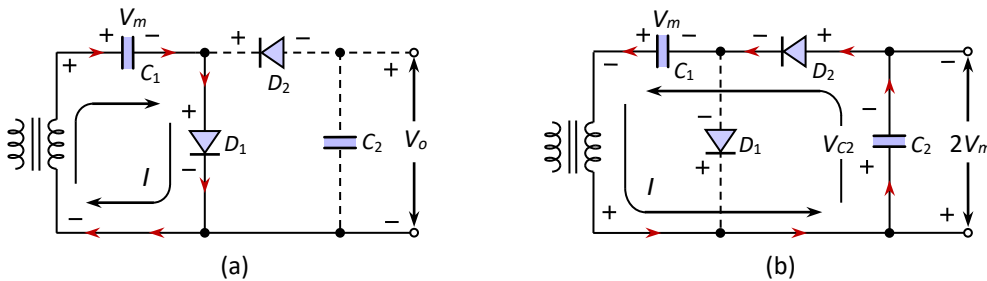


Fig.3-55: Circuit status for: (a) positive half-cycle, and (b) negative half-cycle

D_1 becomes reverse biased [see Fig.3-55(b)]. Now, current flows through D_2 and charges the capacitor C_2 . But this time, the total voltage that is applied across the capacitor is the sum of the peak value of AC voltage (V_m) and V_m voltage stored in the capacitor C_1 . Well, to make it clear we can apply KVL at the outer loop. Moving anticlockwise we get,

$$+V_m - V_{C2} + V_{C1} = 0 \quad \text{or,} \quad V_m - V_{C2} + V_m = 0 \quad [\because V_{C1} = V_m]$$

$$\therefore V_{C2} = 2V_m \quad (3-85)$$

Note the polarity of the output voltage negative at the upper point. We can change this polarity by reversing the direction of the diodes.

During the third positive half-cycle, again D_1 conducts and D_2 remains OFF. At this time the capacitor C_2 discharges through the load resistance. Depending on the value of the load resistance the voltage level of C_2 decreases slightly. But it will be recharged in the next negative half-cycle. In this way the process will be continued.

As shown in Fig.3-56, the output voltage waveform will be like that of a half-wave rectified capacitor filtered waveform. But the average DC voltage level is $2V_m$. PIV across each

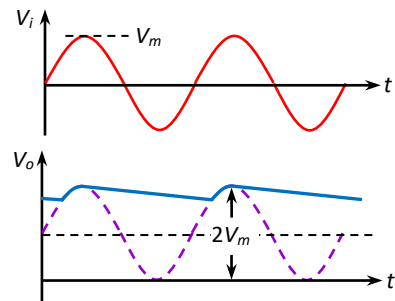


Fig.3-56: Output waveform of the half-wave voltage doubler

diode is $2V_m$. Ripple frequency will be same as the supply frequency. Although the circuit can produce high level of voltage but the current capacity will be low. It has very poor voltage regulation and its ripple factor will also be high.

3.12.2 Full-wave Voltage Doublers

The half-wave voltage doublers supply power to the load only during one half-cycle as discussed above. But the full-wave voltage doublers provide power to the load for both the positive and negative half-cycles. The circuit diagram of a full-wave voltage doubler is shown in Fig.3-57. Like the half-wave voltage doubler, it also consists of two diodes and two capacitors. But here, the output is taken from both the capacitors.

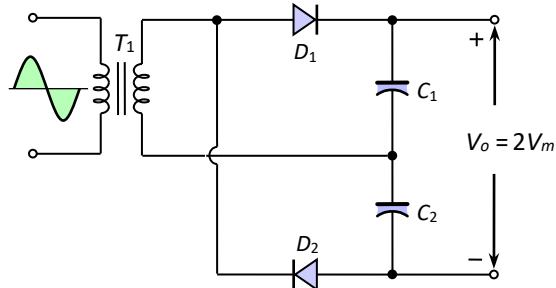


Fig.3-57: Circuit diagram of a full-wave voltage doubler

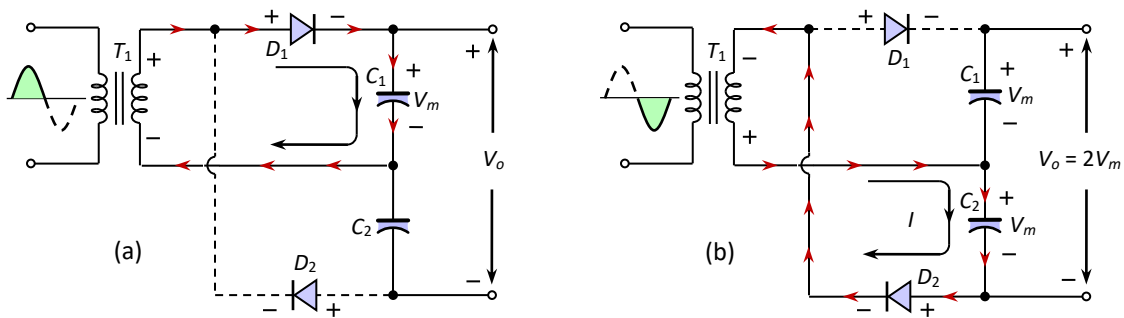


Fig.3-58 : (a) Circuit status for positive half-cycle, and (b) Circuit status for negative half-cycle

During the positive half-cycle of the input AC voltage, diode D_1 is forward biased and D_2 is reverse biased as shown in Fig.3-58(a). Current flows through C_1 and charges it to the peak value of AC supply (V_m). During the negative half-cycle, diode D_2 is forward biased and D_1 is reverse biased as shown in Fig.3-58(b). Current flows through C_2 and charges it to the peak value of AC supply (V_m). As the output is taken across the two capacitors, the output voltage will be the sum of V_{C1} and V_{C2} , that is $2V_m$. However, if a load resistance is connected, output voltage will be less than $2V_m$ depending on the load resistance and capacitances. The waveform of output voltage will be same as that of a full-wave rectifier with a shunt capacitor filter as shown in Fig.3-59.

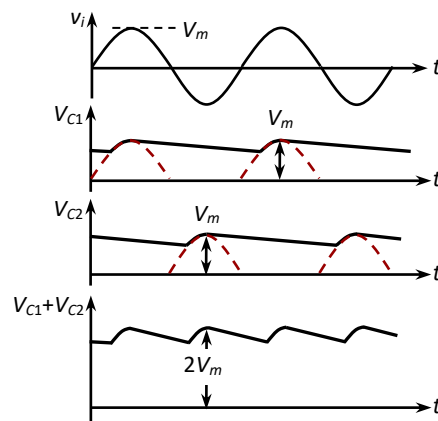


Fig.3-59: Output waveforms of full-wave voltage doubler

For this full-wave voltage doubler, the PIV rating of each diode is $2V_m$. Ripple frequency is twice the frequency AC of supply voltage.

3.12.3 Voltage Tripler and Quadrupler Circuits

The half-wave voltage doubler of Fig.3-57, can be cascaded to obtain any multiple of the peak input voltage (V_m), i.e., $3V_m$, $4V_m$, $5V_m$ and so on. The circuit shown in Fig.3-60 can be used as a doubler, tripler and quadrupler. Theoretically there is no limit to the amount of voltage multiplication that can be generated using this circuit. As the current supplying capacity falls drastically in the higher order voltage multiplier, voltage triplers and quadruplers are practically used. In case of higher order voltage multiplier to increase load current, capacitors with very high capacitance are necessary.

During the first half-cycle, as D_1 is forward biased C_1 charges to V_m via D_1 . During the negative half-cycle D_1 is OFF and D_2 conducts. Thus C_2 is charged to $2V_m$ through D_2 , i.e., to the sum of voltage across C_1 and the peak supply voltage.

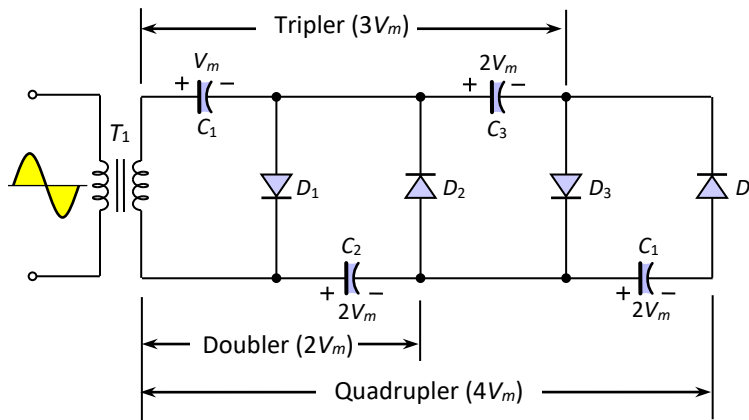


Fig.3-60: Circuit diagram of a voltage doubler, tripler and quadrupler

During the 2nd positive half-cycle, voltage across D_1 is zero ($+V_m - V_m$), D_2 and D_4 are reverse biased by $2V_m$ but only D_3 is forward biased. Thus, capacitor C_3 is charged by the voltage $2V_m$ ($V_m - V_m + 2V_m$). Again, during the 2nd negative half-cycle, if we observe the circuit carefully, we find that all the diodes are reverse biased except D_4 . Thus during this half-cycle capacitor C_4 is charged by $2V_m$ ($V_m + V_m - 2V_m + 2V_m$). Now charging of each capacitor has been completed. The charging process will be repeated for successive half-cycles. During these repeated half-cycles, current will flow through the capacitors and charges them up, if their voltage levels fall by discharging through the load.

If we connect the load across the bottom capacitor C_2 , the output voltage will be $2V_m$. The result is same as a half-wave voltage doubler. If the load is connected across the two upper capacitors, the output voltage will be $3V_m$. That is, the circuit will behave as a voltage tripler. Again, if the load is connected across the two bottom capacitors, the output voltage level will be $4V_m$. That is, the circuit will work as a voltage quadrupler.

The PIV ratings of the diode should be $2V_m$ and the ripple frequency will be twice the frequency of the AC supply voltage.

3.13 Clipper Circuits

Clippers (also called **limiters**) are networks that clip off a portion of an input signal without distorting the remaining part of the input signal (input waveform). The half-wave rectifier discussed in Section 3.7.1, is the example of a simple clipper circuit, because it passes only the positive half-cycle of an alternating waveform and clips off the other half-cycle. Clipper circuits are made using diodes and resistors. Depending on the connection of the diode (polarity), the positive or negative portion of the applied signal is clipped off. That's why clippers are divided into two categories- **positive clippers** and **negative clippers**. Positive clippers remove the positive portion of the applied signal and pass the negative portion. On the other hand, the negative clipper clips the negative region of the applied waveform and keeps the remaining region as original one. Although, an unbiased clipper clips a complete half-cycle, by using suitable bias for the diode less than or more than one half-cycle can be clipped off.

Depending on the position of the diode, clippers are also called either **series clippers** or **parallel clippers**. In series clippers, the diodes (with or without bias voltage) are placed in series with the load resistance, but in parallel clippers, the diode is connected in parallel with the load. In this text, some clipper circuits are described in details considering ideal diode characteristics.

3.13.1 Simple Series Clipper Circuits

These are the unbiased clipper circuits. That is, the diodes are used (without any DC bias voltage) in series with the load resistance. Fig.3-59(a) shows a simple positive series clipper circuit while Fig.3-59(b) is a simple negative series clipper circuit. In the first figure, for positive half-cycle the diode

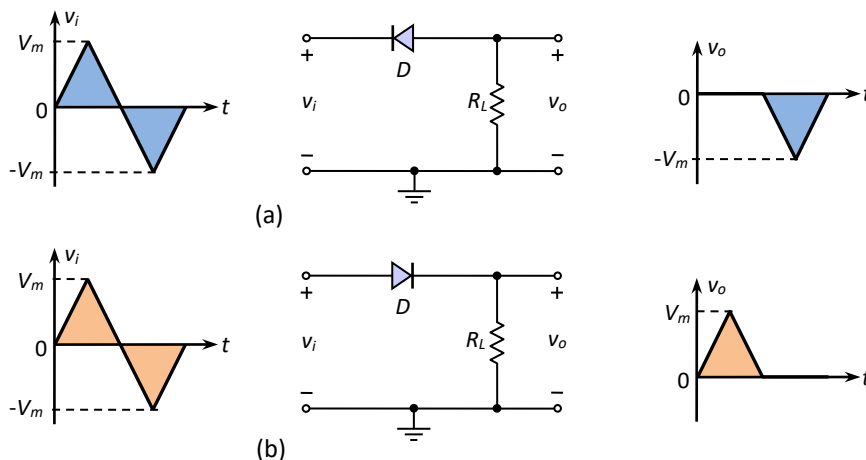


Fig.3-61: (a) Simple series positive clipper circuit, (b) Simple series negative clipper circuit

does not conduct any current through the load, so the output is 0 volt. For the negative half-cycle the diode is forward biased and passes current through the load. Thus, the whole negative half-cycle is developed across the load. As the output has lost the positive half-cycle, it is called positive clipper. Exactly the reverse thing will happen for the circuit of Fig.3-61(b). Here, the output waveform will have no negative half-cycle. That's why it is called negative clipper. For sinusoidal input, the operation has been described in rectifier section. Here, the operation has been shown for triangular waveforms.

3.13.2 Biased Series Clipper Circuit

In these clipper circuits a DC voltage V , which is less than V_m , is used to forward bias or reverse bias the diode. Depending on the bias voltage polarity and the diode direction, total four combinations are possible. Here, two of them are shown in Fig.3-62, and described.

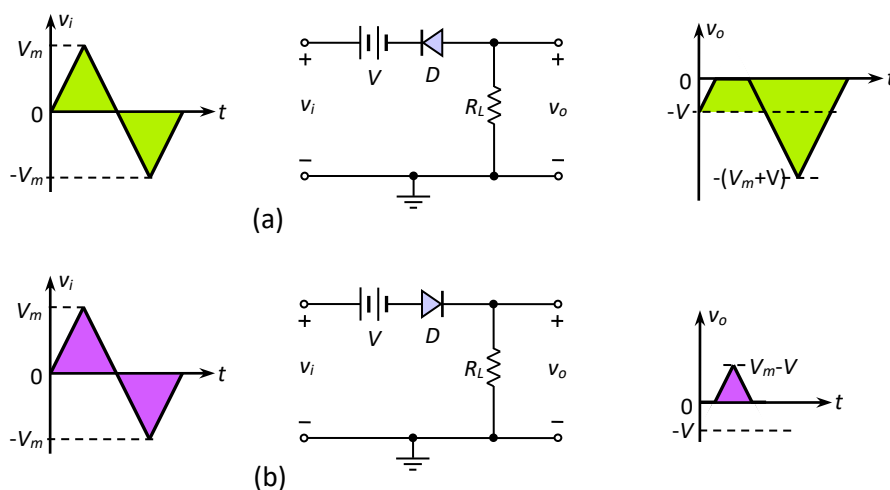


Fig.3-62: (a) Biased series positive clipper circuit, (b) Biased series negative clipper circuit

Description of the first circuit

It is a **biased series negative clipper circuit**. Here, we will consider $0 < V < V_m$ and the operation will be described for an ideal diode. Look at the polarity of the bias voltage (V). It makes the diode forward biased. Hence, even for no applied voltage, that is, if the input terminals are short, the diode is still be forward biased and $-V$ amount of voltage will be available at the output. That's why the output waveform starts from $-V$ volt. Now, if the positive half-cycle of v_i appears, it will gradually decrease the forward bias voltage. It will be clear if we look at the polarity of the positive half-cycle and the polarity of the bias voltage- they oppose each other. For this reason, the output negative voltage will gradually decrease. When the value of v_i is equal to the bias voltage V , the resultant voltage across the diode will be zero, and hence, the diode will not conduct any current through the load. Thus, the output voltage will be zero as shown in v_o of Fig.3-62(a). If the value of v_i exceeds V , the diode will be reverse biased by $(v_i - V)$ volts, and no output again. Thus, the output will be zero as long as $v_i > V$. During the falling edge of the positive half-cycle, when $v_i = V$ the diode starts to conduct again, and we will get negative output voltage. During the negative half-cycle of the input, v_i will be added with V and will bias the diode in forward direction. As we have considered an ideal diode, the total voltage $(-v_i - V)$ will appear across the load. Thus, negative output will be available from $v_i = V$ up to $v_i = -V_m$. For this reason the negative peak will be $-V_m - V = -(V_m + V)$, which is shown in output waveform of Fig.3-62(a).

Description of the second circuit

In this circuit, the polarity of the diode is reversed, and it is reverse biased by V . The positive half-cycle will forward bias the diode. But as the diode is initially reverse biased by V volts, it will remain reverse biased until $v_i = V$. Thus, even for the positive half-cycle output will not be found up to $v_i = V$. When v_i exceeds V , the diode will be forward biased and current will flow through the load

and output voltage will be found. For the negative half-cycle, v_i will be added with V , and the diode remains completely reverse biased. So no output is found. In summary, we can say that the output will be found only for $V_i - V$ portion of the positive half-cycle as shown in Fig.3-62(b).

Simple Parallel Clipper Circuits

These clipper circuits are made by connecting the diode in parallel with the load resistor (across the output terminals) without any bias voltage. Fig.3-63(a) illustrates a simple positive parallel clipper circuit and Fig.3-63(b) illustrates a simple negative parallel clipper circuit, respectively. In the first figure, for positive half-cycle the diode will be forward biased and behaves as short circuit. Thus, no output voltage will be found across the load resistance. On the other hand, for the negative half-cycle the diode is reverse biased and behaves as an open circuit. Now, the negative voltage of the source will be found at the output terminals. In this way, the circuit clips off the entire positive half-cycle of the input signal, and hence it is called positive parallel clipper. Exactly the reverse thing will happen for the circuit of Fig.3-63(b). The diode will be reverse biased for the positive half-cycle. So, positive half-cycle will be developed at the output. For the negative half-cycle of v_i , the diode will be forward biased and will clip off the entire negative half-cycle. That is why, this circuit is a parallel negative clipper.

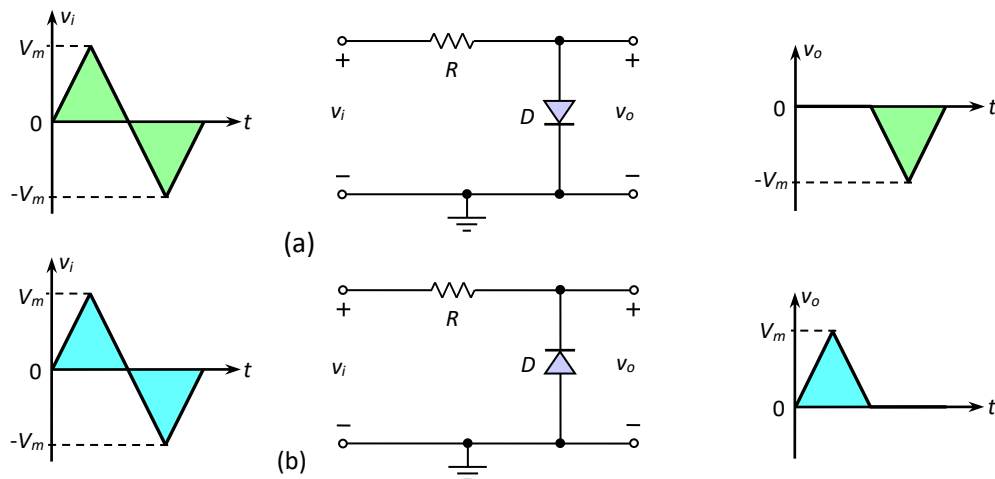


Fig.3-63: (a) Simple parallel positive clipper circuit, (b) Simple parallel negative clipper circuit

Biased Parallel Clipper Circuit

These clippers also use the diodes in parallel with the load resistor, but the diode is either forward biased or reverse biased by a voltage V which is less than the peak value of the input signal (V_m). There are five types of connection arrangements for biased parallel clippers. Only two of them are shown in Fig.3-64 and described here.

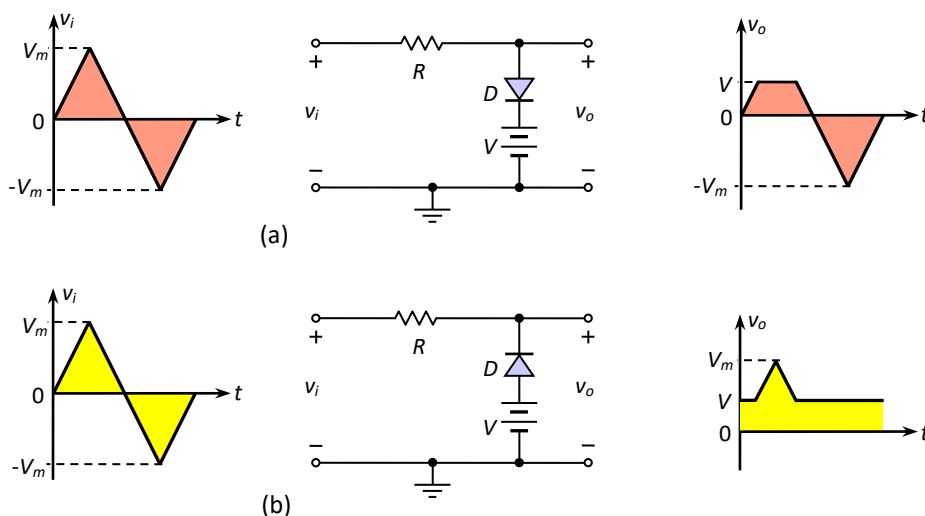


Fig.3-64: (a) Biased parallel positive clipper circuit, (b) Biased parallel negative clipper circuit

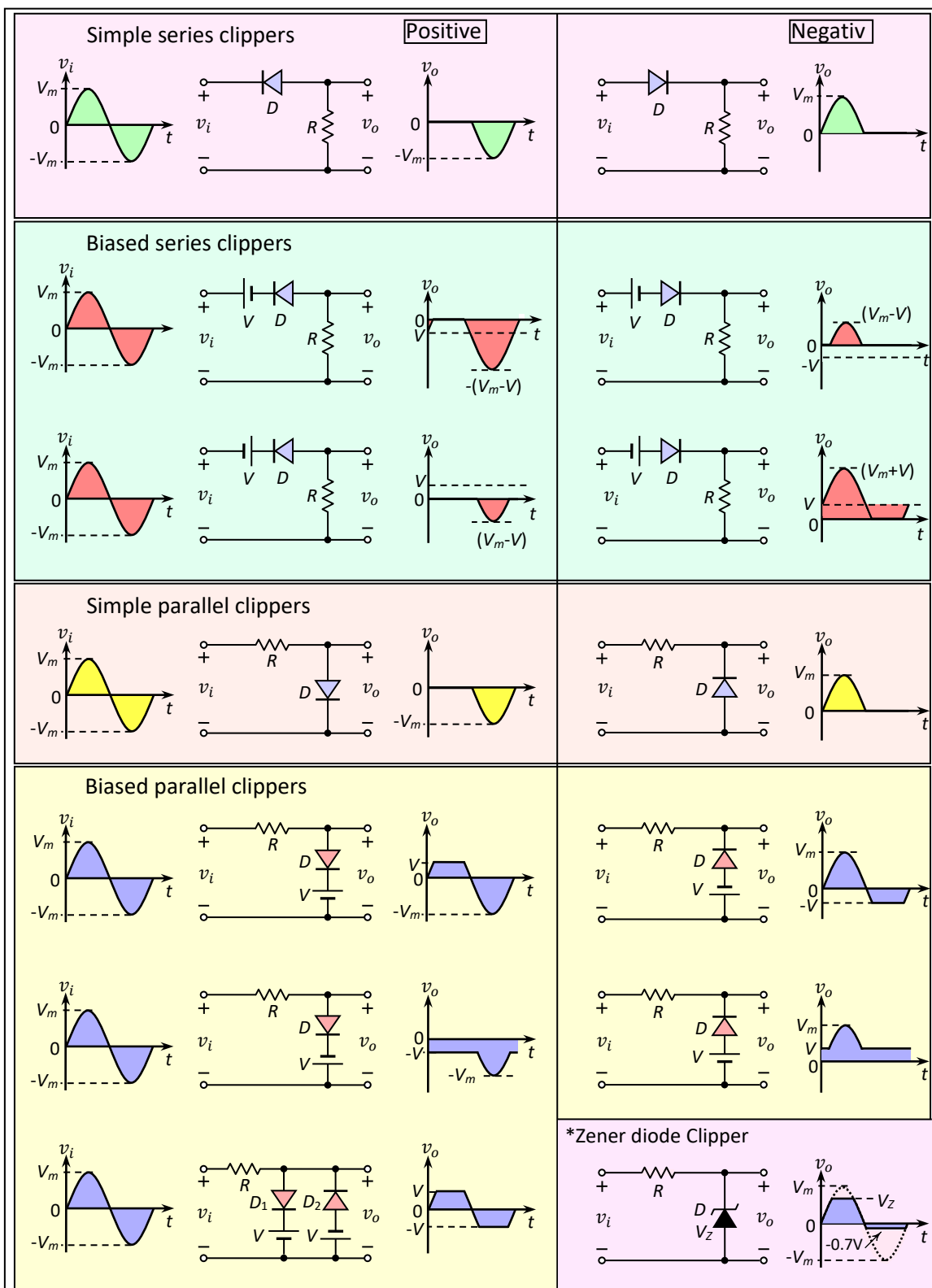
Description of the first circuit

In this circuit, the diode will be forward biased for the positive half-cycle of the input. But the diode has been made reverse biased initially by V volts. Thus, even for the positive half-cycle the diode will remain reverse biased up to $v_i = V$ volts. Therefore, output voltage will be found up to V volts of the positive half-cycle. When the input voltage equals V , the diode will be forward biased and will behave as a short circuit. For this portion as the diode is short, V amount of voltage will be found at the output. On the other hand, for the negative half-cycle, the diode is completely reverse biased. So the whole negative half-cycle will be found at the output. This clipper will clip off only $V_i - V$ portion of the positive half-cycle and rest of the portion of the input signal will remain unchanged.

Operation of the second circuit

For the zero input voltage, that is, if the input terminals are short, the diode will be forward biased by the DC voltage V . Now, the biasing voltage (V) will appear at the output terminals. When $v_i > V$ (for positive half-cycle) the diode will be reversed biased and will behave as open circuit. No current will flow through the diode and the voltage drop across the resistor will be zero (if no load is connected at the output). Hence, the input voltage will appear at the output terminal. Thus, for $V_i = V$, the output will be V volts and for $V_i > V$ the output will follow the input as shown in the Fig.3-64(b). For the negative half-cycle, the diode will be forward biased (and short) and current will flow through the diode and all of the input voltage will be dropped across the resistor. But as the diode is forward biased and short the DC voltage, V , will be found across the output.

All of the clipper circuits described above with some other extra combinations are summarized in the following figure (Fig.3-65).

Fig.3-65: Clipper circuits: +ve clippers in 1st column and -ve clippers in 2nd column

*In the Zener diode clipper circuit shown in Fig.3-65 the barrier voltage (0.7 V) in forward bias is considered. In all other circuits an ideal diode has been considered.

Example 3-23

Fig.3-66 shows a circuit combining a positive clipper and a negative clipper. Determine the output voltage waveform considering Si diodes.

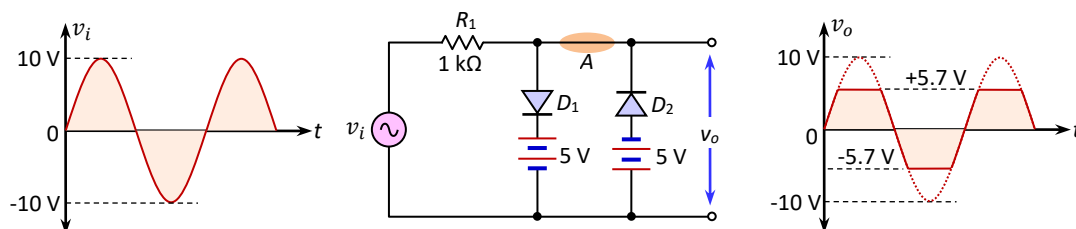


Fig.3-66: Combined positive clipper and negative clipper circuit for Example 3-23

Solution:

The peak value of v_i is 10 V, and the biasing voltage of the diodes is 5 V. For the Si diodes we will consider $V_D = 0.7$ V. Look at the circuit configuration; it combines two parallel clipper circuits. For parallel clippers, the input signal will appear at the output if the diodes are reverse biased. Here, the diodes are initially reverse biased by 5 V. Therefore, for the +ve half-cycle, the diodes will be reverse biased until $v_i = 5.7$ V [$V + V_D$]. So the +ve half-cycle will be found at the output until $v_i = 5.7$ V. When $v_i > 0.7$ V, D_1 will be short and D_2 will be open and the output voltage will be $v_o = 5.7$ V.

Similarly, during the -ve half-cycle, the diodes will not conduct until $v_i = -5.7$ V. Therefore, the portion of the negative half-cycle upto -5.7 V will be found at the output. After that ($v_i < -5.7$ V) the diode D_2 will be short and D_1 will be open, but the output will be -5.7 V.

Therefore, positive voltage above +5.7 V and negative voltage below -5.7 V are clipped off. The resulting output voltage waveform is shown in Fig.3-66. In this graph, the clipped off portion of the input signal is shown by dashed line.

Comments: There is little non-linearity in the rising and falling edges of the output waveform. If this nonlinearity is neglected, we can use this output as trapezoidal wave.

Example 3-144

Draw and describe the output waveform for the sinusoidal signal voltage applied to the given circuit of Fig.3-67.

Solution:

The given circuit is a biased positive parallel clipper. Here, the bias voltage is generated by using a 12 V source and a voltage divider network (R_2 - R_3). The level of voltage applied to the cathode of the diode can be calculated using voltage divider rule as,

$$V_{Bias} = \frac{(12 \text{ V}) \times (200 \Omega)}{100 \Omega + 200 \Omega} = 8 \text{ V}$$

This voltage reverse biases the diode. So the diode will not conduct until the supply voltage reaches to $8\text{ V} + 0.7\text{ V} = 8.7\text{ V}$. When the diode conducts, the output voltage will remain constant at 8.7 V and the rest of the input voltage will be dropped across the resistor, R_1 . For remaining portion of the input, the diode is reverse biased, and hence, the input signal is found at the output. The waveform is shown in Fig.3-67.

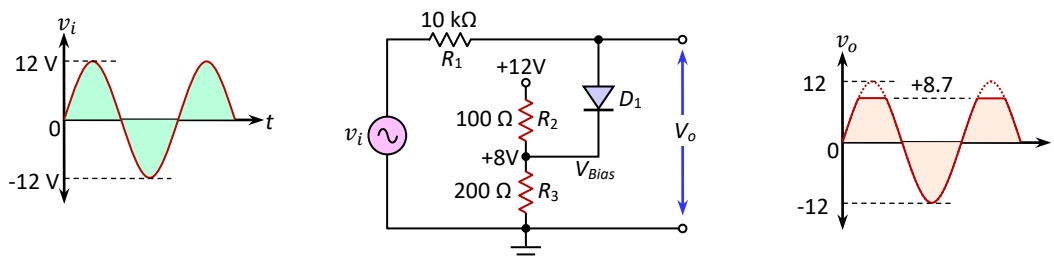


Fig.3-67: (a) Biased parallel positive clipper circuit, (b) Biased parallel negative clipper circuit

Comments: As R_1 is very large compared to R_2 and R_3 , a very small current will flow through the diode when it is ON. For this reason, the Thevenin's resistance of the voltage divider has not been considered. A problem with value of R_1 comparable with R_2 and R_3 is given in Exercise no 44.

Example 3-155

Determine the output waveform of the clipper circuit of Fig.3-66 for sinusoidal input. Assume the barrier voltage of the Si diode is 0.7 V .

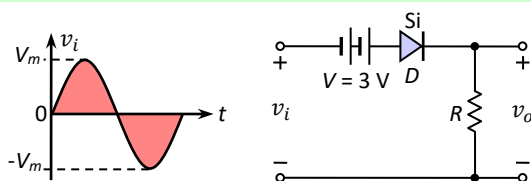


Fig.3-68: Clipper circuit and input waveform for Example 3-25

Solution:

The circuit is called biased series clipper. The output is directly across the resistor R . When the diode is forward biased we will get output, and when the diode is reversed bias the output will be

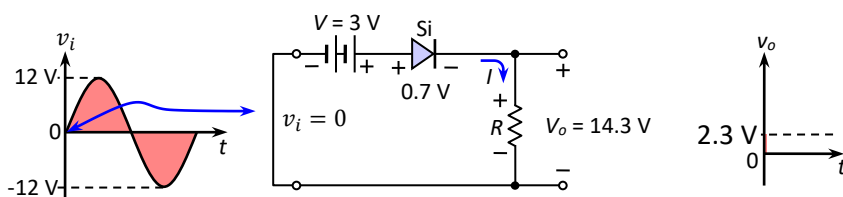


Fig.3-69(a): Circuit state when $v_i = 0$

zero. As the diode is forward biased by $V = 3\text{ V}$, for $v_i = 0$ [as illustrated in Fig.3-68(a)], the output voltage will be:

$$v_o = V - V_D = 3 \text{ V} - 0.7 \text{ V} = 2.3 \text{ V}$$

For the entire positive half-cycle the diode will be forward biased and for $v_i = V_m = 12 \text{ V}$ [as illustrated in Fig.3-67(b)], the output will be,

$$v_o = (V_m + V) - V_D = (12 \text{ V} + 3 \text{ V}) - 0.7 \text{ V} = 14.3 \text{ V}$$

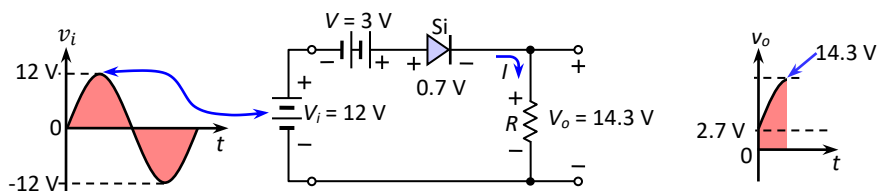


Fig.3-69(b): Circuit state when $v_i = 12 \text{ V}$

The diode will be reverse biased, when the total voltage applied across the diode is less than 0.7 V . When the input voltage becomes -2.3 V , the diode is barely forward biased by 0.7 V , but the output voltage will be 0 . The situation is illustrated in Fig.3-69(c).

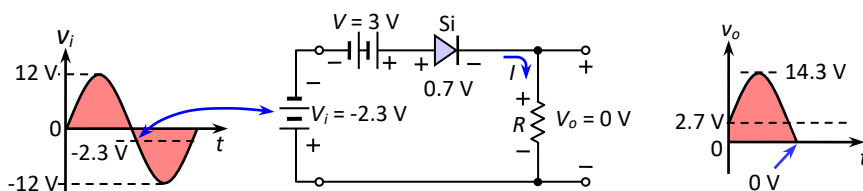


Fig.3-69(c): Circuit state when $v_i = -2.3 \text{ V}$

For $v_i < -2.3 \text{ V}$, the diode will be reverse biased and the output will be zero, and all of the applied voltage will be dropped across the diode as shown in Fig.3-69(d). The overall result is the input signal shifted to positive direction by 2.7 V , and the portion of the input signal below -2.3 V is clipped off.

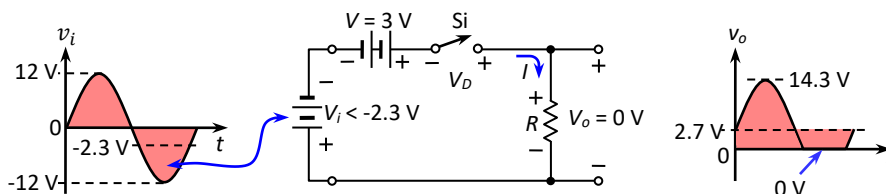


Fig.3-69(d): Circuit state when $v_i < -2.3 \text{ V}$

Comments: This circuit is similar to the 2nd circuit of negative biased series clippers in Fig.3-65, but here diode barrier voltage has been considered.

3.13.3 Clipper Circuits using Zener Diodes

Clipper circuits can be designed using Zener diodes as well. If the current capacity of the source is greater than the knee current I_{ZK} , a Zener diode can be a substitute of a diode-with-biasing voltage. Fig.3-70(a) shows a parallel clipper circuit consisting of two back to back (series) connected

Zener diodes. For the entire positive half-cycle, D_1 is forward biased and D_2 is reverse biased. So, the voltage across D_1 diode will be forward voltage drop, V_{F1} (For Si Zener diode about 0.7 V) and rest of the input voltage will be dropped across D_2 . But, when $v_i = V_{Z2} + 0.7$ V, Zener diode D_2 will breakdown and will behave as a constant voltage source of voltage level V_{Z2} .

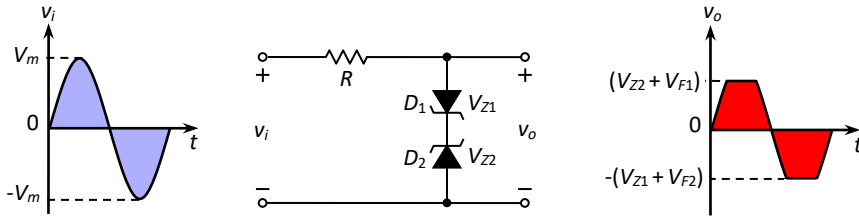


Fig.3-70: (a) Zener diode parallel clipper circuit

Thus, the output voltage will be $(V_{F1} + V_{Z2})$. The situation is illustrated in Fig.3-70(b).

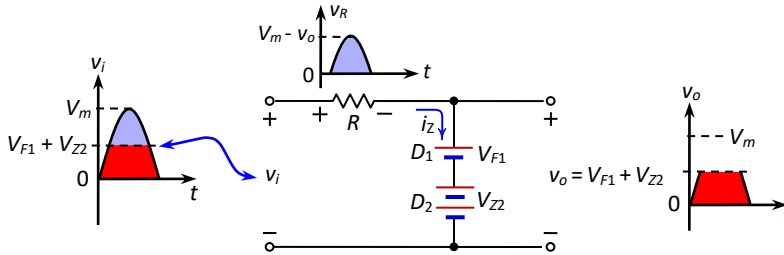


Fig.3-70: (b) Equivalent circuit when $v_i = V_{F1} + V_{Z2}$

Similarly, during the negative half-cycle, D_2 is forward biased and D_1 is reverse biased. For this time the output will be $-(V_{F2} + V_{Z1})$. With Zener diodes of equal voltages, the maximum output voltage swing will be,

$$v_o = \pm(V_F + V_Z) \quad (3-86)$$

The resistor voltage drop is $(v_i - v_o)$, and its maximum value will be $V_m - (V_F + V_Z)$. If a load resistance (R_L) is connected at the output, it will draw load current (i_L). Then the resistor current is $i_{R1} = (i_L + i_Z)$. To ensure breakdown, the minimum level of Zener current should be I_{ZK} . The maximum load current will be $I_{L(max)} = (V_F + V_Z)/R_L$, and then, we can set minimum value of i_Z . Therefore, the maximum value of R_1 should be,

$$R_{1(max)} = \frac{V_m - (V_F + V_Z)}{I_{ZK} + I_{L(max)}} \quad (3-87)$$

If the value of R_1 is set more than $R_{1(max)}$, the circuit will not work, as i_Z will fall below I_{ZK} .

On the other hand, the minimum value of this resistor should be,

$$R_{1(min)} = \frac{V_m - (V_F + V_Z)}{I_{ZM} + I_{L(max)}} \quad (3-88)$$

Here, I_{ZM} is the maximum current rating of the Zener diode. If we set $R_1 < R_{1(min)}$, the Zener diode will draw more current than its maximum rating I_{ZM} and it may be burnt out. We should remember that here, $R_{1(min)}$ has been calculated assuming that the load is always connected at the output. But, if the load is not connected, the minimum value of R_1 should be,

$$R_{1(min)} = \frac{V_m - (V_F + V_Z)}{I_{ZM}} \quad (3-89)$$

Because, in that case, $i_{R1} = i_Z$, and i_Z should not be greater than I_{ZM} .

Example 3-166

Analyze the clipper circuit of Fig.3-71(a) and determine the value of R_1 , different values of voltages and currents. Assume: $V_{Z1} = V_{Z2} = 4.3$ V, $I_{ZK} = 2$ mA, $P_Z = 1/4$ W, and Zener diode forward voltage drop is 0.7 V.

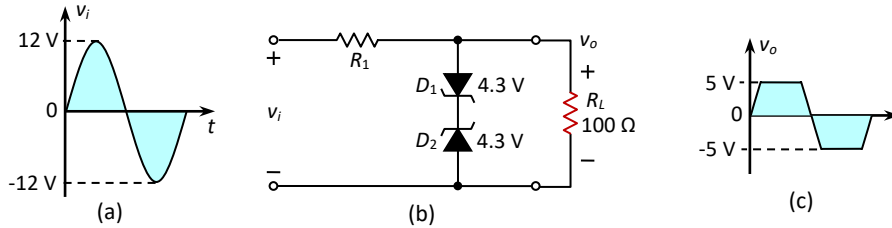


Fig.3-71: (a) Input waveform, (b) Circuit diagram of clipper, and (c) Output waveform

Solution:

The maximum current rating of the Zener diode can be calculated as,

$$I_{ZM} = \frac{P_Z}{V_Z} = \frac{1/4 \text{ W}}{4.3 \text{ V}} \approx 58 \text{ mA}$$

The maximum voltage across the load will be $V_{L(max)} = V_F + V_Z = 4.3 \text{ V} + 0.7 \text{ V} = 5 \text{ V}$

So, the maximum load current will be, $I_{L(max)} = V_{L(max)} / R_L = 5 \text{ V} / 100 \Omega = 50 \text{ mA}$

The maximum voltage drop across R_1 will be $V_m - (V_F + V_Z) = 12 \text{ V} - (0.7 \text{ V} + 4.3 \text{ V}) = 7 \text{ V}$.

Using Equ.(3-87),
$$R_{1(max)} = \frac{V_m - (V_F + V_Z)}{I_{ZK} + I_{L(max)}} = \frac{7 \text{ V}}{2 \text{ mA} + 50 \text{ mA}} \approx 135 \Omega$$

Using Equ.(3-88),
$$R_{1(min)} = \frac{V_m - (V_F + V_Z)}{I_{ZM} + I_{L(max)}} = \frac{7 \text{ V}}{58 \text{ mA} + 50 \text{ mA}} \approx 65 \Omega$$

If the load is not connected, then

Using Equ.(3-89),
$$R_{1(min)} = \frac{V_m - (V_F + V_Z)}{I_{ZM}} = \frac{7 \text{ V}}{58 \text{ mA}} \approx 120 \Omega$$

If we use $R_1 = 12 \Omega$, 2 mA will flow through Zener diode and 50 mA will flow through the load resistor, when $v_i = 12 \text{ V}$. For $v_i < 5 \text{ V}$, one Zener diode will be open, so the output will follow the input. If we use $R_1 = 65 \Omega$, 58 mA will flow through Zener diode and 50 mA will flow through the

load resistor, when $v_i = 12\text{ V}$. In this case, if the load is disconnected, 108 mA will flow through the Zener diode and it may burn out. On the other hand, if we use $R_1 = 120\ \Omega$, 50 mA will flow through the load and the remaining 8 mA will flow through the Zener diode.

Comments: This type of circuit is suitable for lower output current. For higher output current, the design becomes complicated, as we have to consider the power of the Zener diode and the series resistor, R_1 .

3.14 Practical Applications of Clipper Circuits

In communications, when signals reach from one system to another system through wired or wireless medium, it may pick up different types of noises. Using clipper circuits, we can remove these noises. For example, if a signal contains very low level noise (less than 0.7 V) at the ground level, we can remove it using a series clipper circuit as shown in Fig.3-72(a). Here, the upper diode (D_1) passes the positive half-cycle of the input but due to the barrier voltage of the diode 0 to 0.7 V of the signal is blocked. Similarly the bottom diode (D_2) passes the negative half-cycle by clipping 0 to -0.7 V portion of this cycle. In this way, a noise-less signal is found at the output.

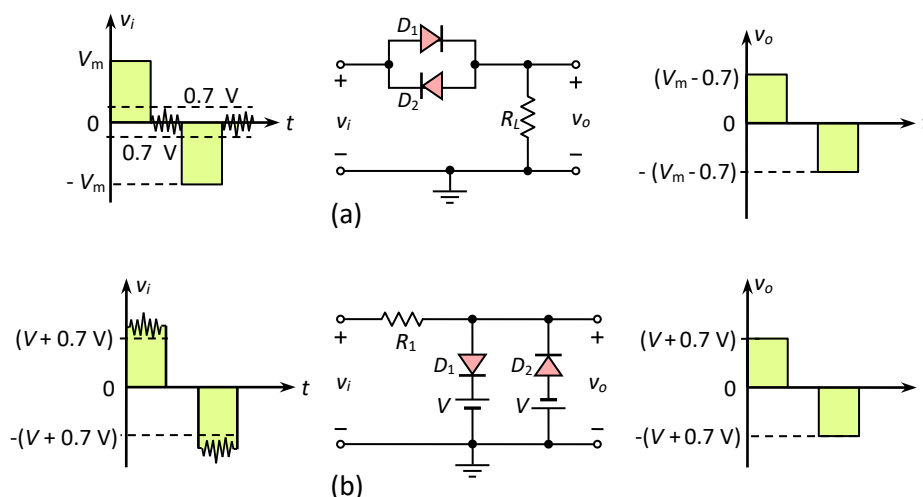


Fig.3-72: Noise elimination using diode clipper circuits

On the other hand, if a signal contains noise at the positive and negative peak levels, this noise can be eliminated using biased parallel clipper as shown in Fig.3-72(b). Diode D_1 , will remove the noise from the positive half-cycle and diode D_2 will remove the noise from the negative half-cycle. As the diodes are reverse biased by V volt, they will be forward biased when the input voltage is higher than $\pm(V + 0.7)\text{ V}$. Here, we have considered Si diodes with forward voltage drop of 0.7 V.

3.15 Clamper Circuits

A **clamper** is a circuit constructed of a diode, a resistor, and a capacitor that shifts a signal to a different DC level without changing the shape of the input signal waveform. That is, a clamper

circuit adds a DC level with the input signal. Depending on the direction of shifting, clamper circuits are classified as – **positive clampers** and **negative clampers**. Both of them may be either biased or unbiased. Thus, there are four types of clampers:

1. Simple negative clamper
2. Simple positive clamper
3. Biased negative clamper
4. Biased positive clamper

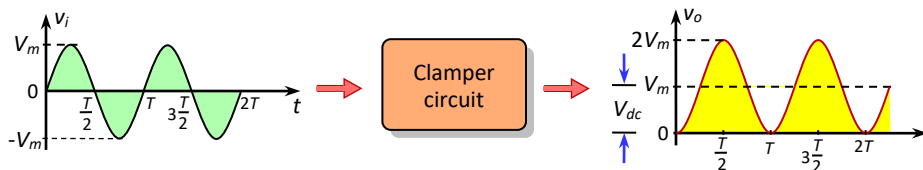


Fig.3-73: Clamper circuit has shifted the input signal to a new level $V_{dc} = V_m$.

The clamper circuit works by charging the capacitor in one cycle and discharging it in another cycle. To get better output waveform, the charging time constant ($\tau_c = r_s C$; r_s is the source resistance) should be very small, and the discharging time constant ($\tau_d = R_L C$; R_L is the load resistance) should be very large compared to the time period (T) of the applied signal. As the internal resistance of a source (r_s) is very small, the first requirement will automatically be fulfilled. By using very large value of R_L , we can ensure the second condition. Here, we will describe the operation considering an ideal diode and $\tau_d = R_L C \gg T/2$.

3.15.1 Simple negative clamper

The circuit diagram of a **simple negative clamper** is shown in Fig.3-74. Here, simple means unbiased, that is, the diode is not biased initially. Operation procedure of unbiased clamper circuits is very simple. To understand the operation of any type of clamper circuits, first we have to consider the half-cycle that forward biases the diode, and then the

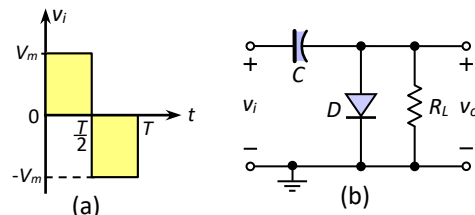


Fig.3-74: Negative clamper circuit (unbiased): (a) Input waveform, and (b) Circuit diagram

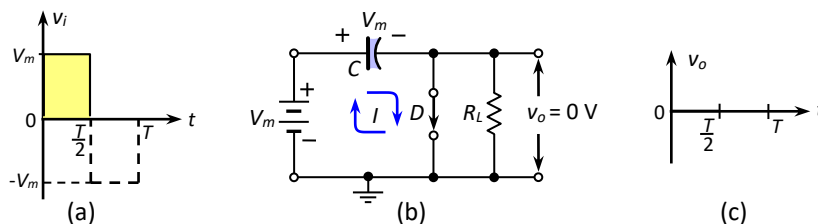


Fig.3-75: Operation for positive half-cycle: (a) Positive half-cycle of input, (b) Circuit state for positive input, and (c) Output waveform

other half-cycles. Here, for the positive half-cycle of the input signal ($0 \rightarrow T/2$), the diode will be forward biased and the capacitor will be charged to the maximum value of input, V_m , with +ve on the left

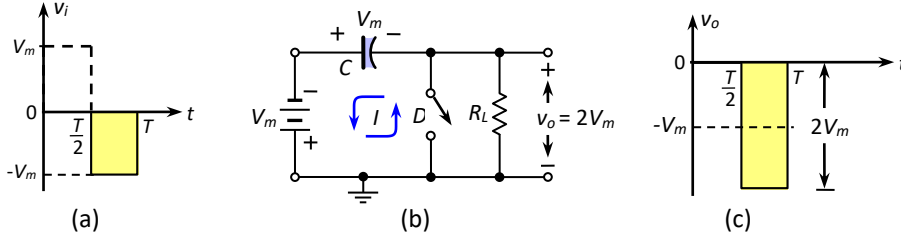


Fig.3-76: Operation for negative half-cycle: (a) Negative half-cycle of input, (b) Circuit state for negative input, and (c) Output waveform

side as shown in Fig.3-75(b). As the diode is short during this positive half-cycle, the voltage across the load resistance will be zero ($V_D = 0.7 \text{ V}$ is neglected). However, during the negative half-cycle ($T/2 \rightarrow T$), the diode is reverse biased and will behave as an open circuit as shown in Fig.3-76(b). The amount of output voltage can be determined using KVL as (moving anticlockwise starting from ground)

$$v_o + V_m + V_m = 0$$

$\therefore$

$$v_o = -2V_m$$

(3-90)

That is, the output voltage will be the sum of the capacitor voltage and source voltage with positive at the bottom. Thus, we will get $v_o = -2V_m$. Observe that, the input signal has been shifted to negative direction by a value of V_m . Or we can say, a DC value of $-V_m$ has been added to the output waveform.

3.15.2 Simple positive clamper

The circuit diagram of a simple positive clamper is shown in Fig.3-77. Here, the output signal has a DC value of V_m , or we can say that the output waveform has been shifted to positive direction. Its operation procedure is similar to that of simple negative clamper and is left for the students.

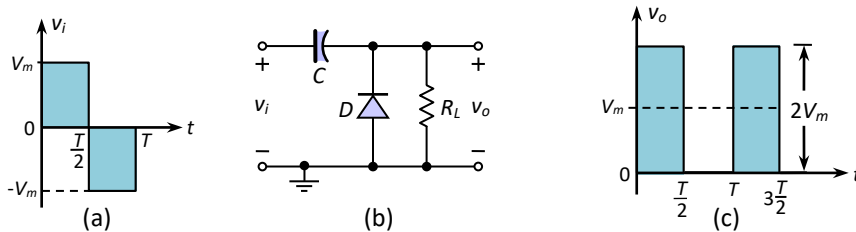


Fig.3-77: Circuit diagram of a positive clamper: (a) Input waveform, (b) Circuit diagram, and (c) Output waveform.

3.15.3 Biased negative clamper

Fig.3-78(b) shows a biased negative clamper circuit. This circuit is almost same as the simple negative clamper. Here, only the difference is- the diode is initially reverse biased by a DC

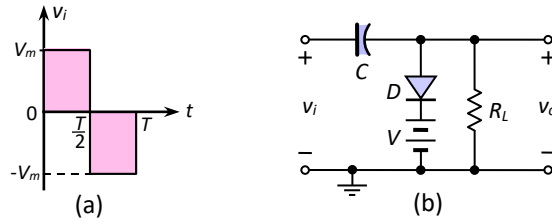


Fig.3-78: Biased negative clamper circuit: (a) Input waveform, and (b) Circuit diagram

voltage of value V . Here, $V < V_m$, and V reverse biases the diode. When $V_m > V$, during the positive half-cycle ($0 \rightarrow T/2$) the diode is forward biased and acts as a short circuit. So, V amount of voltage will be available at the output as shown in Fig.3-79(c). For this time, current will flow clockwise and the capacitor will be charged to $(V_m - V)$ volts.

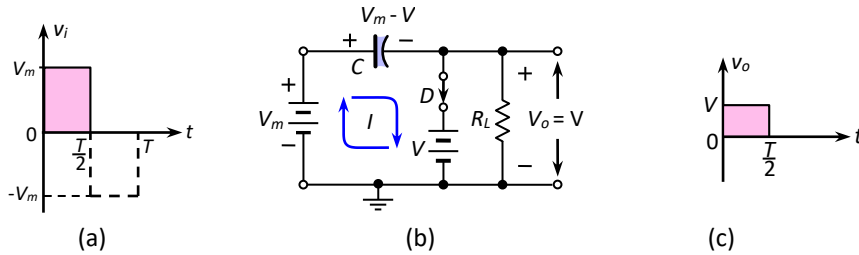


Fig.3-79: Operation for positive half-cycle: (a) Input waveform, (b) Circuit diagram, and (c) Output waveform.

During the negative half-cycle ($T/2 \rightarrow T$), the diode is reverse biased and acts as open circuit as shown in Fig.3-80. The value of the output voltage can be found using KVL as (starting from ground),

$$-V_m - (V_m - V) - v_o = 0$$

$\therefore$

$$\boxed{V_o = -(2V_m - V)} \quad (3-91)$$

That is, the output voltage will be the sum of the capacitor voltage and source voltage, therefore, $v_o = -(2V_m - V)$.

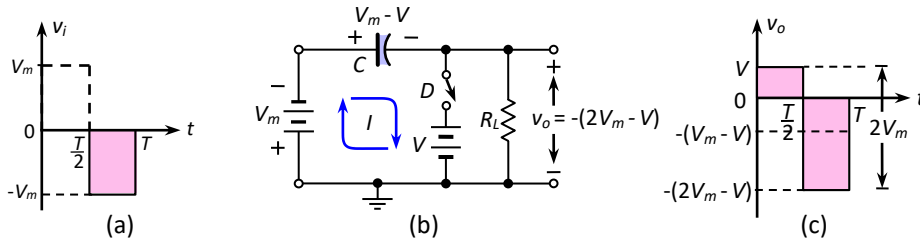


Fig.3-80: Operation for negative half-cycle: (a) Input waveform, (b) Circuit diagram, and (c) Output waveform

Observe Fig.3-80(c), the input signal has been shifted to negative direction by a value of $(V_m - V)$. Or we can say a DC value of $-(V_m - V)$ has been added to the output waveform.

3.15.4 Biased positive clamper

The circuit diagram of a biased positive clamper is shown in Fig.3-81. Here, the diode is initially forward biased by a DC voltage of value V . The operation of the biased negative clamper is described elaborately in Section 3.15.4. But now, the biased positive clamper circuit is described briefly. During the negative half-cycle the diode will be forward biased and work as short circuit. So V amount of voltage will be available at the output. For this time, current will flow anticlockwise and the capacitor will be charged to $(V_m + V)$ volts. During the positive half-cycle, the diode will be reverse biased and will act as open circuit. The value of the output voltage can be found using KVL as

$$-V_m - (V_m + V) + v_o = 0$$

$$\therefore v_o = (2V_m + V) \quad (3-92)$$

That is, the output voltage will be the sum of the capacitor voltage and source voltage with negative at the bottom. So, the value of output voltage we will be $v_o = (2V_m + V)$. Observe that, the input signal has been shifted to positive direction by a value of $(V_m + V)$. Or we can say a DC value of $(V_m + V)$ has been added to the output waveform. The state of the circuit for positive half-cycle and for negative half-cycle can easily be drawn as before. So, it is left for the students.

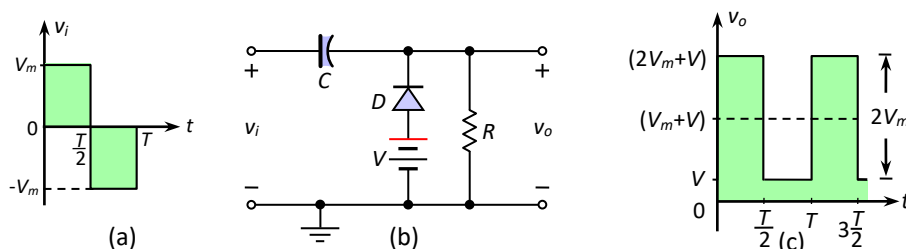


Fig.3-81: Biased positive clamper: (a) Input waveform, (b) Circuit diagram, and (c) Output waveform.

Up to this, we have discussed three types of clamping circuits. There are some other clamper circuits. All of the clamper circuits are summarized in Fig.3-91. From the operation of the clamper circuits, discussed so far, we can conclude the following points regarding their operation.

- The output waveform will be shifted in the same direction as the arrow-head of the diode.
- If there is no DC bias, the whole half-cycle (+ve or -ve) will be shifted for ideal diode.
- If the diode is forward biased by a DC voltage V , output will be shifted V volts more.
- If the diode is reverse biased by a DC voltage V , output will be shifted V volts less.
- For a practical Si diode, we have to consider 0.7 volts more in forward bias.

Example 3-17

Determine the output waveform of the network of Fig.3-82 for the given input waveform. Consider $V_D = 0.7$ V for a Si diode.

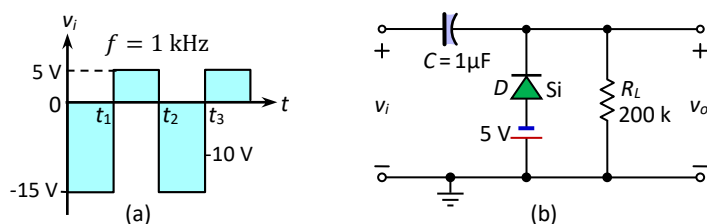


Fig.3-82: (a) Input waveform, and (b) Clamper circuit for Example 3-27

Solution:

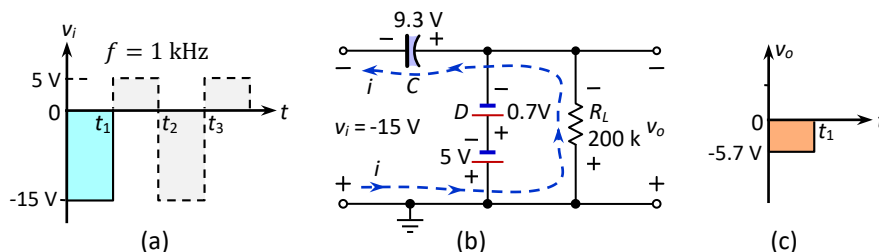
When the capacitor discharges through the load resistor (R_L), the effective time constant will be,

$$\tau_d = R_L C = 200 \text{ k}\Omega \times 1 \mu\text{F} = 0.2 \text{ s}$$

but the time period of the input signal is,

$$T = \frac{1}{f} = \frac{1}{1 \text{ kHz}} = 0.001 \text{ s}$$

That is, τ_d is very larger than T .

Fig.3-83: Circuit operation for 1st -ve half-cycle: (a) Input waveform, (b) Circuit diagram, and (c) Output waveform

During the negative half-cycle of input (0 to t_1), the diode will be forward biased and being a Si diode it will drop 0.7 V as shown in Fig.3-83. So the output voltage across the load (R_L) will be $v_o = -(5 \text{ V} + 0.7 \text{ V}) = -5.7 \text{ V}$. As $v_i = -15 \text{ V}$, the capacitor will be charged by 9.3 V ($15 - 5.7$) with +ve on right side. Now, the charging current will flow through the forward biased diode not R_L . So, the charging time constant, τ_c , will be very small (only the internal resistance of D and V will be effective) and the capacitor will be charged instantly.

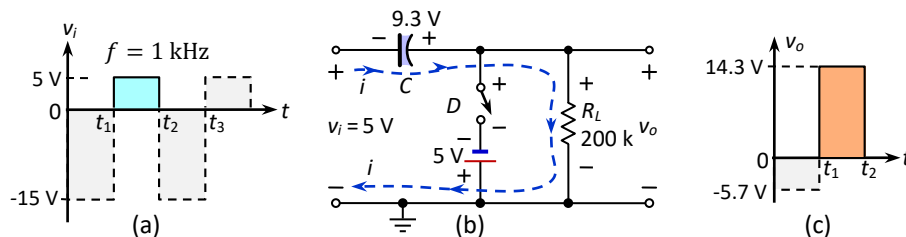


Fig.3-84: Circuit operation for +ve half-cycle: (a) Input waveform, (b) Circuit diagram, and (c) Output waveform

For the positive half-cycle the diode will be reverse biased (open), so the current will flow through the load (R_L). As illustrated in Fig.3-84(b), the output voltage will be the sum of input voltage and capacitor voltage. Therefore, $v_o = v_i + v_C = 5 \text{ V} + 9.3 \text{ V} = 14.3 \text{ V}$. During the reverse biased state of the diode, the capacitor will discharge through the load (R_L). As R_L is very large, τ_d will also be very large compared to T . So, the change in capacitor voltage (v_C) due to discharge will be very small and can be neglected.

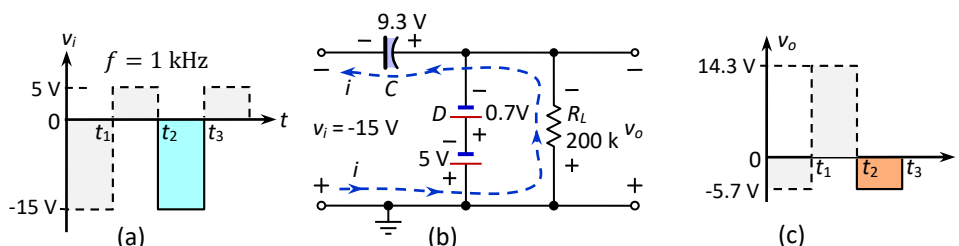


Fig.3-85: Circuit operation for the 2nd -ve half-cycle: (a) Input waveform, (b) Circuit diagram, and (c) Output waveform

Again for the negative half-cycle, the diode will be forward biased and the circuit operation is shown in Fig.3-84. This operation is exactly same as Fig.3-83. In this way, the operation will be repeated and the overall output waveform will be as shown in Fig.3-86.

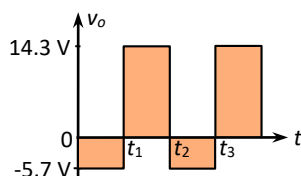


Fig.3-86: Complete output waveform of the given circuit

Comments: As the diode is reverse biased by 5 V and we have considered the 0.7 V drop of the diode, so the circuit will shift the input signal to $V_m - 5.7 \text{ V}$.

3.15.5 Clamper Circuit with Sinusoidal Input

So far we have discussed the operation of different clamper circuits with square wave input. Here, we will discuss the operation of a clamper circuit for sinusoidal input as shown in Fig.3-87. For proper functioning of a clamper circuit, the discharging time constant ($\tau_d = R_L C$) of the circuit must be much much greater than the time period (T) of the input signal. A reasonable consideration is $\tau_d > 10 T$.

Though we should start description, with the negative half-cycle (that forward biases the diode), here we will start from the positive half-cycle to get more detailed operation. During the positive half-cycle of the input, the diode will be reverse biased and will work as an open circuit. So current

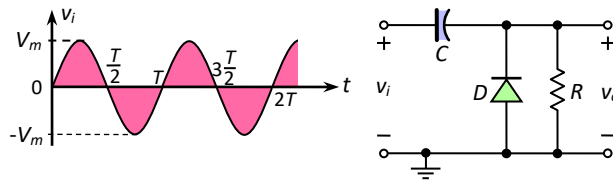
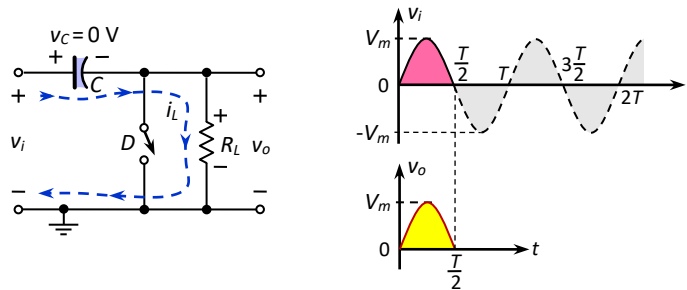
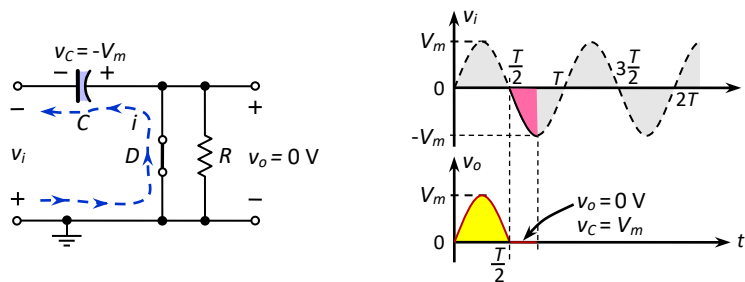


Fig.3-87: Circuit diagram of a positive clamper with sinusoidal input

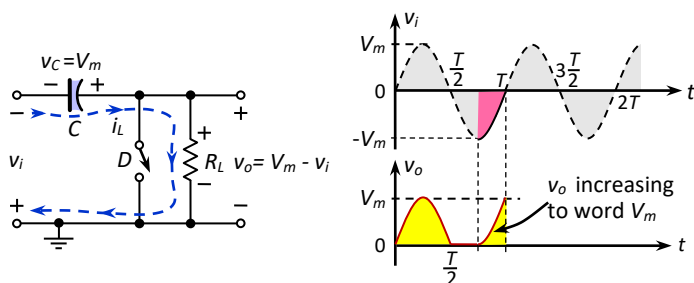
will flow through the capacitor and the load resistance and the capacitor will be charged up. But as the time constant ($\tau_d = R_L C$) is very larger than the time period of the input signal the charge accumulated in the capacitor will be very small and so does its voltage. Thus, the entire positive half-cycle will appear across the load, as shown in Fig.3-88(a).

Fig.3-88(a): Circuit operation for first positive half-cycle (0 to $T/2$)

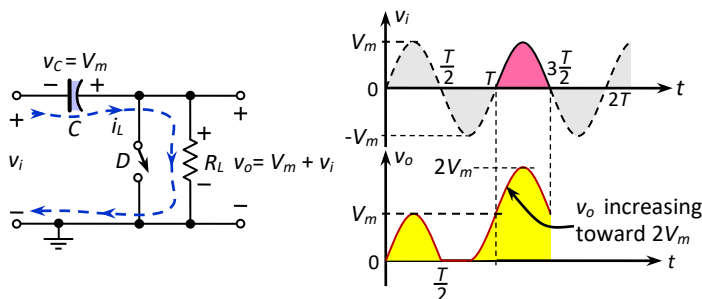
During the negative half-cycle, the diode will be forward biased and will work as a short circuit (neglecting V_D). So current will flow through the short path and the capacitor, but not through the load resistance. The voltage across the load (that is the output voltage) will be zero as long as the

Fig.3-88(b): Circuit operation for half of the first negative half-cycle ($T/2$ to $3T/4$)

diode behaves as short circuit [Fig.3-88(b)]. When the diode is short, the time constant is almost zero ($\tau_c = 0$) due to the short path ($R = 0$). Thus, the capacitor voltage will follow the input voltage and will be charged to $-V_m$, immediately when the input voltage falls to $v_i = -V_m$. After this peak point, the input (v_i) will gradually increase and the diode will be reverse biased. To understand this, look at the polarity of v_C and v_i of Fig. Fig.3-88(c). v_C reverse biases the diode, but v_i forward biases it. Now, $|v_i|$ is less than $|v_C|$, so the diode will be reverse biased. The current will flow through the load and the entire voltage will be dropped across the load. From Fig.3-88(c), it is also clear that as input increases from $-V_m$ to 0 V, the output voltage will increase from 0 V to V_m . When $v_i = 0$, $v_o = V_m - 0 = V_m$.

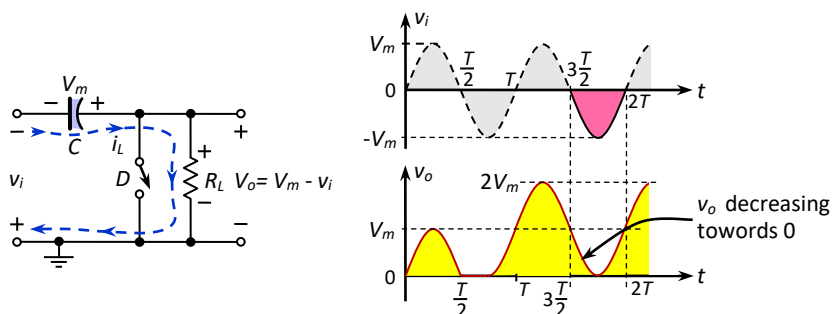
Fig.3-88(c): Circuit operation for from $3T/4$ to T

As shown in Fig.3-87(d), for the next positive half-cycle, the output voltage will be $v_o = v_C + v_i = V_m + v_i$. Here also the diode will remain reverse biased. As the input increases from 0 V to V_m , the output voltage will increase from V_m to $2V_m$. Actually, the positive half-cycle of input voltage will be added up with the constant voltage of the capacitor (V_m).

Fig.3-88(d): Circuit operation for from T to $3T/2$

Again for the next negative half cycle, the input voltage will be subtracted from the capacitor voltage as shown in Fig.3-87(e). So, the output voltage will be, $v_o = V_C - v_i = V_m - v_i$. When $v_i = -V_m$, the output voltage will be $v_o = V_m - V_m = 0$.

This process will be repeated and a periodic output voltage will be produced that has an average DC voltage of V_m .

Fig.3-88(e): Circuit operation for from $3T/2$ to $2T$

3.15.6 Practical Clamper Circuits

Operations of the above clamper circuits are described neglecting the diode forward voltage drop and the capacitor discharge slope. But in practical case these parameters will be effective.

The effect of the diode forward voltage drop is to decrease the shifting level. That is, for the simple clamper circuits, the value of level shifting will be $(V_m - 0.7 \text{ V})$ instead of V_m . For the forward biased clamper circuits, the input signal will be shifted to $(V_m + V - 0.7 \text{ V})$ instead of $(V_m + V)$. Similarly, for the reverse biased clamper circuits, shifting level will be $(V_m - V - 0.7 \text{ V})$, in place of $(V_m - V)$.

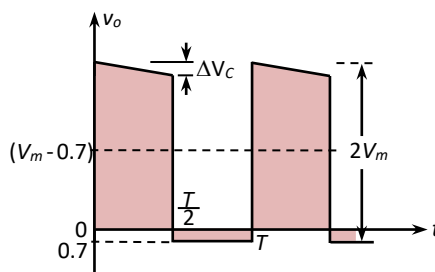


Fig.3-89: Output of a practical clamper

During the time when the diode remains reverse biased, the capacitor discharges through the load resistor. Thus, a slope is produced at the output waveform. So long as the load current is very small, it can be calculated as

$$I_C = I_L = \frac{v_o}{R_L} \quad (3-93)$$

As the capacitor is discharged during a half-cycle ($t = T/2$), capacitor voltage drop can be found as,

$$\Delta V_C = \frac{I_C \times t}{C} = \frac{I_C \times T}{2C} \quad (3-94)$$

Considering these effects the output waveform of a simple positive clipper circuit is shown in Fig.3-89.

Example 3-18

For the clamper circuit of Fig.3-90, draw the output waveform. Consider a Si diode and discharging of the capacitor.

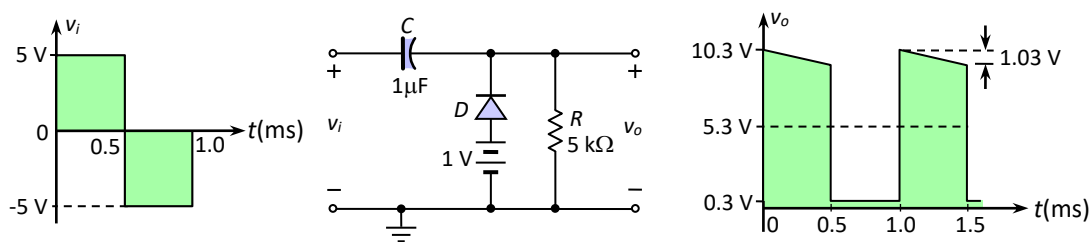


Fig.3-90: Clamping circuit for Example 3-28

Solution:

During the negative half-cycle the diode will be forward biased and the output voltage will be $(1 - 0.7) \text{ V} = 0.3 \text{ V}$. The capacitor will be charged by $5 - 1.7 = 5.3 \text{ V}$. During the positive half-cycle the diode is open so output voltage will be $v_i + V_C = 5 \text{ V} + 5.3 \text{ V} = 10.3 \text{ V}$. Thus, the output waveform will be shifted towards +ve direction by 5.3 volts. The minimum output voltage will be $V - V_D = 1 \text{ V} - 0.7 \text{ V} = 0.3 \text{ V}$ (forward biasing voltage -diode forward voltage drop).

The discharging current of the capacitor will be

$$I_C \approx \frac{v_o}{R} = \frac{10.3 \text{ V}}{5000} = 2.06 \text{ mA}$$

Discharging time of the capacitor is

$$\frac{T}{2} = \frac{1 \text{ ms}}{2} = 0.5 \text{ ms}$$

Therefore, change in capacitor voltage is

$$\Delta V_C = \frac{I_C \times t}{C} = \frac{2.06 \text{ mA} \times 0.5 \text{ ms}}{1 \mu\text{F}} = 1.03 \text{ V}$$

Comments: The time constant of this circuit is $\tau_d = RC = 5 \text{ k}\Omega \times 1 \mu\text{F} = 5 \text{ ms}$. The discharging time of the capacitor is only 0.5 ms. For this reason, the discharging slope is shown linear, but practically it will be exponential.

3.15.7 Summary of Clamper Circuits

So far we have discussed different types of clamper circuits for different types of input waveforms. To get the summary of the clamper circuits at a glance, all types of clamper circuits with square wave inputs are given in Fig.3-91.

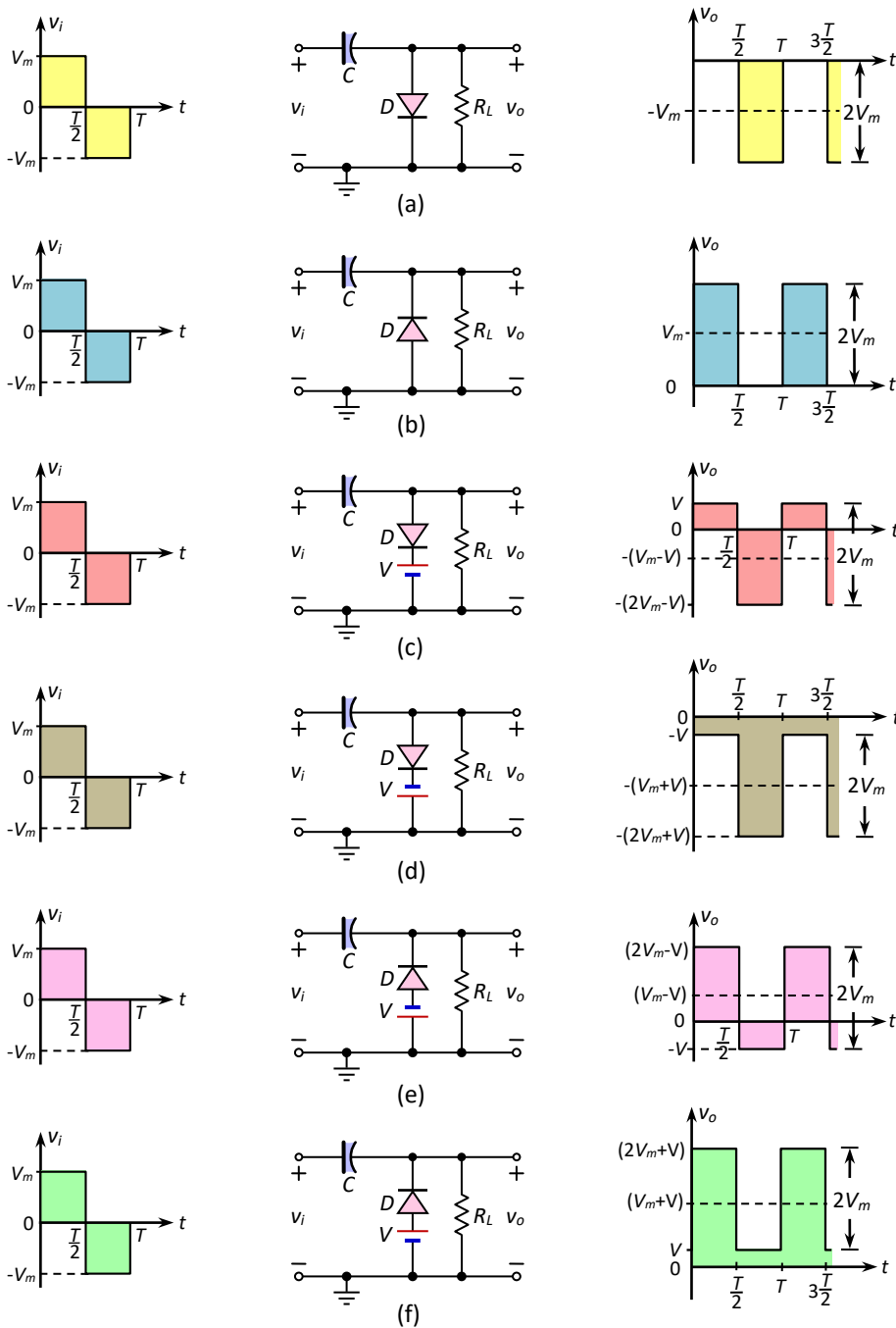
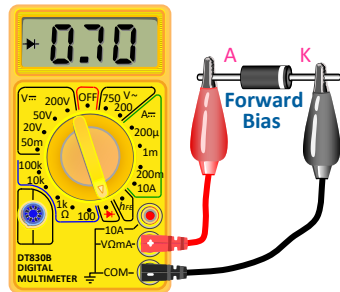


Fig.3-91: Different types of clamper circuits

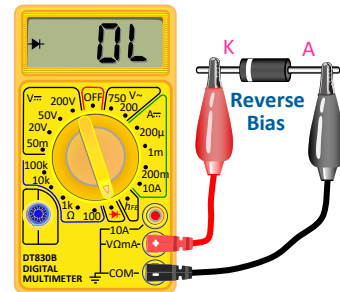
3.15.8 Applications of Clamper Circuits

The applications of clamper circuits are summarized below,

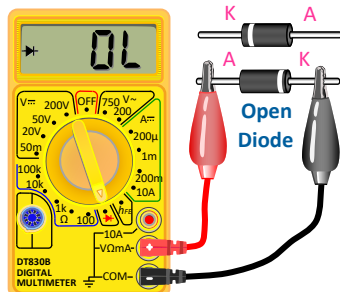
- It is used as a voltage multiplier.
- It is used to clamp the waveforms to a fixed DC level, and hence it is called DC restorer
- It is used for improving the reverse recovery time.
- It is used for removing the distortion in the signal.
- It is also used as test equipment
- For improving the overdrive recovery time clampers are used.



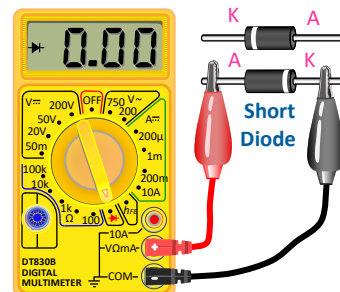
(a) Forward-bias test of a good diode shows '0.70' reading



(b) Reverse-bias test of a good diode shows 'OL' reading



(c) Forward and reverse-bias tests for an open diode gives 'OL' reading.



(d) Forward and reverse-bias tests for a short diode give '0' reading.

Fig.3-92: Diode testing using digital multimeter

3.16 Testing a Diode

A multimeter can be used as a fast and simple way to check a diode out of the circuit (not connected in circuit). A good diode will show an extremely high resistance (ideally open) with reverse bias and a very low resistance with forward bias. A defective diode may be either open or short. An open diode will show an extremely high resistance (ideally infinite) for both forward and reverse bias. A defective short diode will show zero or a low resistance for both forward and reverse bias. Most commonly a defective diode becomes open.

Many digital multimeters (DMMs) have a diode test function that provides a convenient way to test a diode. A typical DMM, as shown in Fig.3-92, has a small diode symbol where the function switch or selector switch has to be set. When set to this position, the meter provides an internal voltage sufficient to forward-bias and reverse-bias a diode. This internal voltage may vary among different

DMMs, but 2.5 V to 3.5 V is a typical range of values. The meter provides a voltage reading or other indication to show the condition of the diode under test.

3.16.1 Testing a Good Diode

In Fig.3-92(a), the red (positive) lead of the meter is connected to the anode and the black (negative) lead is connected to the cathode to forward bias the diode. If the diode is good, we will get a reading between 0.5 V and 0.9 V, with 0.7 V being typical for forward bias of Si diodes.

In Fig.3-92(b), the diode is turned around for reverse bias. If the diode is working properly, we will typically get a reading of 'OL' (Over Load). Some DMMs may display the internal voltage for a reverse-bias condition.

3.16.2 When the Diode is Defective

When a diode is open, we get an out-of-range 'OL' indication for both the forward-bias and the reverse-bias conditions, as illustrated in Fig.3-92(c). If a diode is shorted, the meter reads 0 V in both forward and reverse bias tests, as indicated in Fig.3-92(d).

3.16.3 Checking a Diode with the OHMs Function

DMMs that do not have a diode testing function can also be used to check a diode by setting the function switch on Ohm range (Ω). For a forward-bias check of a good diode, we will get a resistance reading that can vary depending on the meter's internal battery. Many meters do not have sufficient voltage on the Ohm setting to fully forward bias a diode and we may get a reading from several hundred to several thousand Ohms. For the reverse-bias check of a good diode, we will get an out-of-range indication such as "OL" on most DMMs because the reverse resistance is too high for the meter to measure.

Even though we may not get accurate forward and reverse resistance readings on a DMM, the relative readings indicate that a diode is functioning properly, and that is usually all we need to know. The out-of-range indication shows that the reverse resistance is extremely high, as we expect. The reading of a few hundred to a few thousand ohms for forward bias is relatively small compared to the reverse resistance, indicating that the diode is working properly. The actual resistance of a forward-biased diode is typically much less than 100 Ω .

3.17 Sample Questions

1. Explain how we can analyze a diode circuit using different models of diodes.
2. Explain the operation of a forward biased diode circuit when an AC signal is added in the circuit. What will happen if the diode is not initially forward biased?
3. What is load line and operating point? Explain how we can determine the load line and operating point of a diode circuit, with suitable figures.
4. What will happen to the load line and operating point, if the supply voltage is increased, and what will happen if the load resistance is increased?
5. Explain the operation of a diode as a switch.

6. Mention some examples of applications of diode as switches.
7. Draw the circuit diagram of an OR gate using diodes and resistors and explain its operation.
8. Draw the circuit diagram of an AND gate using diodes and resistors and explain its operation.
9. What is a rectifier circuit? Why do we need rectifier circuits?
10. Draw the circuit diagram of a half-wave rectifier circuit and describe its operation.
11. What will be the PIV rating of the half-wave rectifier circuit?
12. Derive the expression for the average value of the output of a half-wave rectifier.
13. Derive the expression for the RMS value of the output of a half-wave rectifier.
14. Draw the circuit diagram of a full-wave two-diode rectifier and describe its operation with suitable diagrams.
15. Determine the PIV rating of this type of rectifiers?
16. Draw the circuit diagram of a full-wave bridge rectifier and describe its operation with suitable diagrams.
17. Determine the PIV rating of the full-wave bridge rectifier.
18. Derive the expression for the average value of the output of a full-wave rectifier.
19. Derive the expression for the RMS value of the output of a full-wave rectifier.
20. What is ripple and ripple factor in rectifier circuits?
21. Derive the expression for the ripple factor of the output of a half-wave rectifier.
22. Derive the expression for the ripple factor of the output of a full-wave rectifier.
23. What do you mean by the efficiency of a rectifier? Determine the efficiency of the half-wave and full-wave rectifiers.
24. What is filter? Why we need filters in rectifiers?
25. With suitable circuits describe the operation process of a shunt capacitor filter.
26. Derive the expression for the average output voltage and ripple factor of the half-wave rectifier after filtering by a shunt capacitor filter.
27. Derive the expression for the average output voltage and ripple factor of the full-wave rectifier after filtering by a shunt capacitor filter.
28. With suitable circuits describe the operation process of a series inductor filter.
29. Derive the expression for the average output voltage and ripple factor of the half-wave rectifier after filtering by a series inductor filter.
30. Derive the expression for the average output voltage and ripple factor of the full-wave rectifier after filtering by a series inductor filter.
31. With suitable circuits describe the operation process of an LC filter.

32. Derive the expression for the average output voltage and ripple factor of the half-wave rectifier after filtering by an LC filter.
33. Derive the expression for the average output voltage and ripple factor of the full-wave rectifier after filtering by an LC filter.
34. What is critical value of L in an LC filter? Calculate the value of this critical value of the inductor.
35. What is bleeder resistance in power supply? Why do we need a bleeder resistance?
36. What is voltage regulation, source regulation and load regulation?
37. Explain the operation of Zener diode voltage regulator.
38. Explain how to design a Zener diode voltage regulator, if the maximum load current is given.
39. Explain how to design a Zener diode voltage regulator, if the maximum load current is not given.
40. Discuss about the performance of a Zener diode voltage regulator.
41. Draw the schematic diagram of a complete DC power supply and describe the function of every stage with waveforms.
42. What is voltage multiplier? What are the applications of voltage multipliers?
43. Draw the circuit diagram of a half-wave voltage doubler and describe its operation.
44. Draw the circuit diagram of a full-wave voltage doubler and describe its operation.
45. Draw the circuit diagram of a voltage tripler and describe its operation.
46. Draw the circuit diagram of a voltage quadrupler and describe its operation.
47. What is a clipper circuit? Give the classifications of the clipper circuits.
48. Describe the operation of a simple positive series clipper circuit with suitable diagrams.
49. Describe the operation of a simple negative series clipper circuit with suitable diagrams.
50. Describe the operation of a biased series positive clipper circuit with suitable diagrams.
51. Describe the operation of a biased series negative clipper circuit with suitable diagrams.
52. Describe the operation of a simple positive parallel clipper circuit with suitable diagrams.
53. Describe the operation of a simple negative parallel clipper circuit with suitable diagrams.
54. Describe the operation of a biased parallel positive clipper circuit with suitable diagrams.
55. Describe the operation of a biased parallel negative clipper circuit with suitable diagrams.
56. Explain how clipper circuit can be constructed using Zener diodes.
57. Explain how the noises induced in communication signals can be removed using clipper circuits.

58. What is clamper circuit? What is the use of a clamper circuit?
59. Give the classifications of the clamper circuits.
60. Draw the circuit diagram of a simple negative clamper and describe its operation.
61. Draw the circuit diagram of a simple positive clamper and describe its operation.
62. Draw the circuit diagram of a biased negative clamper and describe its operation.
63. Draw the circuit diagram of a biased positive clamper and describe its operation.
64. Give the summary of the operation of different types of clamper circuits.
65. Describe the operation of a clamper circuit with sinusoidal input.
66. Describe the effect of forward voltage of diode and the discharging effect of the capacitor on the operation of clamper circuits.
67. Describe how to test a diode whether it is OK or not, using multimeter.
68. Describe how to test a diode whether it is OK or not, using Ohmmeter.

3.18 Exercise:

1. For the circuit of Fig.3-93, determine the operating point of the diode considering three types of equivalent circuits discussed in Section 3.2. Assume that the diode is made of Si and hence its barrier voltage is $V_B = 0.7 \text{ V}$. Sketch the load line and operating point for all the three cases. [For the 3rd equivalent circuit use graphical method with a typical I - V characteristic curve of Fig.3-95.]

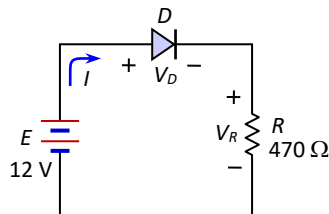


Fig.3-93: Basic diode circuit
for Exercise no. 1

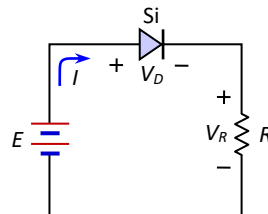


Fig.3-94: Basic diode circuit
for Exercise no. 2

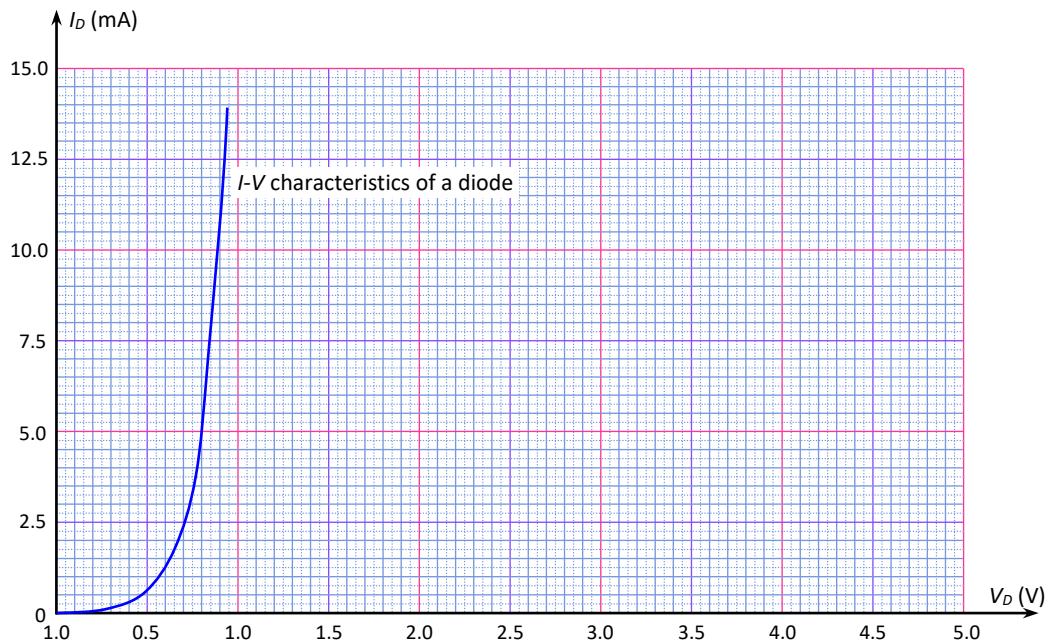


Fig.3-95: I-V characteristics of a diode

2. Determine the value of R for the circuit of Fig.3-94 that will result in a diode current of 10 mA, if $E = 10\text{ V}$. Use the characteristic curve of Fig.3-95 for the diode.
3. For the circuit of Fig.3-96, calculate the values of I_L , I_{R1} , V_o , and the exact diode voltage drop, V_D . Assume the bulk resistance of the diode $r_s = 10\ \Omega$, $I_s = 10^{-9}\text{ mA}$, and $T = 27^\circ\text{C} = 300\text{ K}$.
4. Determine V_o and I_D for the series circuit of Fig.3.97. The voltage drop across a red LED is 1.8 V.

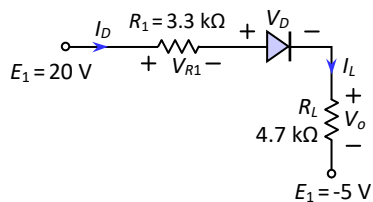


Fig.3-96: Diode circuit for Exercise 3

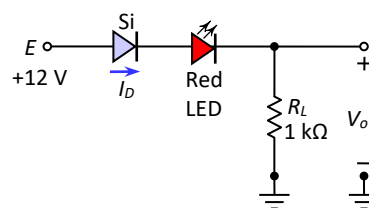


Fig.3-97: Diode circuit for Exercise 4

5. As shown in the Fig.3-98, a silicon diode is biased with 12 V DC and then an AC signal $e(t) = 5 \sin(\omega t)$ is applied. The bulk resistance of the diode is $2\ \Omega$, calculate the current through and the voltage drop across the diode. Sketch the current versus time graph.

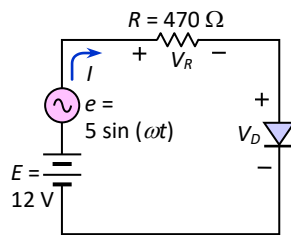


Fig.3-98: Circuit for Exercise 5

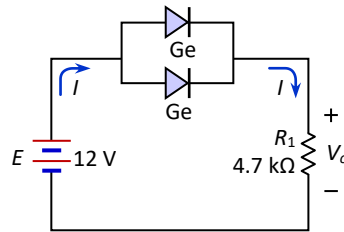


Fig.3-99: Circuit for Exercise 6

6. For the diode circuit of Fig.3-99, calculate the output voltage (V_o) considering the barrier voltages of the diode.
7. Repeat problem 6 considering two Si diodes.
8. Fig.3-100 shows a (12 V) battery charging circuit. v_i is a sinusoidal voltage with 24 V peak amplitude. Find the portion of each cycle during which the diode conducts and the battery gets charge. Also, find the peak value of the diode current and the maximum reverse-bias voltage that appears across the diode. Assume the internal resistance of the battery and source is zero.

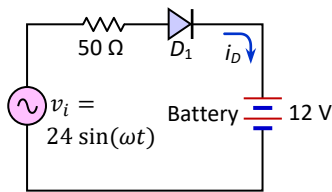


Fig.3-100: Battery charging circuit for Exercise 8

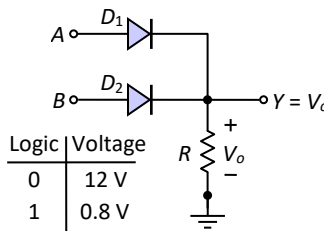


Fig.3-101: Circuit diagram for Exercise 9

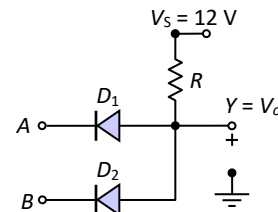


Fig.3-102: Circuit diagram for Exercise 10

9. For the logic gate implemented by diodes in Fig.3-101, determine the value of output voltage for the input logic combinations $A = 0, B = 1$ and $A = 0, B = 0$. Assume $V_D = 0.7$ V for each forward biased diode. What type of logic gate is it?
10. Repeat problem no. 9 for the logic gate of Fig.3-102.
11. Fig.3-103 shows an AND gate, with a voltage source ($E = 6$ V) in series with the output load. The logic level of this circuit is: HIGH = 12 V to 11 V and LOW = 0 to 1 V. Determine the output level for the input levels shown in the figure. Also determine the current flowing through the load resistance.

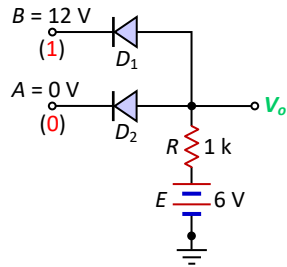


Fig.3-103: AND gate for Exercise 11

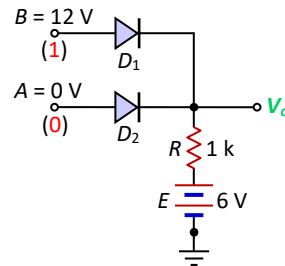
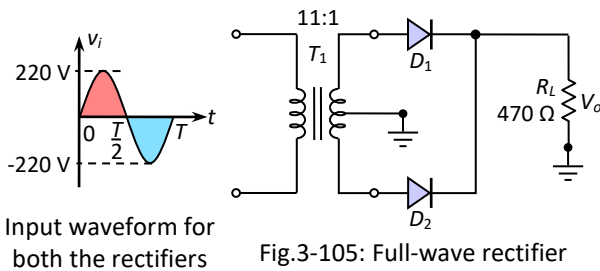


Fig.3-104: Circuit for given logic states

12. Repeat exercise no. 11 for the logic circuit of Fig.3-104.
13. A half-wave rectifier circuit is designed using a 9 V transformer. If the transformer secondary resistance (r_s) is $5\ \Omega$, diode internal resistance (r_D) is $60\ \Omega$ and the load resistance is $470\ \Omega$, calculate the DC output voltage and RMS output voltage i) Neglecting V_D and ii) Considering V_D for a Si diode.
14. For the full-wave rectifier of Fig.3-105, i) Calculate the peak output current, and the PIV rating of the diodes. ii) Draw the input waveforms to the diodes and the output waveforms. For both cases assume Si diodes with $V_B = 0.7\text{ V}$.



Input waveform for both the rectifiers

Fig.3-105: Full-wave rectifier for Exercise no. 14

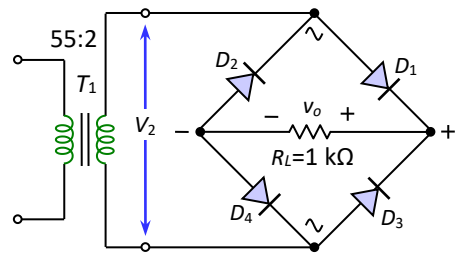


Fig.3-106: Full-wave rectifier for Exercise no. 15

15. For the full-wave rectifier of Fig.3-106, i) Calculate the peak output current, and the PIV rating of the diodes. ii) Draw the input waveforms to the rectifier and the output waveforms. For both cases assume Si diodes with $V_B = 0.7\text{ V}$.
16. A full-wave bridge rectifier is shown in Fig.3-107. If the diode resistance $r_D = 50\ \Omega$, and the source resistance $r_s = 50\ \Omega$, and $V_D = 0.7\text{ V}$ calculate the RMS and average output voltage i) With considering all parameters, and ii) Considering ideal condition.

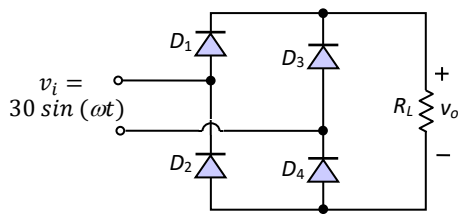


Fig.3-107: Full-wave bridge rectifier for Exercise no. 16

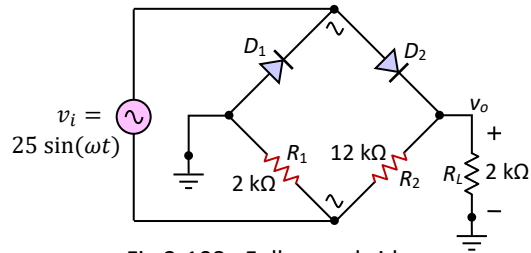


Fig.3-108: Full-wave bridge rectifier for Exercise no. 17

17. Find the output voltage across the load resistance, R_L , in the bridge circuit of Fig.3-108. Also determine the PIV rating for each diode.
18. In the rectifier circuit of Exercise 14, if a $330 \mu\text{F}$ capacitor is connected across the output, calculate the DC output voltage, ripple voltage, and the ripple factor considering an ideal diode and 60 Hz AC supply voltage.
19. Repeat Exercise no. 18 for the circuit of Fig.3-106 and Fig.3-107.
20. For the full-wave rectifier of Fig.3-107, calculate the value of capacitance required so that the ripple factor is not more than 1 % of average value.
21. For the full-wave rectifier of Fig.3-107, calculate the value of capacitance required so that the ripple factor is not more than 0.5 % of average value.

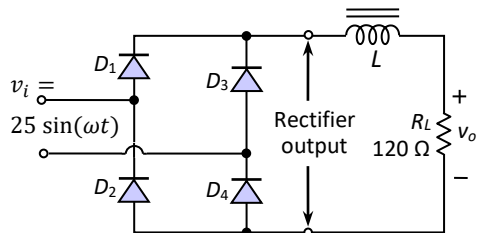


Fig.3-109: Full-wave bridge rectifier for Exercise 22

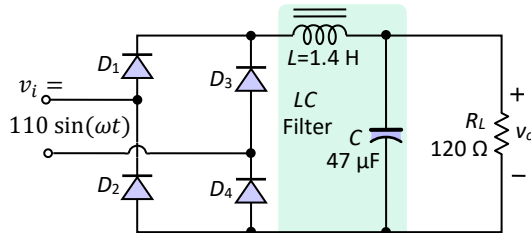


Fig.3-110: LC filter for Exercise no. 23

22. What should be the value of inductance to be used in the inductor filter of Fig.3-109 operating at 50 Hz, if the ripple factor is not to exceed 2 %.
23. Fig.3-110 shows an LC filter at the output of a full-wave rectifier. Calculate the value of DC output voltage and the ripple factor.
24. A single phase full wave rectifier uses a centre tapped 220-0-220 V, 50 Hz transformer. For a load current of 100 mA, design an LC filter so that the ripple factor is not more than 2 %.

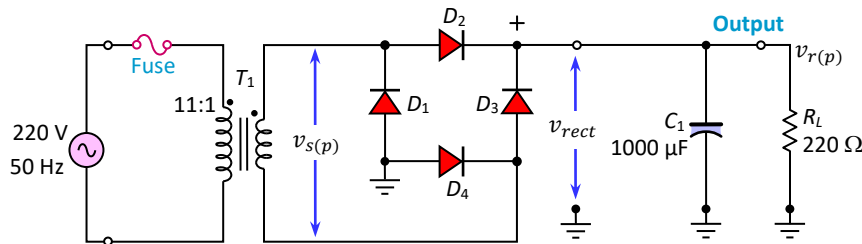


Fig.3-111: Full-wave bridge rectifier for Exercise 25

25. To solve the following problems, use the full-wave bridge rectifier given in the above Fig.3-111.
- Calculate the minimum value of the capacitor required so that the ripple factor of the DC output is not more than 2 % of its peak value.
 - Determine the peak value of the output rectified waveform. Consider the diode forward voltage drop, $V_F = 0.7 \text{ V}$ for each diode.
 - Calculate the peak-to-peak value of the output ripple, and ripple factor of the output voltage considering $C = 2200 \mu\text{F}$.
 - How much is the DC voltage applied to the load and the DC power dissipated in the load resistance considering $C = 1000 \mu\text{F}$.
26. (a) Determine V_L , I_L , I_Z and I_R for the network of Fig.3-112, if $R_L = 180 \Omega$.
- (b) Repeat part (a) if $R_L = 470 \Omega$.
- (c) Determine the value of R_L that will establish maximum power conditions for the Zener diode.
- (d) Determine the minimum value of R_L to ensure that the Zener diode is in the breakdown state.

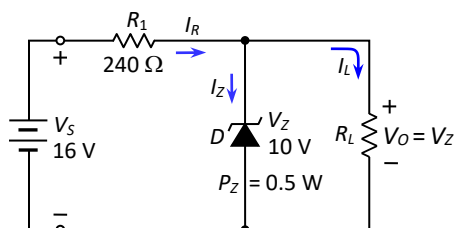


Fig.3-112: Circuit for problem no. 26

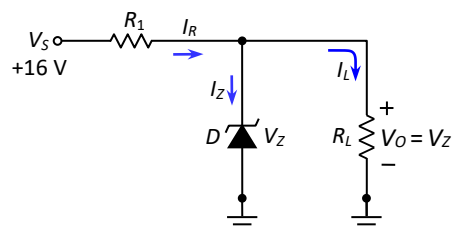


Fig.3-113: Circuit for problem no. 27

27. (a) Design the network of Fig.3-113 to maintain $V_L = 10 \text{ V}$ and for a variable load current of $I_L = 0$ to 120 mA .
- (b) Determine $P_{Z(max)}$ for the Zener diode of part (a).
28. i) For the network of Fig.3-114(a), determine the range of V_S that will maintain V_L at 8 V and not exceed the maximum power rating of the Zener diode.

- ii) For the network of Fig.3-114(b), determine V_L , I_L , I_Z and I_R . Determine the minimum power rating for the resistor and the Zener diode.

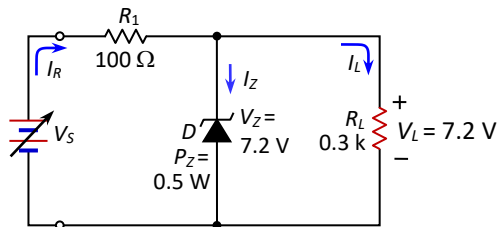


Fig.3-114: (a) Zener diode circuit for Exercise

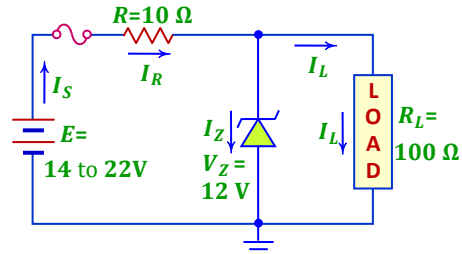


Fig.3-114: (b) Zener diode circuit for Exercise

29. Design a voltage regulator that will maintain an output voltage of 20 V across a load of 1 kΩ with an input that varies from 30 V to 50 V. [Determine R_1 , V_Z , power rating of R_1 and the Zener diode]

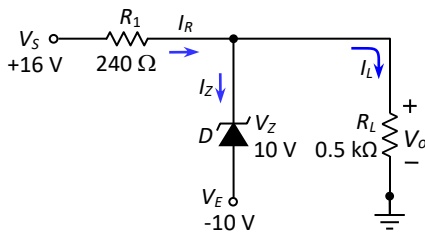


Fig.3-115: Zener diode circuit for Exercise no. 30

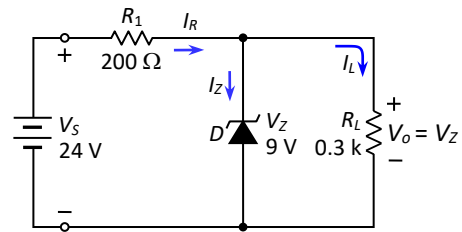


Fig.3-116: Zener diode circuit for Exercise no. 31

30. For the voltage stabilizer circuit of Fig.3-115, determine the value of i) I_R , I_Z , and I_L , ii) V_{R1} , V_O and V_Z , and iii) P_{R1} , P_Z , and P_L [Power ratings of different components]
31. Repeat the problem no. 30 for the Zener diode circuit given in Fig.3-116.
32. Fig.3-117 shows a half-wave rectifier circuit with a step down transformer. The turn ratio of the transformer is 0.1, and the input AC voltage of the transformer is 220 V. The Si diode has a barrier voltage of 0.7 V. Determine: (i) Peak value, (ii) Average value and (iii) RMS value of the output voltage.
33. Repeat the process if the turn ratio of the transformer is 0.5.

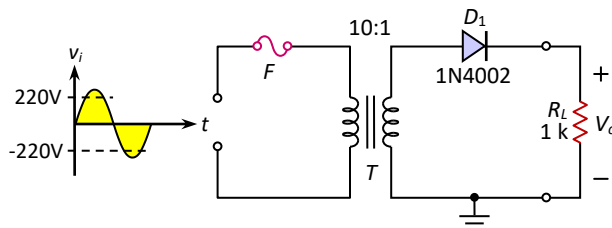


Fig.3-117: Half-wave rectifier for Exercise no. 32 and 33

34. Using the given I - V characteristics of the diode, determine the current and the voltage across the diode and the resistor as shown in Fig.3-118.

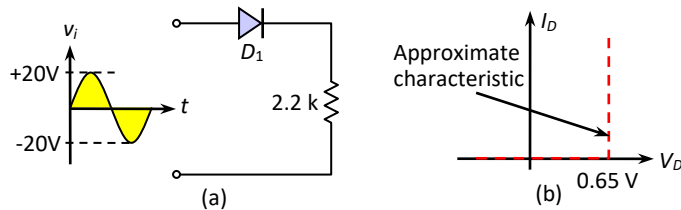


Fig.3-118: Diode circuit for Exercise 34 (a) Diode circuit
(b) Approximate I - V characteristics of the diode

35. Using the given I - V characteristics of the diode, determine the current and the voltage across the diode and the resistor as shown in Fig.3-119.

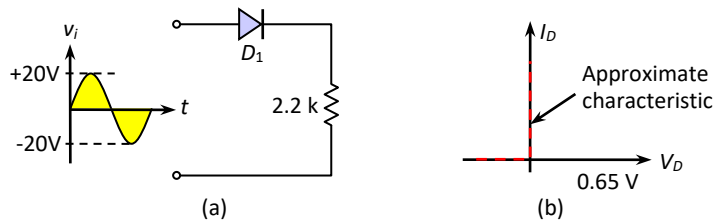


Fig.3-119: Diode circuit for Exercise 35 (a) Diode circuit
(b) Approximate I - V characteristics of the diode

36. For the given waveforms of Fig.3-120, determine the RMS (V_{rms}) and average voltage (V_{dc}).

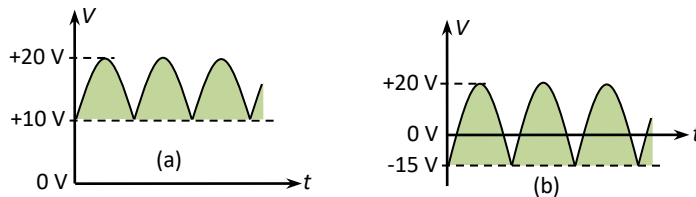


Fig.3-120: Output waveforms for Exercise 36

37. A rectifier filter produces a DC output voltage of 50 V with a peak-to-peak ripple voltage of 0.4 V . Calculate the ripple factor.
38. A full-wave rectifier has a peak output voltage of 20 V and output frequency of 120 Hz . A 40 mF capacitor-filter is connected to the rectifier. Calculate the peak-to-peak ripple and the DC output voltage across a $500\text{ }\Omega$ load resistance.
39. A half-wave rectifier is to supply 24 V to a load of $500\text{ }\Omega$. The peak-to-peak ripple voltage is not to exceed 10% of the average output voltage and the input AC voltage frequency is 50 Hz . Calculate the capacitance of the filter capacitor. [Ans. $2000\text{ }\mu\text{F}$]

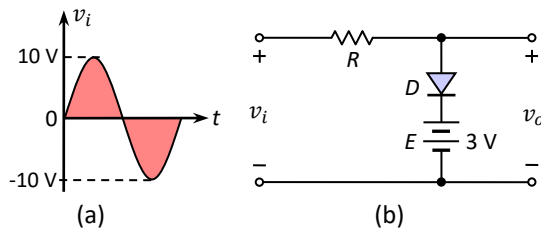


Fig.3-121: (a) Input signal, and
(b) Clipper circuit for Exercise 40

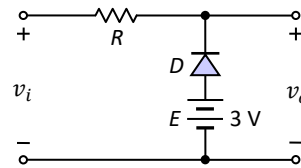


Fig.3-122: Clipper circuit
for Exercise 41

40. For the given input waveform and clipper circuit in Fig.3-121 determine and sketch the output waveform assuming i) An ideal diode and ii) A practical Si diode.
41. Repeat problem 40 for the clipper circuit of Fig.3-122.

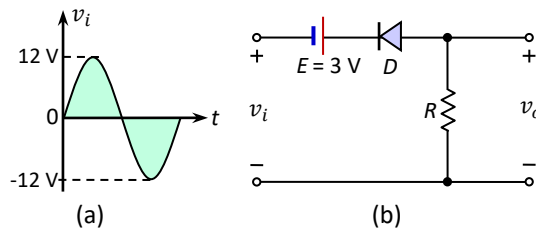


Fig.3-123: (a) Input signal, and
(b) Clipper circuit for Exercise 40

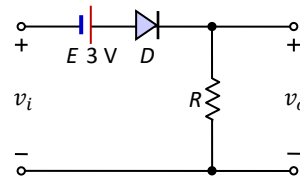
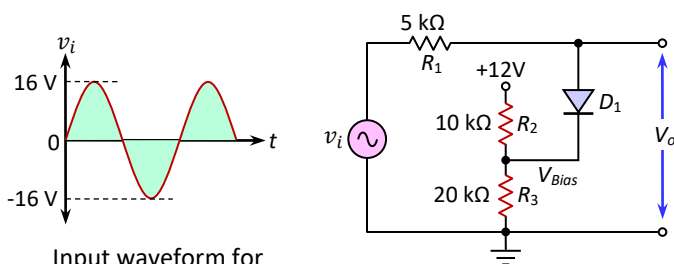


Fig.3-124: Clipper circuit
for Exercise 41

42. For the given input waveform and clipper circuit in Fig.3-123 determine and sketch the output waveform assuming i) An ideal diode and ii) A practical Si diode.
43. Repeat problem 40 for the clipper circuit of Fig.3-124.
44. Draw and describe the output waveform for the sinusoidal voltage applied to the given circuit of Fig.3-125.



Input waveform for
Exercise no. 44 and 45

Fig.3-125: Circuit for Exercise no. 44

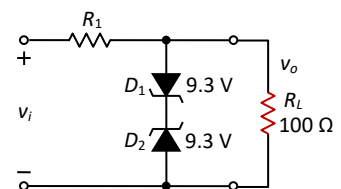


Fig.3-126: Circuit for
Exercise no. 45

45. Analyze the clipper circuit of Fig.3-126 and determine the value of R_1 , different values of voltages and currents. Assume: $V_{Z1} = V_{Z2} = 4.3$ V, $I_{ZK} = 2$ mA, $P_Z = 1/4$ W, and Zener diode forward voltage drop is 0.7 V.
46. Repeat problem no. 45 considering $R_L = 1$ kΩ.

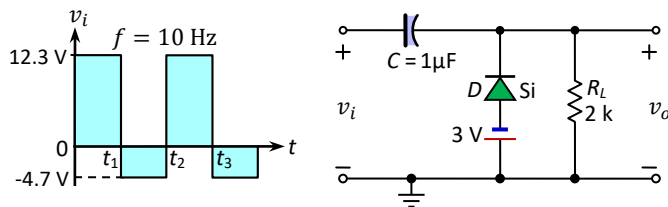


Fig.3-127: (a) Input waveform, and (b) Clamper circuit for Example 3-

47. Determine the output waveform of the network of Fig.3-127 for the given input waveform. Consider a Si diode with $V_F = 0.6 \text{ V}$.

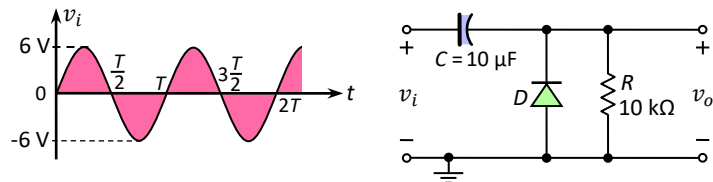
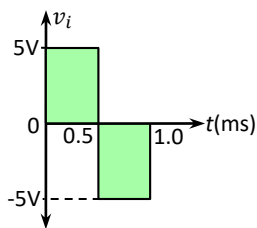
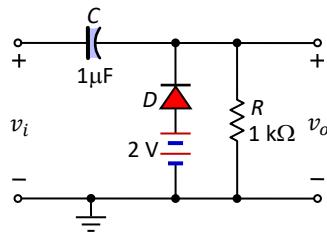
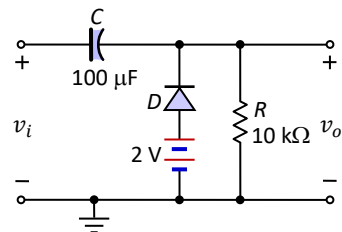


Fig.3-128: Circuit diagram and input waveform for Exercise no. 48

48. Determine the output waveform of the network of Fig.3-128 for the given input waveform.

Input waveform for
Exercise no. 49 and 50Fig.3-129: Clamping circuit
for Exercise no. 49Fig.3-130: Clamping circuit
for Exercise no. 50

49. For the clamping circuit of Fig.3-129, draw the output waveform. Consider a Si diode and discharging effect of the capacitor.
50. Repeat problem no. 49 for the clamper circuit of Fig.3-130.